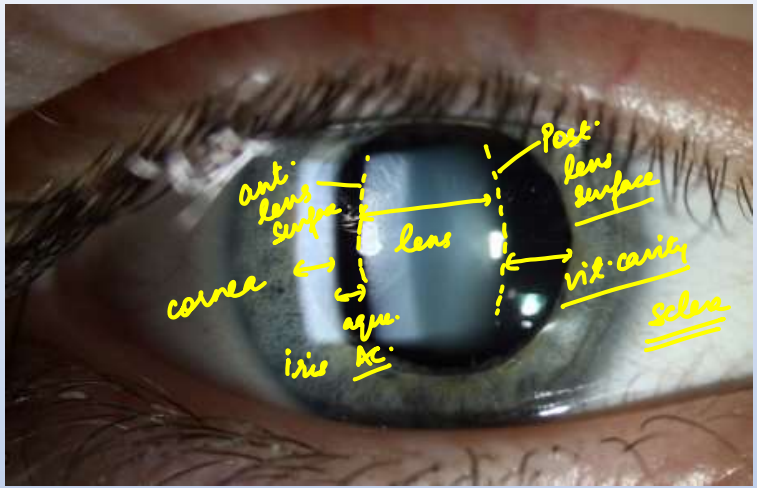
SUBJECT - OPHTHALMOLOGY
REVISION VIDEO

Revision Ophthalmology 2026

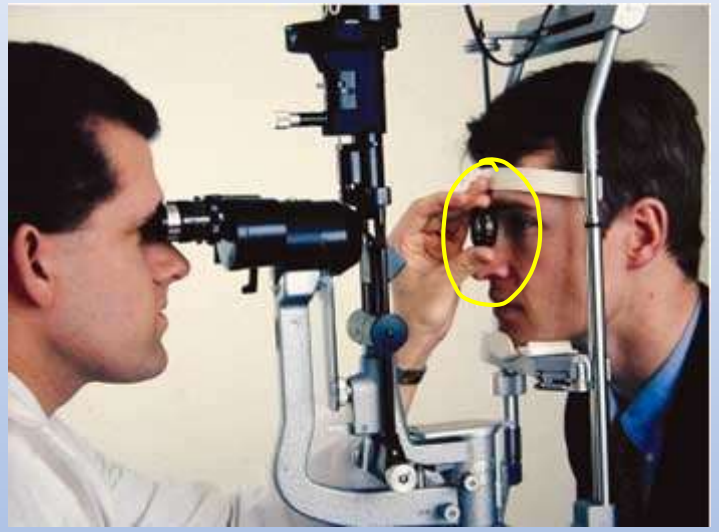
Anuradha Dhawan

Exam Techniques

Slit lamp
Bimicroscope


Post. lens surface
Ant. lens surface
Lens
cornea
vit. cavity
aque.
Sclera
iris
AC.

Fundus Examination Using Slit lamp



+90D
Lens

Direct Ophthalmoscope

For Central Retina (Disc and macula)

Magnification 15 X

Virtual Erect Image

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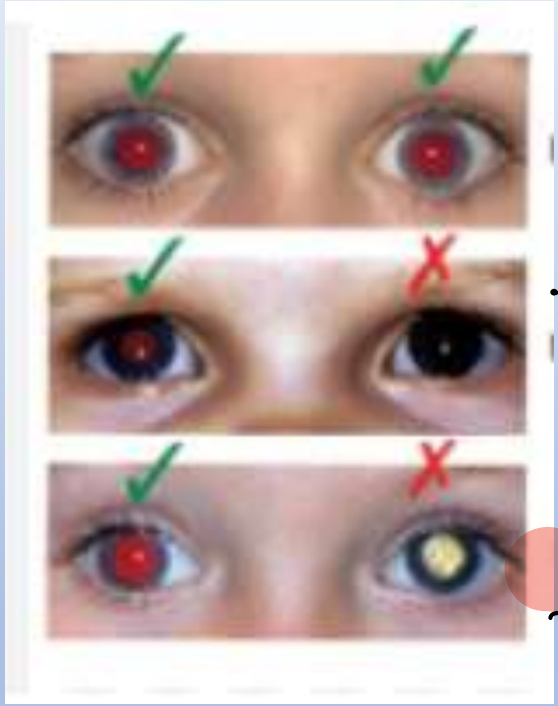
Distant Direct Ophthalmoscopy DDO

at distance of 25 cm

Distant Direct Ophthalmoscopy



Red glow

- dense corneal opacity
- mature cataract

no glow

white pupillary reflex

Leucocoria

Retinoblastoma
Total RD

Indirect Ophthalmoscope

Peripheral Retina

Magnification 3-5 X

Real Inverted Image

Binocular stereoscopic view

condensing lens:
+20D

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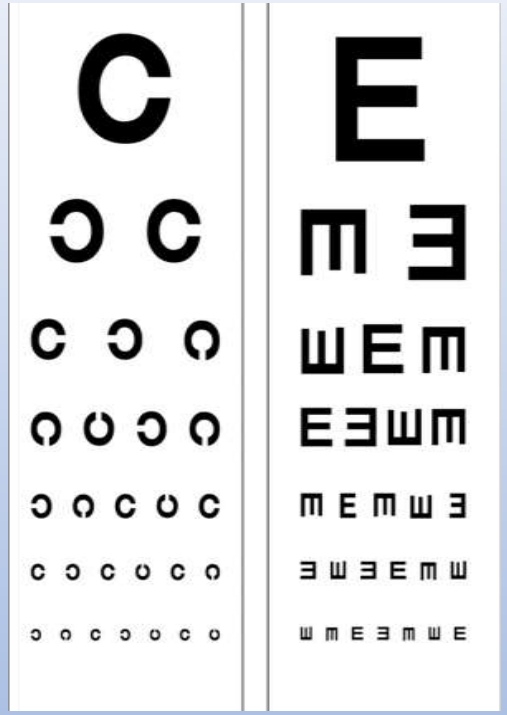
Snellen's distance vision chart.



Landolt's C and E chart.

Testing Distance 6m

6/6

- ETDRS : Early Treatment of Diabetic Retinopathy study
- Research Trials
- Testing distance : 4m

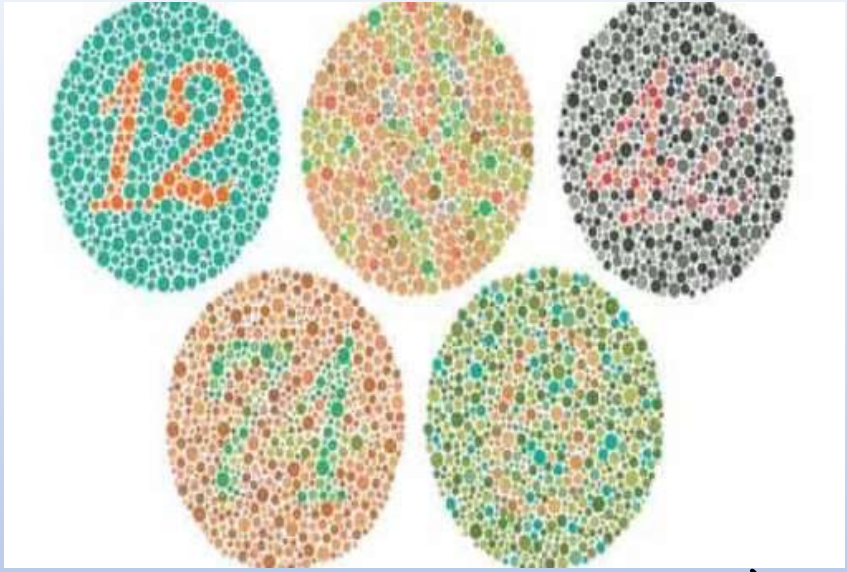
Pelli Robson Chart (contrast sensitivity)

Visual Acuity Tests for preverbal Children (upto age of 1 year)

(objective test)

- Optokinetic Nystagmus
- Preferential Acuity Test (Teller's)
- Visual Evoked Potential

Older children: Irony Ball Test
coin test
Allen's picture cards
Cardiff " "



Ishihara color vision Test : most common screening Test

Most specific color vision Test

↓

FM 100 hue Test

Humphrey's Visual Field Test

Perimetry

Static Perimeter

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Embryology of the Eye: Important points

1. **Structures developing from Surface Ectoderm:**

Lens, Lacrimal Gland, Epithelium of Conjunctiva & Cornea

2. **Structures developing from Neuroectoderm:**

Retina, Optic nerve, Secondary Vitreous, Epithelium of Iris & Ciliary body, Ciliary zonules, Sphincter pupillae & Dilator pupillae muscles

Embryology of the Eye: Important points

3. **Structures developing from Neural Crest Mesenchyme:**

Sclera, Corneal layers except epithelium, Iris stroma, Choroid, Trabecular meshwork, Primary Vitreous

4. **Structures developing from Mesodermal mesenchyme:**

Ciliary body, extraocular muscles, Iris stroma, Choroid

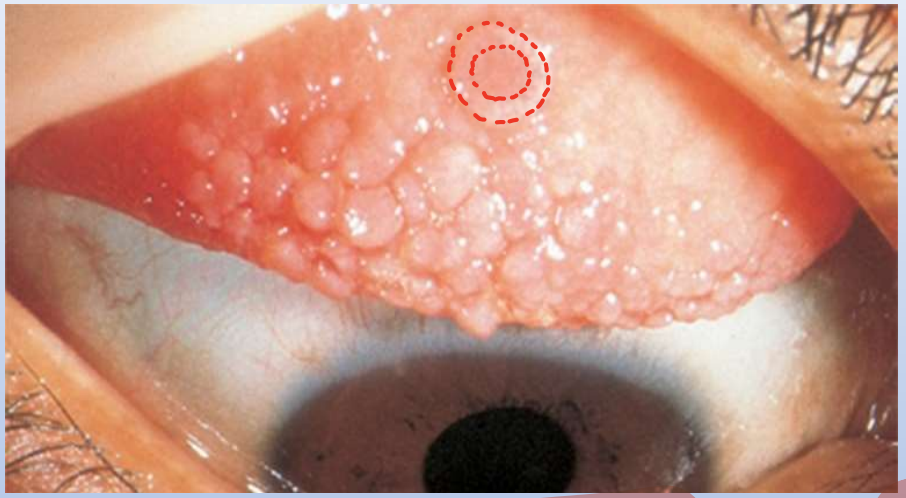
5. **Uveal Melanocytes: Neural Crest Cells**

Conjunctiva

VERNAL KERATOCONJUNCTIVITIS

VKC

(Spring Catarrah)



Giant papillae

Cobblestone appearance

Clinical Case

7 y old boy

itching, redness

Ropy discharge

worsening of symptoms in Summer

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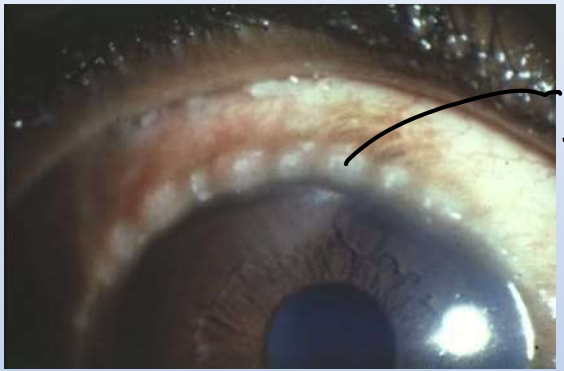
VERNAL KERATOCONJUNCTIVITIS

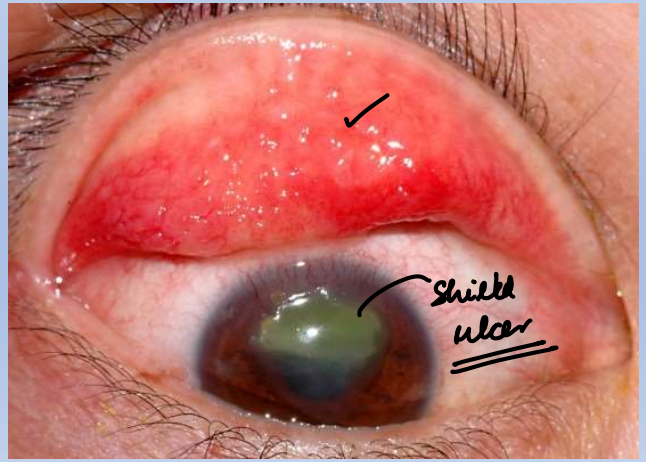
- Type 1 hypersensitivity
- Itching with ropy discharge
- Cobblestone papillae
- Horner Tranta's spots
- Shield ulcer
- Pseudogerontoxon (cupid's bow)
  peripheral corneal opacity

VERNAL KERATOCONJUNCTIVITIS



Rx: Topical (eyedrops)

- Antihistaminics
- Mast cell stabilizers
- Mild steroids



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Giant Papillary Conjunctivitis

GPC

use of soft contact lenses

clinical Q

20y girl

no association with season

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PHYLCTENULAR CONJUNCTIVITIS

- TYPE IV hypersensitivity reaction
- Secondary to TB, staph
- Corneal involvement: Fascicular ulcer

Clinical Case: malnourished

Tx: - Topical steroids

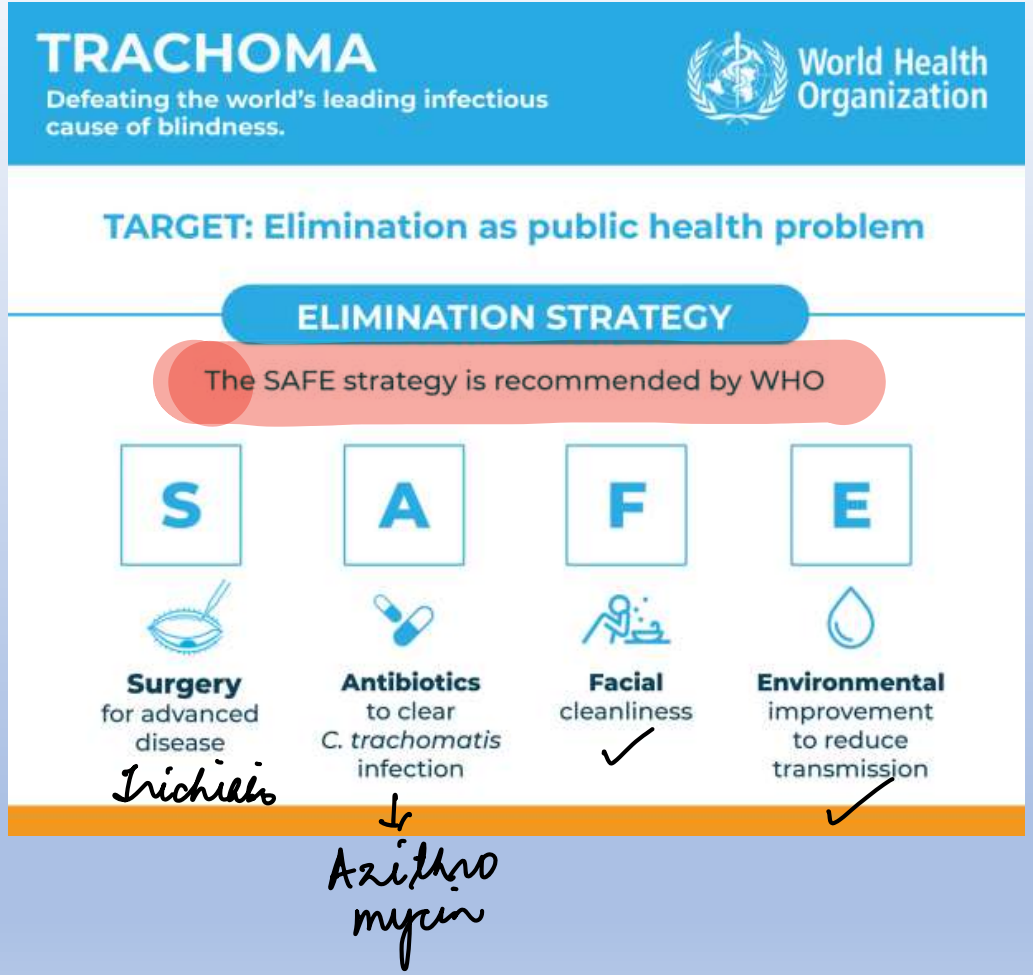
TRACHOMA

- By Chlamydia trachomatis A, B, Ba, C
- Sago grain follicles in upper tarsal conjunctiva
- Herbert's follicles on limbus
- Arlt's line subtarsal fibrosis
- Herbert's pits

healing stage

TRACHOMA

- Trichiases
- corneal ulceration
- corneal opacity


TRACHOMA
Defeating the world's leading infectious cause of blindness.
World Health Organization
TARGET: Elimination as public health problem
ELIMINATION STRATEGY
The SAFE strategy is recommended by WHO
S
A
F
E
Surgery for advanced disease
Antibiotics to clear C. trachomatis infection
Facial cleanliness
Environmental improvement to reduce transmission
Trichiasis
Azithromycin

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Swimming Pool Conjunctivitis

- Chlamydia trachomatis Serotypes D to K

Adult inclusion conjunctivitis

- Adenovirus

Viral Conjunctivitis

- Follicular Conjunctivitis COVID-19.
- Adenovirus: Epidemic Keratoconjuncitivitis – Nummular (cornea) lesions, lymphadenopathy

OPHTHALMIA NEONATORUM

Conjunctivitis in a neonate (within 28 days) of birth

matted eye lashes
discharge in newborn.

CHEMICAL: within hour — Silver nitrate sol- used as prophylaxis. ×

GONOCOCCAL: most dangerous
corneal ulcer

CHLAMYDIAL: most common

STAPH AUREUS

HERPES SIMPLEX — antiviral

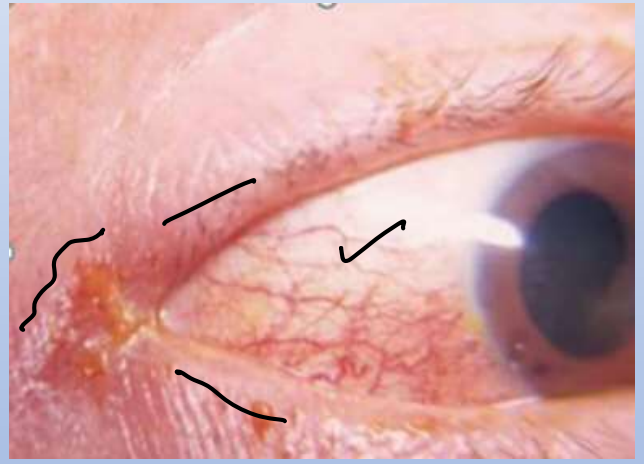
Prophylaxis: Povidone-Iodine 5%.
Tetracycline oint.

Diplobacillary conj

Angular Conjunctivitis - Moraxella axenfeld

Inner & outer canthi

Excoriation of skin due to a proteolytic enzyme.



Clinical Case.

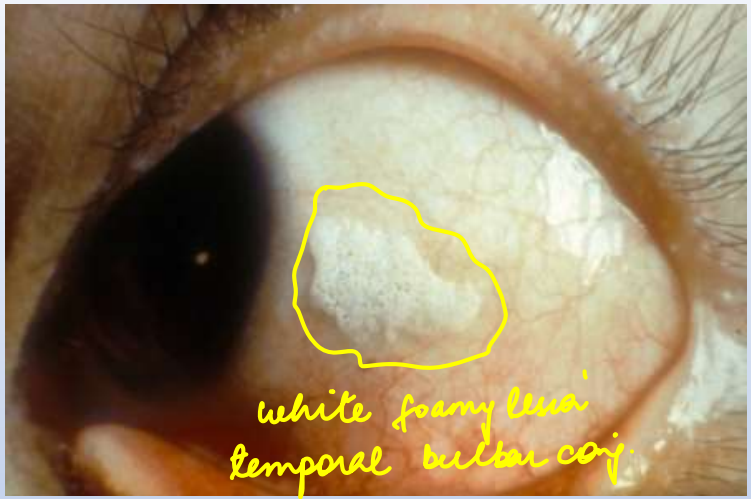
Add Zinc oxide

Vitamin A deficiency. loss of goblet cells.

Xerophthalmia

XN	Night blindness ✓ 1st symptom
X1A	Conjunctival xerosis - 1st sign.
X1B	Bitot spot ✓
X2	Corneal xerosis ✓
X3A	Corneal ulceration, less than 1/3
X3B	Corneal ulceration, more than 1/3
XS	Corneal scar
XF	Xerophthalmic fundus : pigmentary changes.

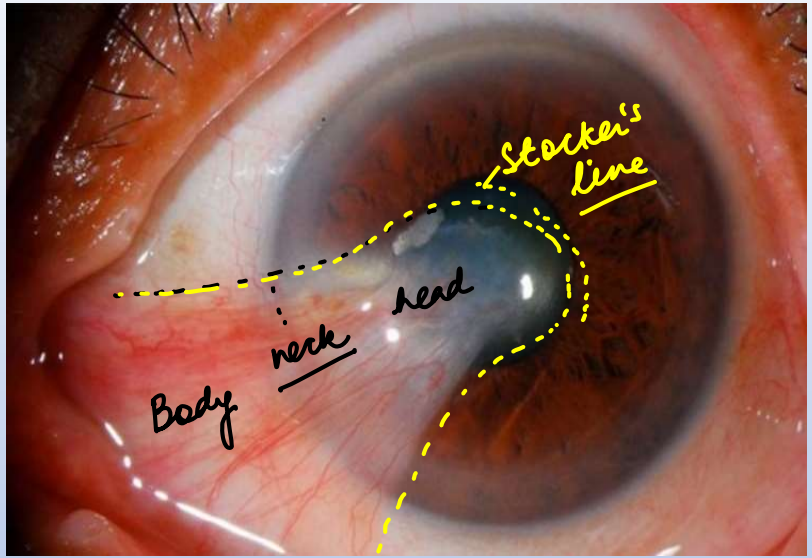
] Keratomalacia



Bitot spot

- Vit A supplementation:
  Day 0, 1, 14.
- Children < 1yr: oral dose of 1 lac IU
- Children>1yr: oral dose of 2 lac IU

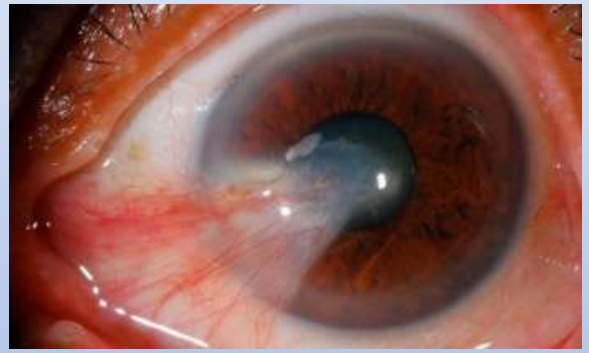
Clinical Case : Farmer / field work



Pterygium

PTERYGIUM

- Elastotic degeneration, UV rays , dusty environment
- Head, neck, Body: Stocker's line : line of iron deposition



simple excision
+
V. high Recurrence

★ ↑Ideal/Best:

- Excision with Conjunctival autograft, use of MMC
  Amniotic membrane graft

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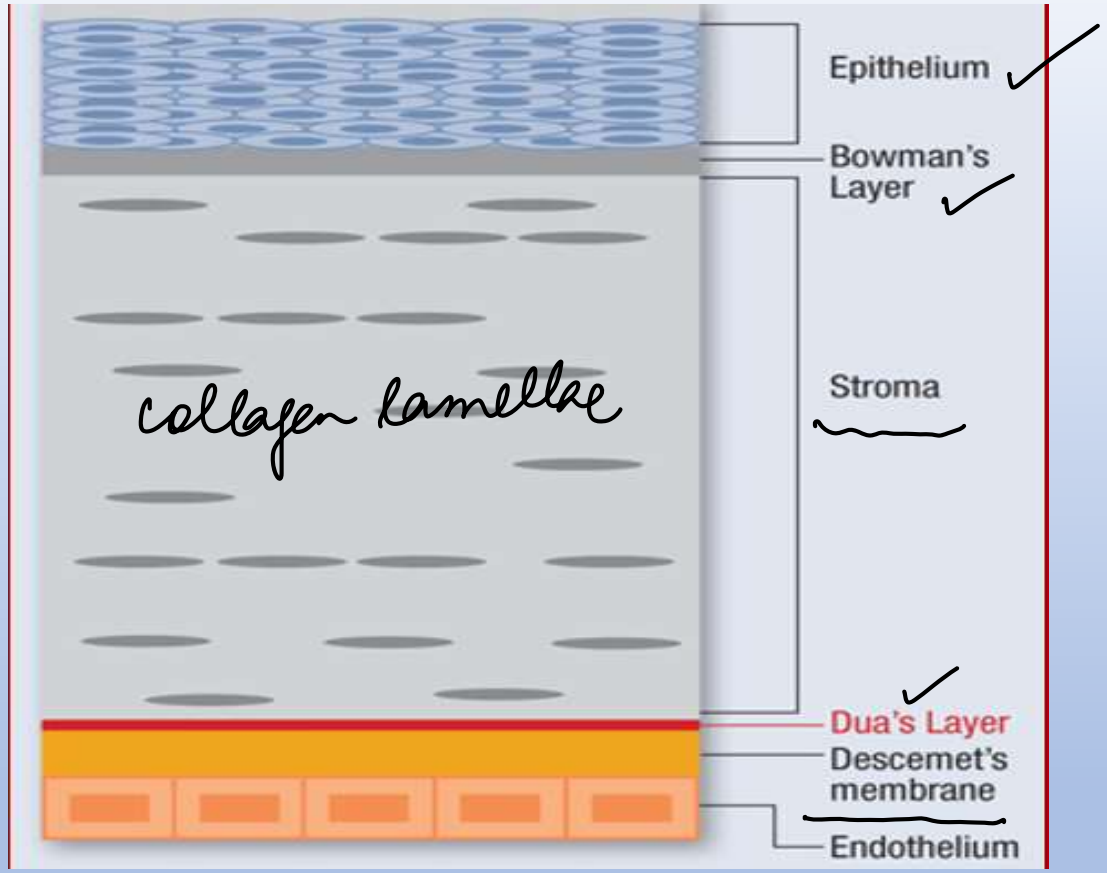
Cornea


Epithelium
Bowman's Layer
collagen lamellae
Stroma
Dua's Layer
Descemet's membrane
Endothelium

CORNEAL LAYERS – 6 layers

1. Latest Corneal layer Discovered – Dua's layer

2. Most Metabolically Active Layer of Cornea – ENDOTHELIUM
   ↓ Na-K+ ATPase pump.

3. Layer Maximally responsible for Corneal Hydration/Transparency – ENDOTHELIUM

4. BOWMAN Membrane layer can not regenerate.

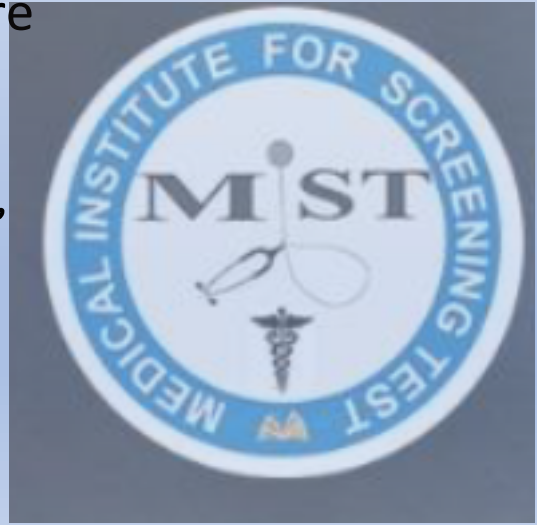
5. Descemet Membrane – Most elastic Layer, can regenerate

6. Dua's Layer – Toughest Layer

7. corneal Epithelium = Barrier function
   ↓ Palisade of vogt (limbal stem cells)

CORNEAL ENDOTHELIUM

- ENDOTHELIAL CELLS SHAPE – Hexagonal single layer
  ↳ don't regenerate.
- Na-K ATPase pump present in ENDOTHELIUM
- Normal Endothelial cell count: 2500-3000 cells/ mm square
- Below 500 cells/mm2 in endothelium: Permanent corneal decompensation → corneal Edema



1. PACHYMETRY – Measurement of Corneal Thickness

2. KERATOMETRY – Measurement of Central Corneal Curvature

3. CORNEAL TOPOGRAPHY - Mapping of entire anterior Corneal surface

4. CORNEAL TOMOGRAPHY – Mapping of entire anterior & posterior Corneal surface

5. SPECULAR MICROSCOPY – Study of Endothelial Cell Count + Morphology

Test for Corneal Sensations

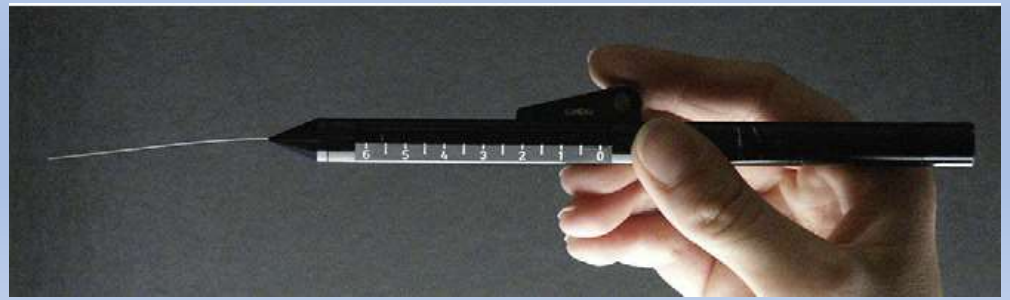
1. Corneal Reflex – Cotton Wisp Test

Sensory 5th N.
(afferent)

Efferent: 7th N.
(orbicularis oculi → Blink)

2. Corneal Aesthesiometer – Cochet Bonnet aesthesiometer

Quantitative measurement.



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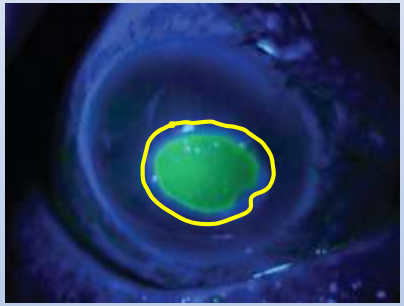
Vital Dyes for Staining

1. • Sodium Fluorescein dye → cobalt blue filter

orange colored dye which stain the area of corneal abrasion (loss of corneal epithelium)



2. • Rose Bengal — Red colored green dye

3. • Lissamine green — green dye

white light

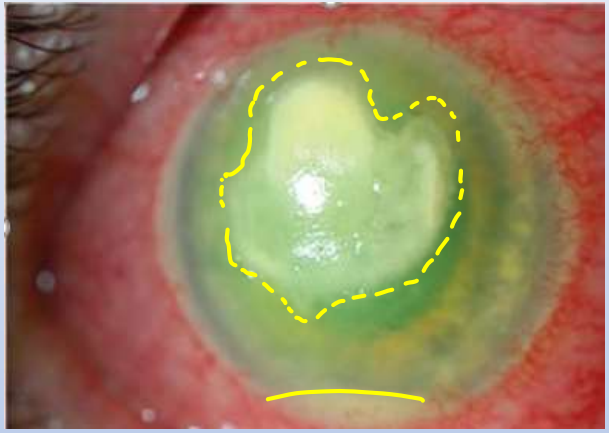
stain the area of conj. & cornea which devitalized epithelial cells

Dry eye staining

Corneal Ulcers: Bacterial

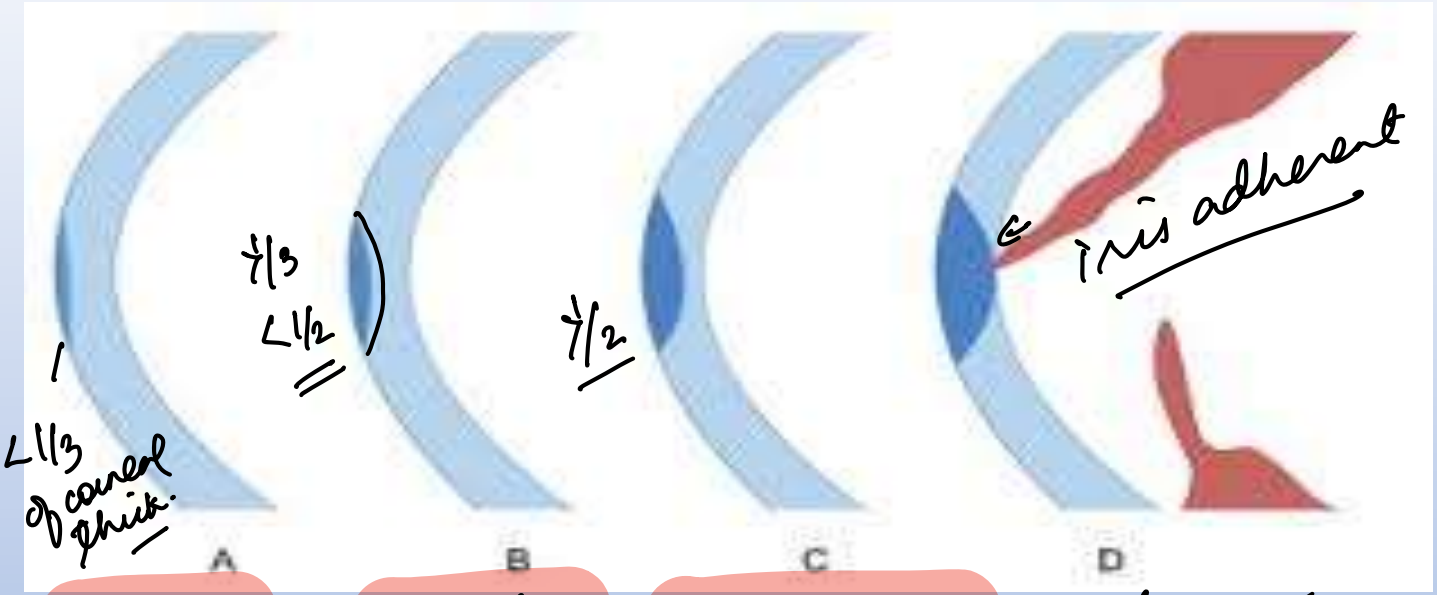
- Developed countries – Contact lens use (Most common Risk Factor)
- Developing countries – Trauma
- Photophobia, Redness, Pain, Decreased vision, Discharge
- Staph, Pneumococcus

Bacterial Corneal Ulcer

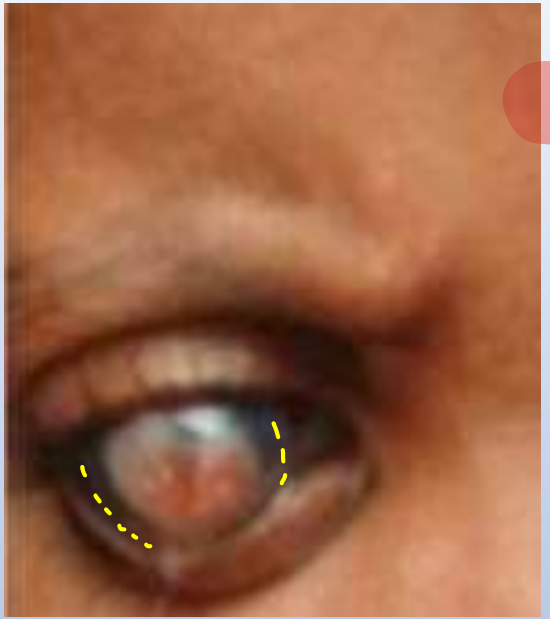


- Wet Infiltrate, Ulcer margins defined,
- Hypopyon is sterile, Mobile,
- Early Vascularization (+), Perforation (+)

Sequelae of the corneal ulcer:



A	B	C	D
Nebular opacity	macular opacity	Leucomatous corneal opacity	Adherent Leucoma
↓	↓	↓	
± vision disturbance • see iris details	iris details not visible	iris not visible	



Anterior Staphyloma

Pseudocornea
↓
ectasia with
iris incarceration

clinical case Q: : malnourished child

Corneal Ulcers: Fungal

Most common Cause – Filamentous Fungi

- (Fusarium & Aspergillus)

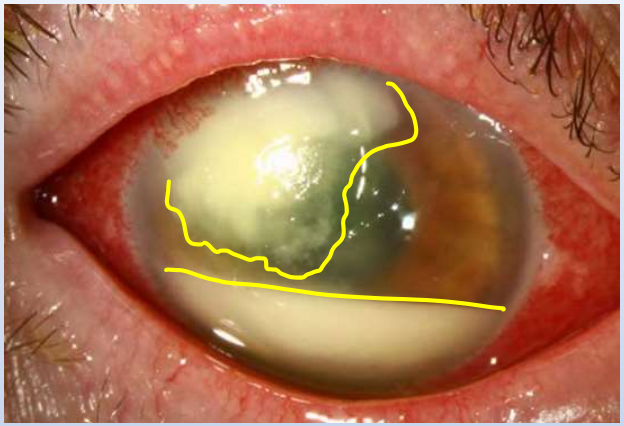
- Candida: In Immunocompromised / Diabetic

Trauma with organic & vegetative matter

Farmers, gardeners, fields
↳
tree twig.

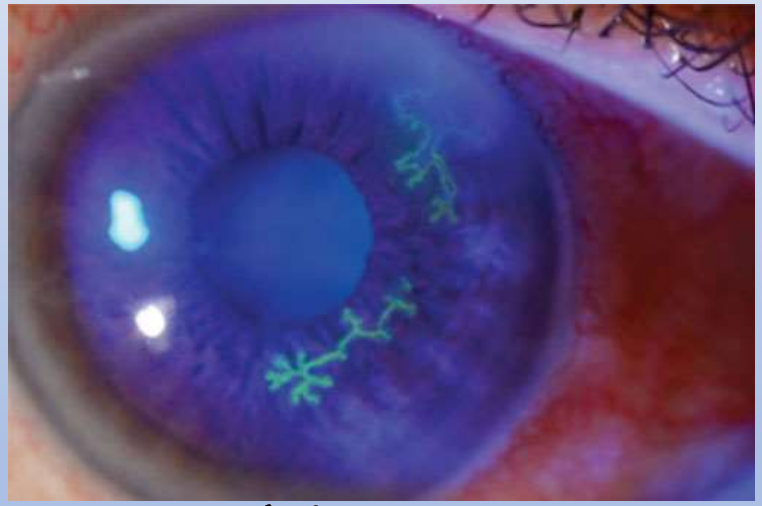
Corneal Ulcers: Fungal

- Dry, Raised infiltrate, Feathery, ill defined margins
- Satellite lesions
- Hypopyon is Large, Dome shaped
- Non Sterile, Fixed
- Vascularization & Perforation Uncommon



Viral Keratitis : HSV, HZV

- Reduced Corneal Sensations * HSV keratitis



Dendritic ulcer HSV.

Herpes simplex keratitis

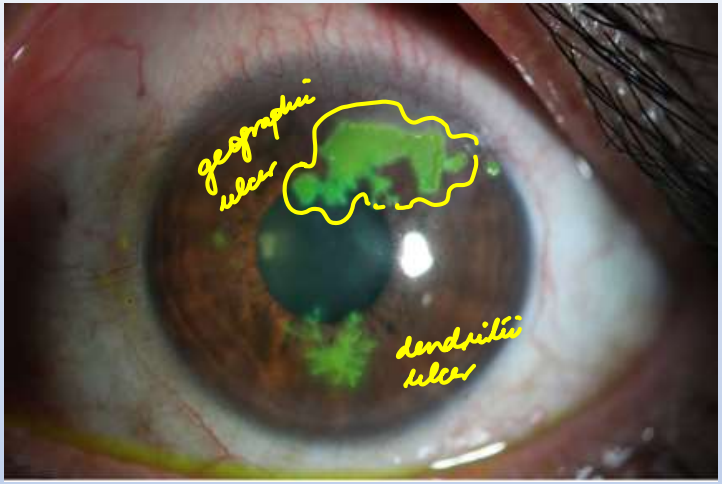
- Punctate Epithelial Keratopathy
- Dendritic ulcer → Geographic Ulcer
- Stromal keratitis, Disciform Keratitis = Immune mediated (no active infection or epi inv.)
- Metaherpetic keratitis

neurotrophic ulcer

↓

no active infect..

Viral Keratitis



Dendritic ulcer
Tm: Acyclovir ointment

Disciform keratitis

Topical
Rx: Steroids
+ acyclovir

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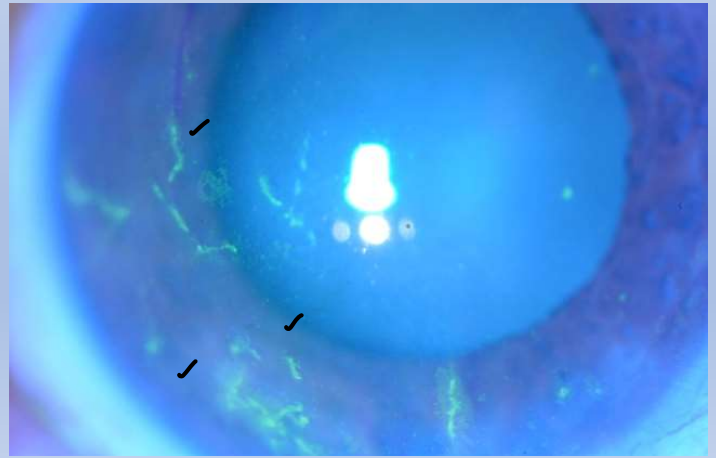
Herpes zoster Keratitis

Immunocompromised patients

- vesicular skin eruptions
- Fever.

Hutchinson's sign.

Pseudodendritic keratitis



Rx: Topical Acyclovir
systemic acyclovir

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Acanthamoeba Keratitis

- Contact Lens Use with poor Hygiene, Sharing of CL

↓ tap water / River water

- Highly Painful ulcer
- Ring shaped ulcer

→ Keratoneuritis.

clinical case

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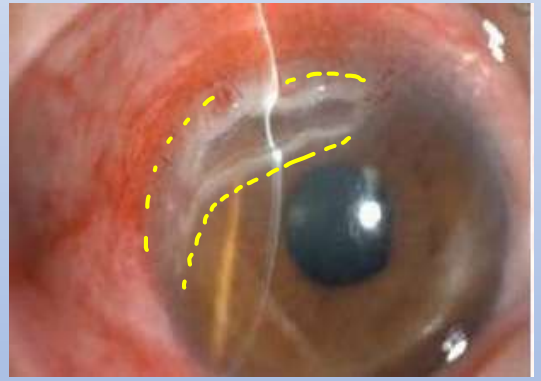
Mooren's Ulcer

- Idiopathic Peripheral Ulcerative Keratitis
- Rodent Ulcer
- Auto-immune



Rx:
Steroids, Immunosupp.
: Keratoplasty

- Drug contraindicated in infective corneal ulcer – steroids

- Drug given in all infective corneal ulcer – Cycloplegics: Atropine

relieve ciliary spasm: reduce pain, inflammation

- Antimicrobial

- Neuroparalytic Keratitis

7th nerve palsy

↓

orbicularis oculi weakness

painful corneal ulcer

→ Lagophthalmos

↓

exposure keratitis

- Neurotrophic Keratitis

↓

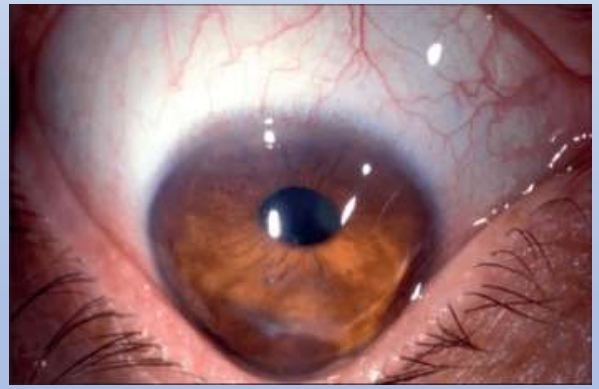
Reduced corneal sensation

5th nerve:

1. 5th nerve palsy, Tumors, CVA
2. Leprosy, DM.
3. Herpes.

Keratoconus

progressive thinning of corneal stroma.



Munson's sign

Clinical case

irregular astigmatism.

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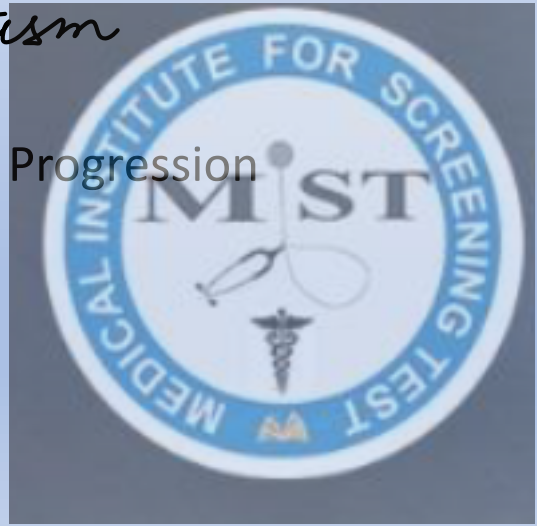
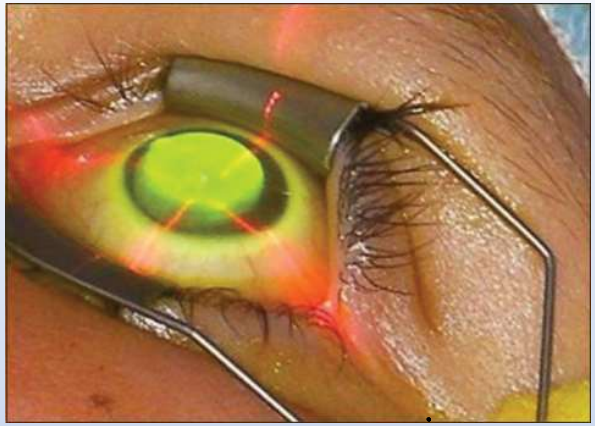
Keratoconus

- Irregular Astigmatism – Scissoring Reflex on Retinoscopy – RIZZUTI SIGN
- Oil Droplet reflex – DDO
- Irregular Cornea on Placido Disc
- Munson's sign
- Stromal Thinning on Slit lamp
- Vogt's Stria – Vertical in Descemet's membrane
- Fleisher Ring – Iron Ring in Epithelium

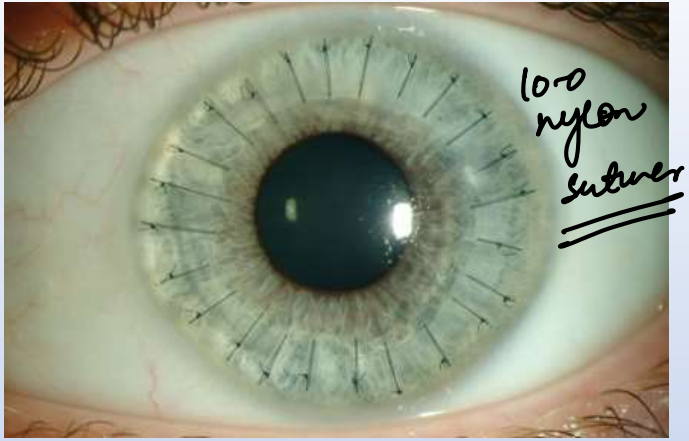
Keratoconus

- CORNEAL HYDROPS - Sudden Rupture of Descemet's membrane, Corneal Scarring
  ↓↓ painful decrease in vision
  corneal edema
- IOC: Corneal Tomography
- RGP contact lenses in Mild cases irregular astigmatism
- Collagen Cross linking C3R- Riboflavin Dye, UV-A Ray's, Reduce Progression
- INTACS (Intra Stromal Corneal Ring Segments) PMMA.
- KERATOPLASTY

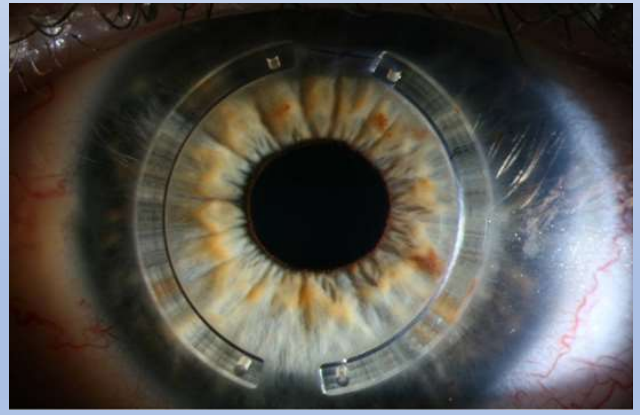




Collagen Cross linking



Penetrating Keratoplasty

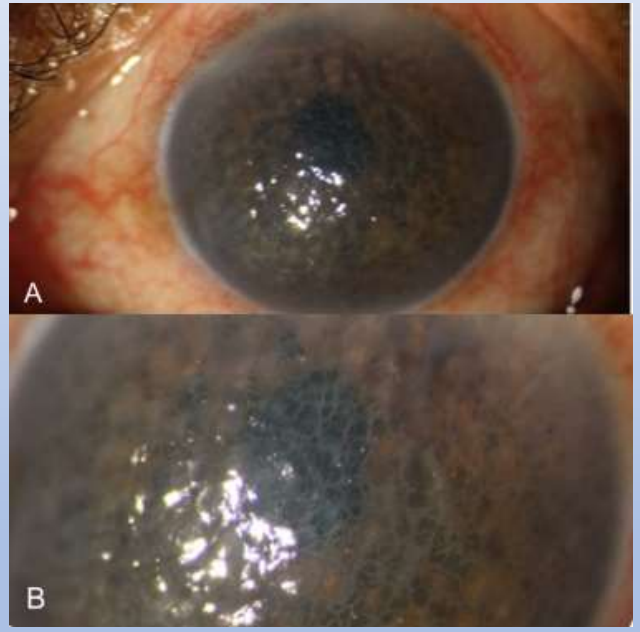


INTACS

Fuch's Endothelial Dystrophy

Elderly patients

Clinical Case



Early cases: corneal guttata
- hypertonic Salaine
- Keratoplasty

Advanced cases: epithelium bullae

Bullous keratopathy

*: Cataract surgery = further loss of end. cells
&
Do specular microscopy

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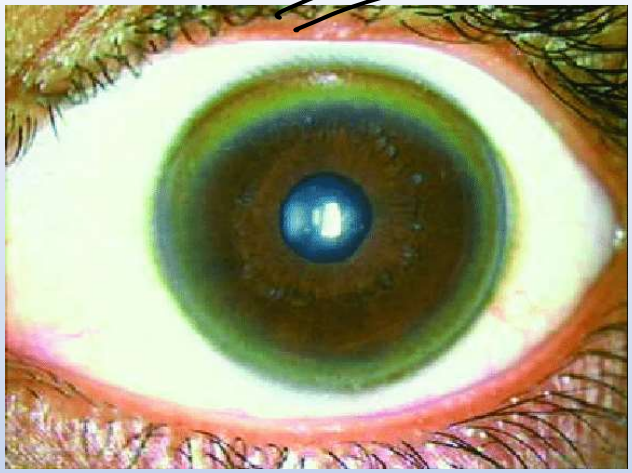
CORNEAL DEPOSITS

- ARCUS SENILIS - Lipid deposition, Starts in superior and Inferior part
- ARCUS JUVENILIS → asso. with hyperlipidemia
- Band Shaped Keratopathy- Calcium deposition
  - chronic uveitis JRA, phthisis bulbi, Silicone oil
  - multiple myeloma.
- Iron Deposition
  - Stocker's line, Fleischer ring
- Vortex Keratopathy – Chloroquine, Amiodarone, Tamoxifen, Indomethacin, Chlorpromazine, Netrasudil, Fabry's disease

cornea verticillata

CORNEAL DEPOSITS

Clinical Case:



Hepatolenticular disease ↑

Keyser Fleisher (KF) ring

- Copper deposition = Wilson's Disease
- Chalcosis = copper deposition

Gonioscopy – Best method for early KF ring

Not present in all patients of Wilson disease

Neurological ds → 90% will have KF ring

Hepatic ds → 50% will have KF ring

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Keratoplasty: Methods of Corneal Preservation

1. Short Term Storage:

a. **Moist Chamber method**: whole globe is stored at 4*C upto 24 hours.

b. **McCarey Kaufman (M-K) medium**: Donor corneoscleral button is stored upto 96 hours.

- Most commonly used medium.
- Consists of Tissue culture-199, 5% Dextran-40, HEPES buffer and gentamycin.

Methods of Corneal Preservation

2. Intermediate Term Storage:

- Storage upto 2 weeks.
- Contains chondroitin sulphate.
- **Most common : Optisol GS, Cornisol.**

3. Long-term preservation:

*Glycerine
*Organ culture method
*Cryopreservation

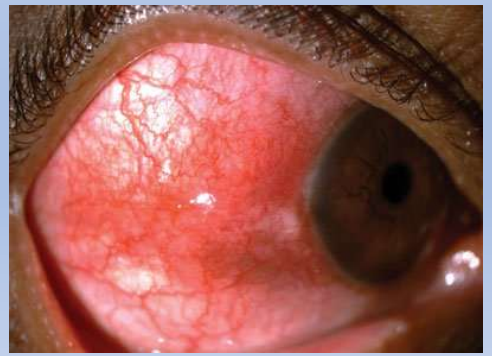
rupture in Blunt trauma

- Thinnest part of sclera: Posterior to insertion of extraocular muscles
- Most common association of Scleritis is Rheumatoid arthritis

- Episcleritis

mild discomfort
self limiting
2.5% Phenylephrine eye drops

- Scleritis

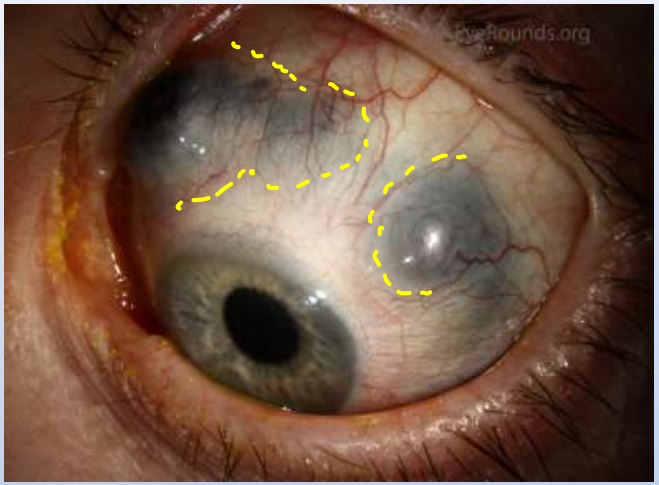


Severe pain
Steroids

necrotizing scleritis

- Scleromalacia perforans

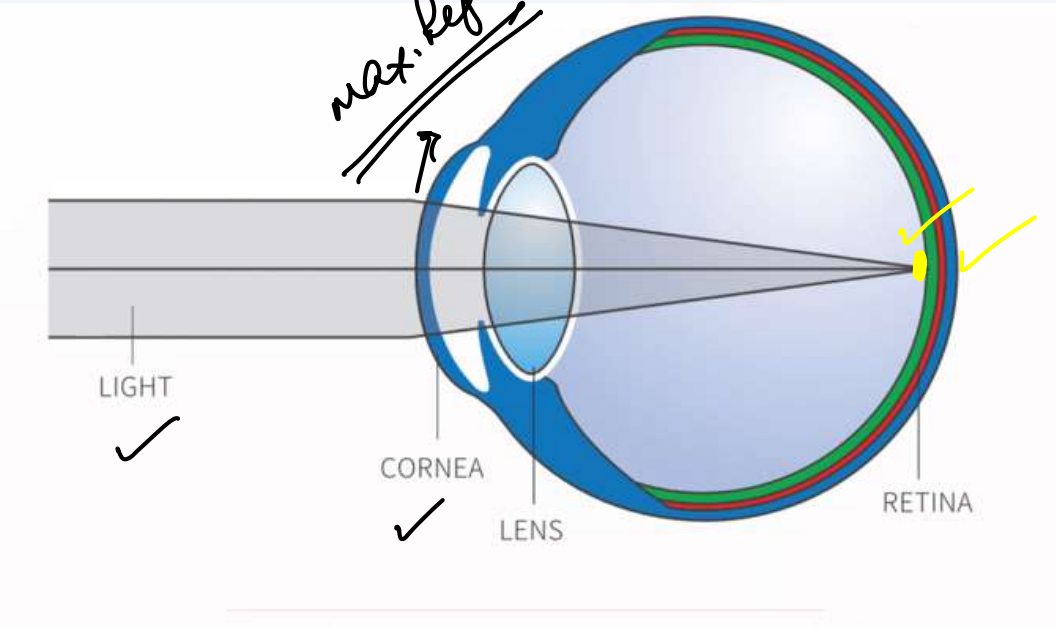
seen in long standing RA



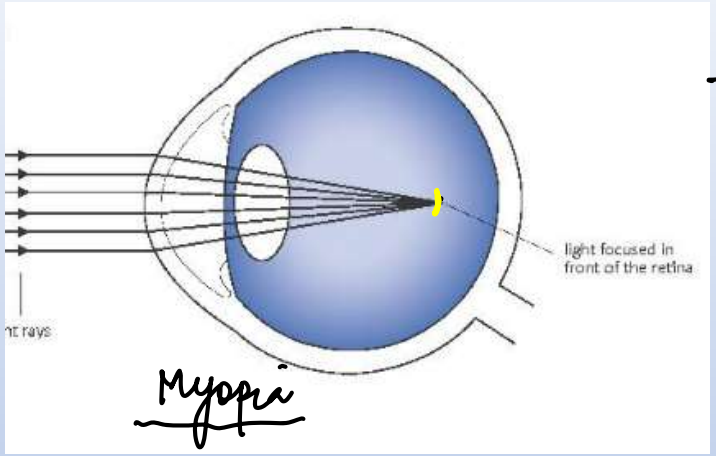
- Posterior Scleritis:

T-sign on USG Bscan (fluid in subtenon area)
Exudative retinal detachment

Refractive Errors

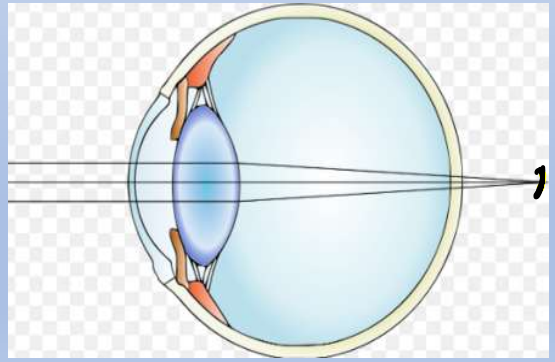


- Total refractive power of the eye +60 D
- Refractive power of the cornea +45 D
- Refractive power of the lens +15 D



Myopia

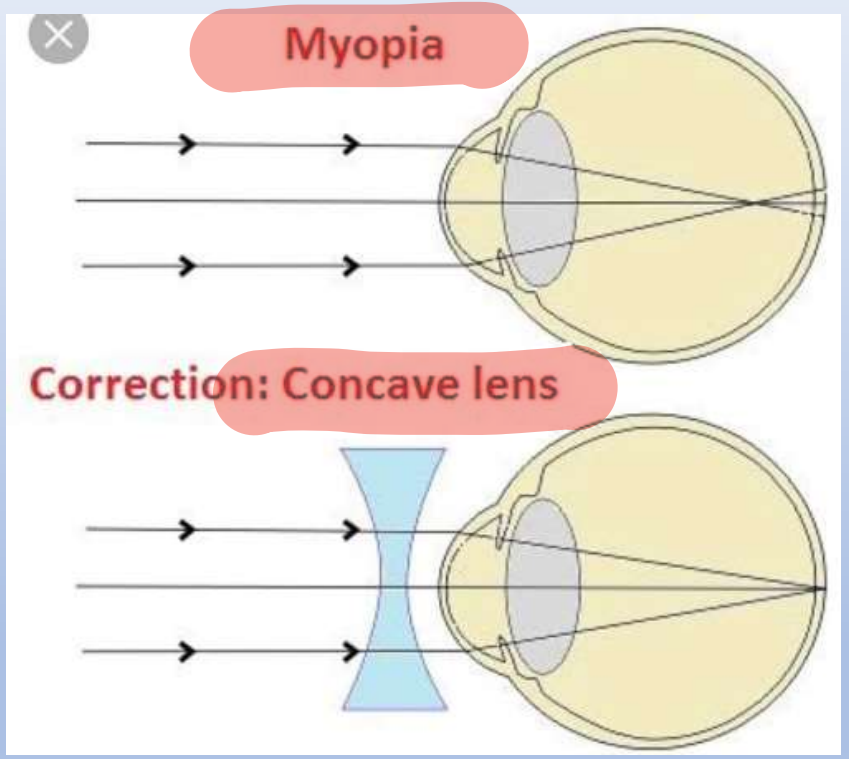
Myopia
difficulty
in distance



Hypermetropia

Difficulty in
near vision

Correction of Myopia

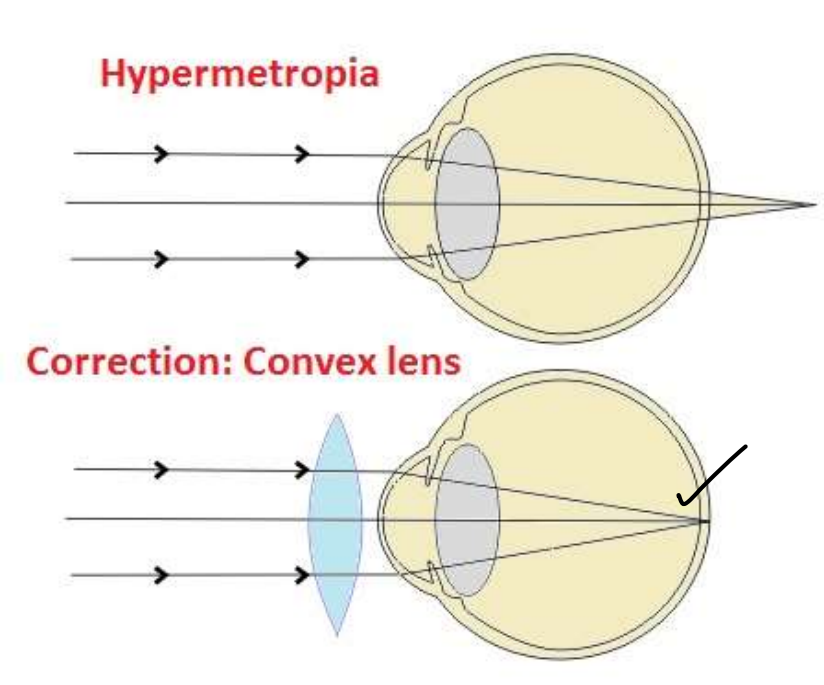


more converging power

↓

Minus concave lens

Correction of Hypermetropia



less converging power
↓
Plus lens
Convex-lens

Far point of Normal Eye is Infinity

Near point of Normal Eye is 25 cm

Power of lens P = 1/focal length in m

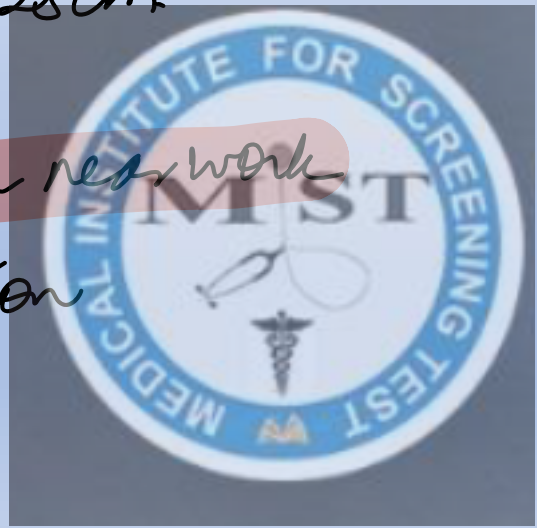
Myopia : Far point is reduced

Hypermetropia : Near point is increased >25cm

Presbyopia → corrected by Plus lens for near work

↓

- physiological decrease in accomodation
- difficulty in near vision >40-45yrs

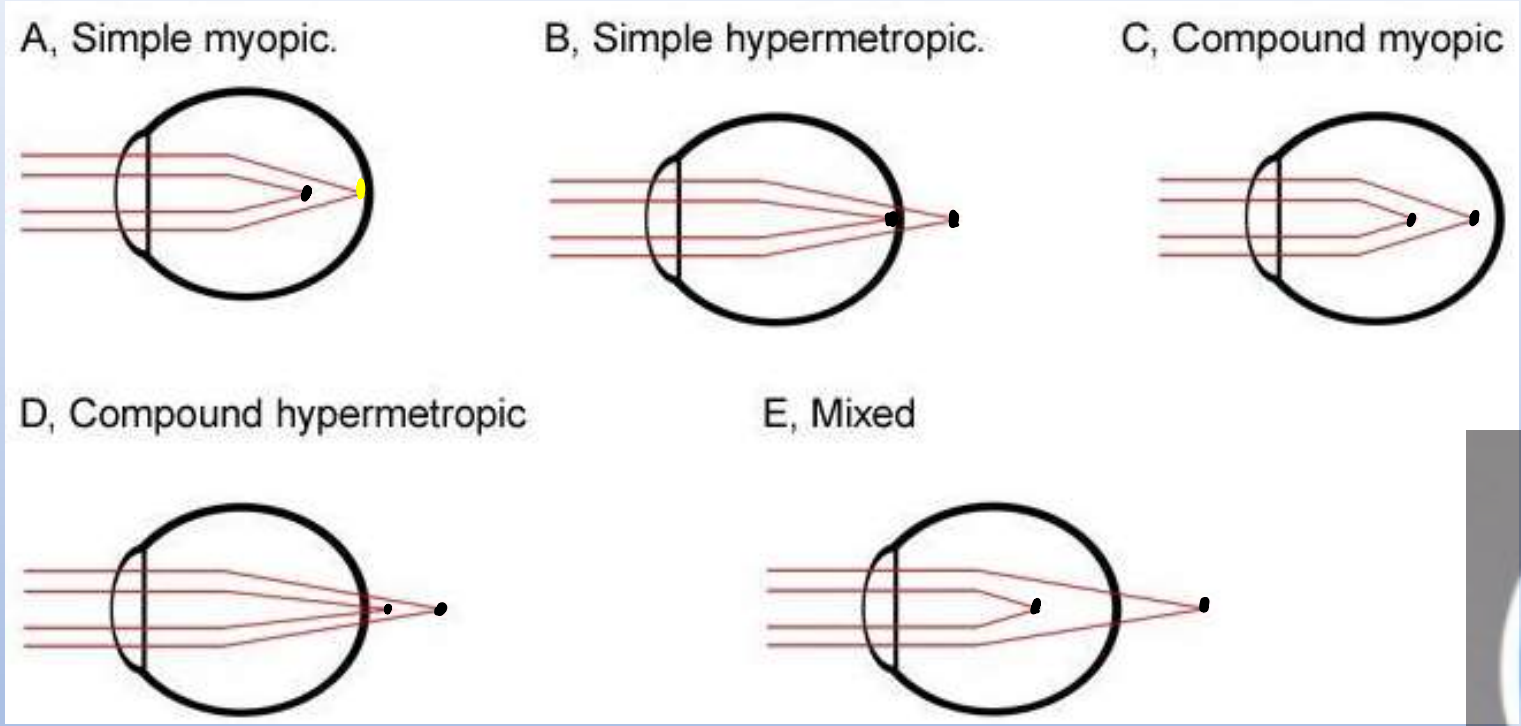


Pathological Myopia → high myopia (7-6 D) + chorioretinal degeneration

- Temporal Crescent, Tesselated fundus.
- Posterior Staphyloma →
- Foster Fuch's spot haemorrhage in foreal area
- Lacquer cracks rupture of Bruch's memb
- CNVM
- Peripheral retinal degenerations (Lattice degeneration, holes)
- Rhegmatogenous Retinal detachment

Astigmatism

Regular Astigmatism — 2 principal meridia



With The Rule Astigmatism WTR:

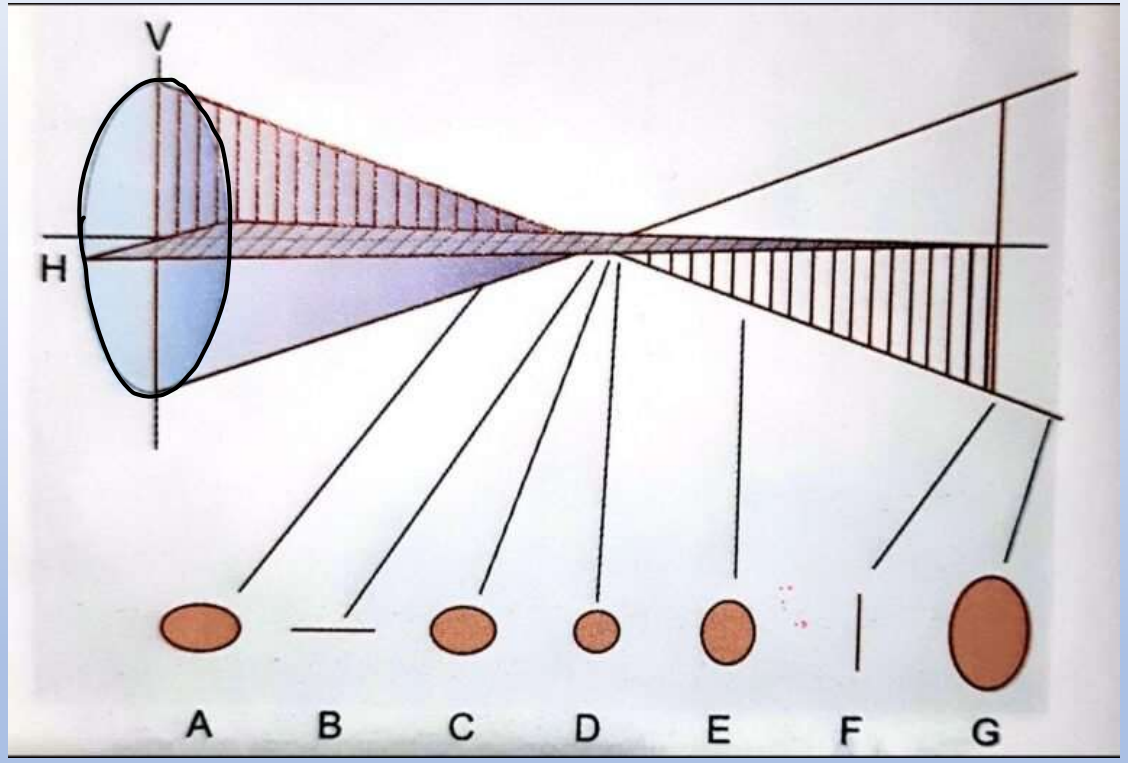
Vertical Meridian is Steeper than Horizontal Meridian V > H
(Minus at 180deg axis in GP is WTR)

Against The Rule Astigmatism ATR

horizontal is steeper than vertical meridian
in GP minus at 90° deg ATR

Irregular Astigmatism : Keratoconus Scarred cornea

Strum's Conoid



optical diagram of Regular Astigmatism

Pinhole

The diameter of the central hole is 1mm

Allows only central rays to pass through

Improvement in visual acuity indicates refractive error

Worsening of visual acuity indicates central pathology

- central corneal opacity
- macular disease

Correction of Aphakia

Absence of lens

High Hypermetropia

Disadvantages of Aphakic glasses:

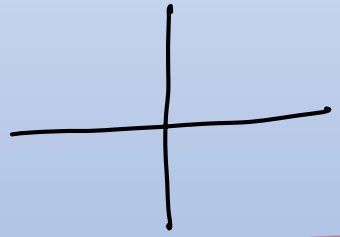
- Image magnification
- Narrow field of vision
- Anisokenia : difference in size of image formed between 2 eyes
- Ring scotoma
- Pin Cushion effect, Jack in the box scotoma

- Alternatives:
  - Secondary IOL implantation
  - contact lenses

Correction of Refractive Error

- Retinoscopy

. objective measurement of Ref Error
. Both horizontal & vertical meridian



. Convert retinoscopy into glass prescription.

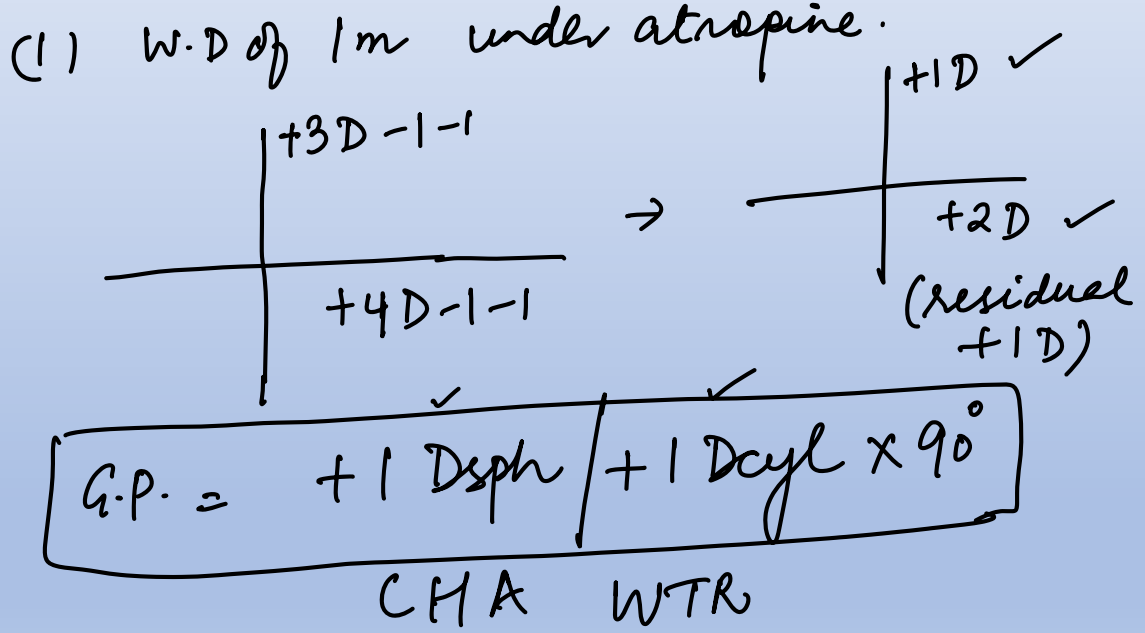
- Subtract W.D. correction
- Subtract for cycloplegia.

Cycloplegic Drugs → children & young adults.

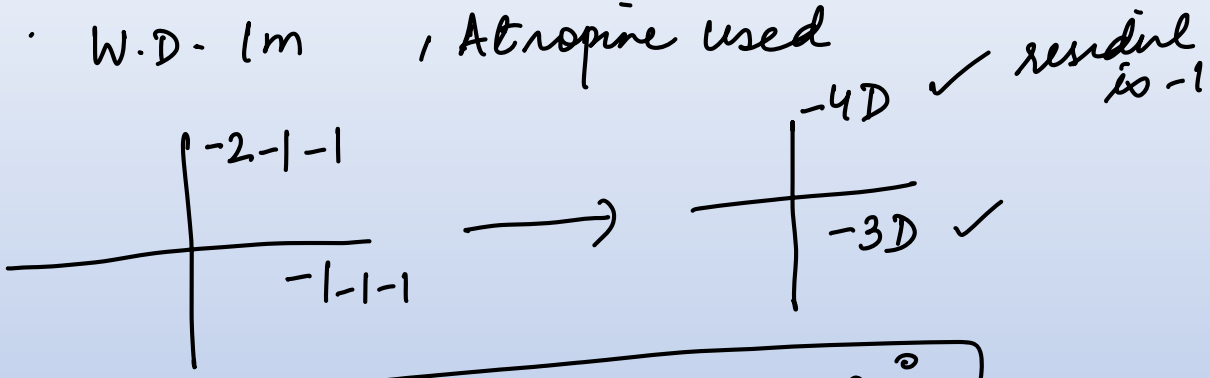
- Atropine 1% ointment preferred in children, squint.
- Homatropine 5-8 y
- Cyclopentolate 7 8 y

Correction of Refractive Error

- Glass Prescription from Retinoscopy



Glass Prescription from Retinoscopy



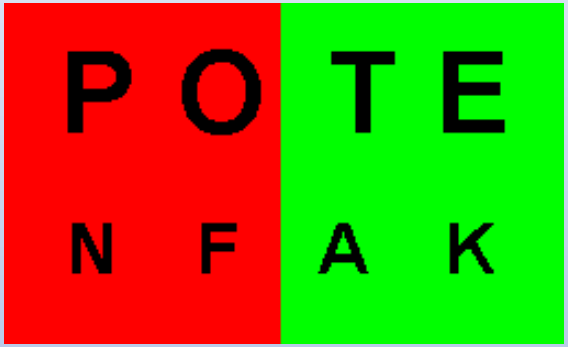
G.P. = -3 Dsph / -1 Dcyl × 180°

CMA WTR

Subject verification of GP

Duochrome Test

↓ refining spherical correction



Jackson's Cross Cylinder

refining the power and axis of cylindrical correction

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Contact Lenses

Preferred in cases of

(1) Anisometropia : diff. in Refernce of 2 eyes
- $\geq 2.5D$

(2) Anisokenia : different size of images formed by 2 eyes.

(3) Aphakia

(4) Irregular astigmatism

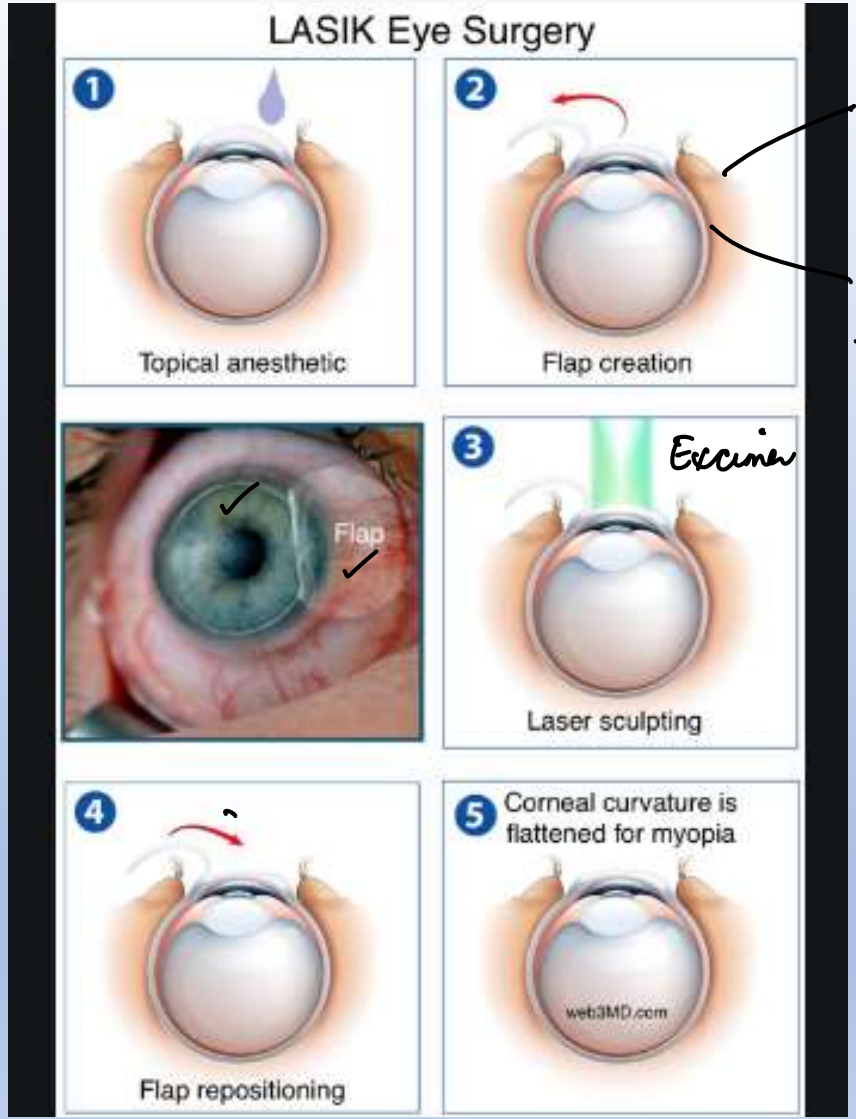
Disadv :
- GPC
- corneal ulcer : Pseudomonas
  - Acanthamoeba

Refractive Surgeries

Criteria
- Age 7, 18 y
- Stable Ref for 1 y
- corneal thickness > 450 µ.

upto -12D
- LASIK : Excimer laser
- RELEX/SMILE : No flap is raised, femtosecond laser

- Phakic IOL - ICL
  - myopie 7 -12D
  - Thin cornea



Blade

Femto second laser

Excimer

Lens

- Most common type of congenital cataract – Blue dot cataract not visually significant

- Most common type of visually significant congenital cataract – Lamellar/Zonular Cataract

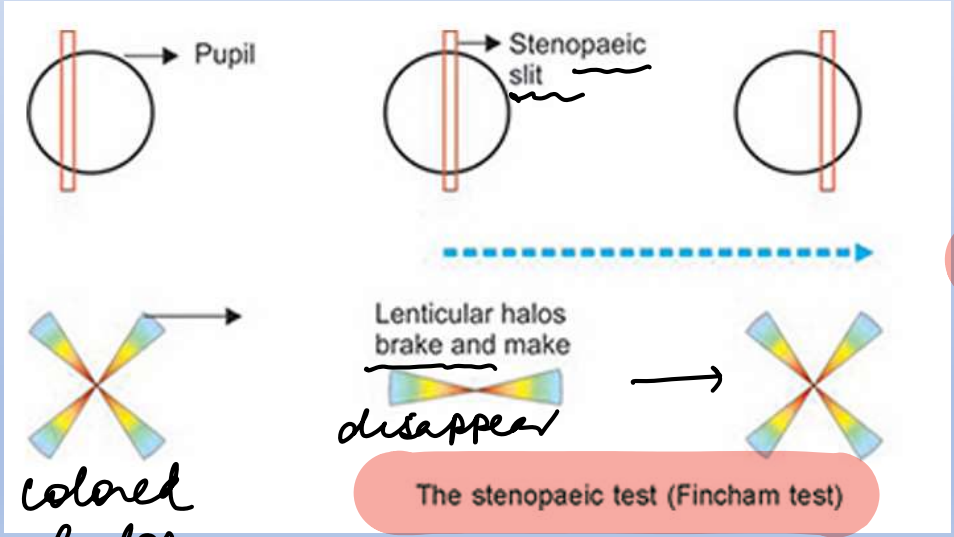
- Most common type of acquired cataract – Senile cataract
- Index Myopia/Second Sight of old age seen in – Nuclear Sclerosis
- Cataract causing Glare/early vision disturbance – Posterior subcapsular Cataract
- Cataract causing vision difficulty in dim light – Cuneiform Cortical Cataract
- Colored Halos – Early Cortical Cataract

- COLORED HALOS

D/D 1. mucopurulent conjunctivitis.

2. Immature senile cataract (cortical cataract)

3. Acute congestive glaucoma.



if halos disappear then it is Cataract.

Nuclear Sclerosis

Second sight old age

↑ Refractive Index of nucleus → Index myopia

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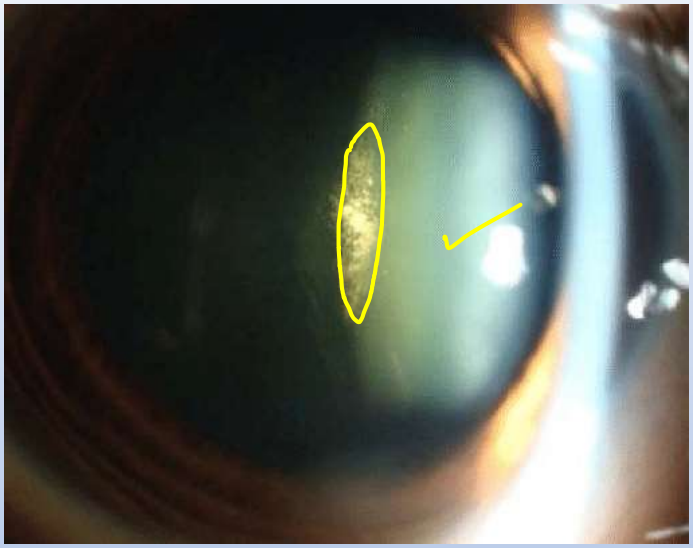
Posterior subcapsular Cat.



- glare
- decreased vision at early stages

Mature Intumescent Cataract

Secondary angle closure
↓
increase IOP

Phacomorphic glaucoma.

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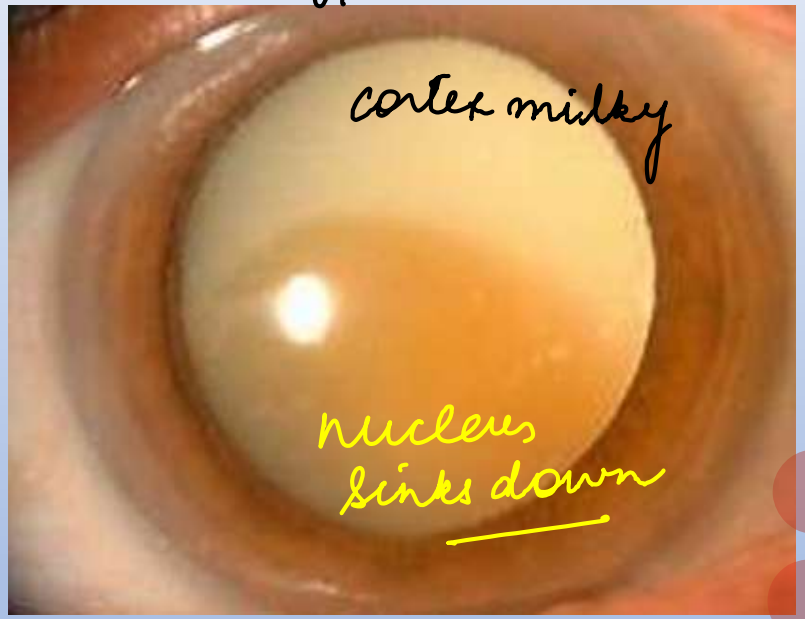
Morgagnian Cataract

Hypermature Cat.



cortical matter leaks out
↓
blocks trabecular meshwork
↓
Phacolytic glaucoma.

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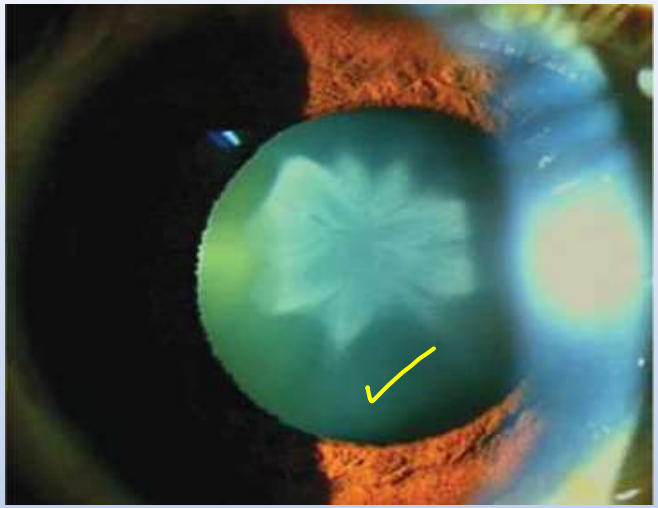
- Phacomorphic glaucoma is due to – Intumescent Mature Cataract

- Complication of Morgagnian hypermature cataract – Phacolytic glaucoma

- Complication of Sclerotic Hypermature Cataract- Dislocation of lens

- Snowflake Cataract – Diabetes
- Rosette Cataract – Blunt Trauma
- Oil droplet Cataract – Galactosemia
- Reversible Cataract - Galactosemia
- Sunflower Cataract – Wilson's Disease Copper deposition
- Christmas tree Cataract- Myotonic Dystrophy



Rossette Cat.

Blunt

Trauma

Snowflake Cataract

deposition of sorbitol:

glucose → sorbitol

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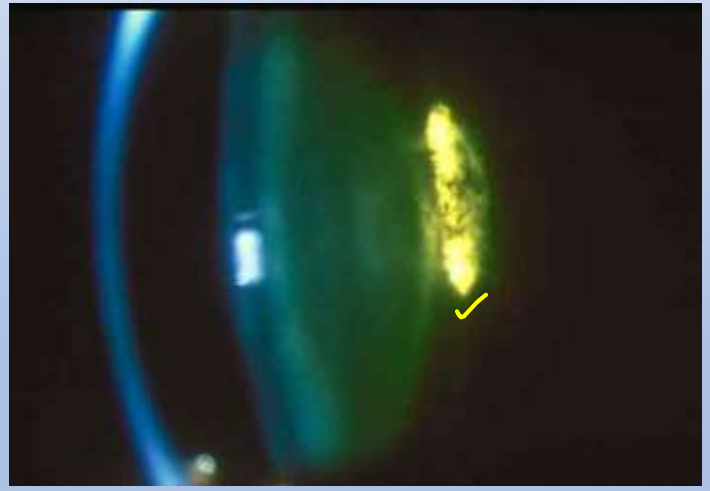
Complicated Cataract → uveitis, RP, High myopia

Polychromatic lustre/Breadcrumb appearance ✱

PSC



In cases of acute congestive glaucoma ↓ glaucomaflecken opacities in ↓ anterior lens

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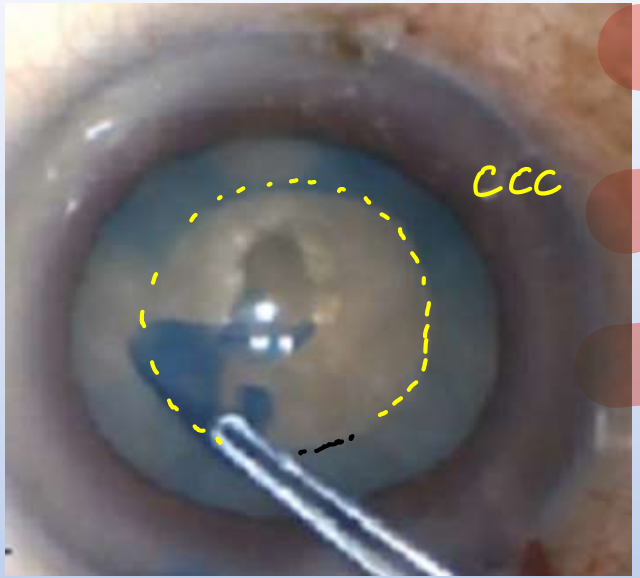
- Glass Blower's cataract – True exfoliation of lens capsule
- Steroid induced cataract – Posterior subcapsular cataract
- Ectopia lentis – superotemporal dislocation – Marfan's syndrome
- Ectopia lentis – inferonasal dislocation – Homocystinuria

• Ectopia lentis : Weil Marchesani
Ant. dislocated lens

- Laser used in Cataract Surgery – Femtosecond Laser
- Ideal site of IOL implantation – In the capsular bag
- Flexible IOL material – Acrylic, Rigid IOL – PMMA
- Formula IOL power calculation – SRK II Formula
- IOL power calculation in post-refractive surgery cases- Haigis formula, Holladay II formula, Barrett TrueK



Capsulorrhexis
or
Capsulotomy

Trypan Blue
Stain for
the ant.
capsule.

Cataract surgery and IOL implantation in children

Lens aspiration/Phaco-aspiration with primary posterior capsulotomy with anterior vitrectomy

↓

age is < 2 y.

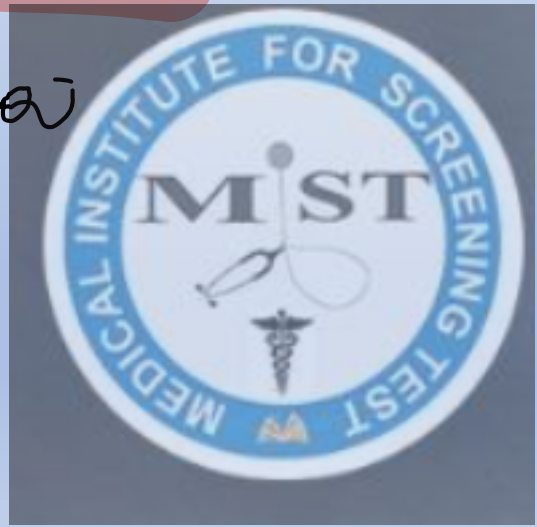
Under-correction of IOL power in children due to Myopic shift

- < 2 years of age: 80% IOL power
- 2-8 years of age: 90% IOL power
- >8 years: 100% IOL power

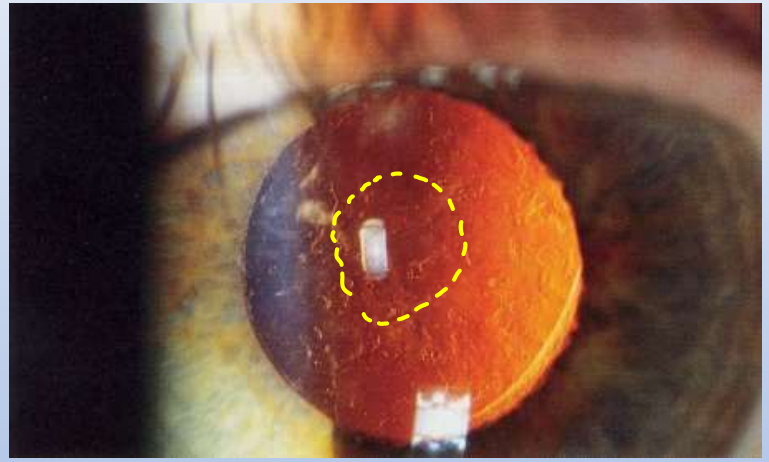
] under correction of IOL power

↓

no undercorrect



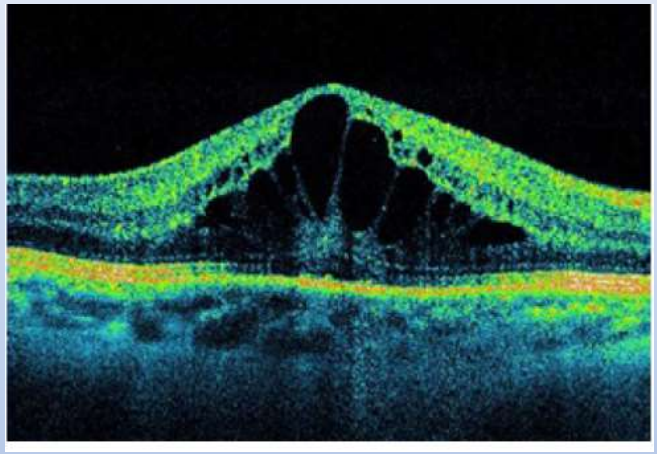
- Most common late complication of Cataract Surgery – After Cataract/Posterior Capsular opacification



- Treatment of After cataract – Nd Yag Laser Capsulotomy

- IRVINE GASS Syndrome

case of



cystoid macular edeme after cataract surgery

OCT macule

FFA: Flower petal leakage

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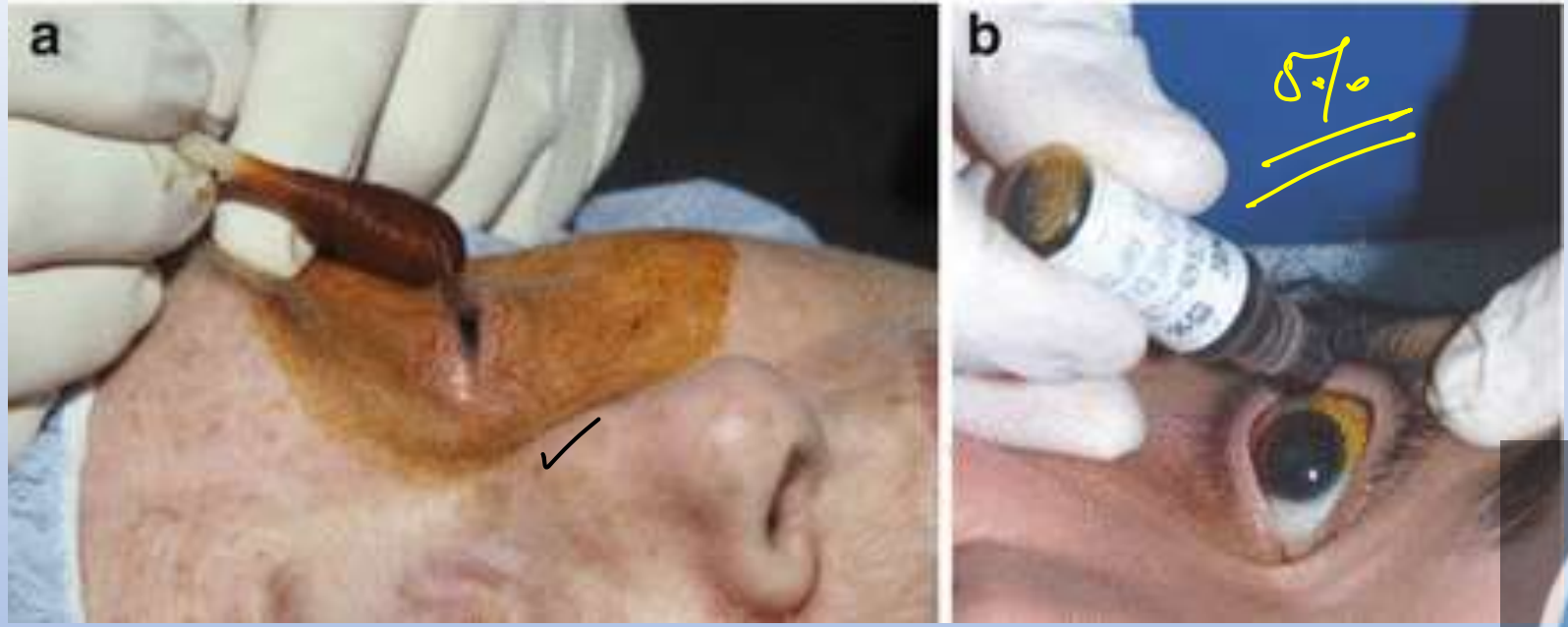
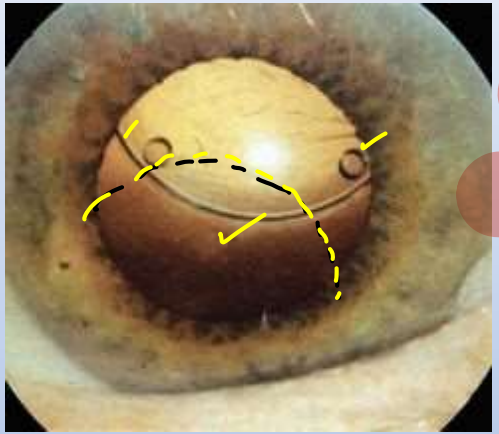
- Most dangerous complication of Cataract Surgery – Endophthalmitis
- Most common cause of acute endophthalmitis – Staph epidermidis (within 1 week)
- Most common cause of late endophthalmitis – Propionibacterium acne, Fungi
- Treatment of Endophthalmitis – Intravitreal Antibiotics/Pars plana vitrectomy

Preoperative Povidone-Iodine Antisepsis



Dislocated PCIOL

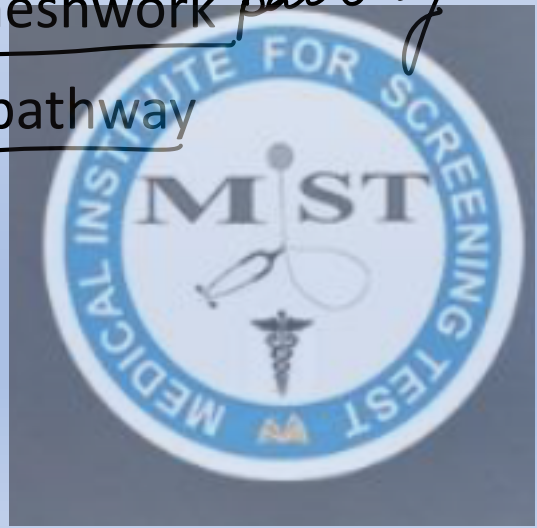


dislocated upwards:
Sun Rise syndrome

dislocated downwards
↓
sunset appearance.

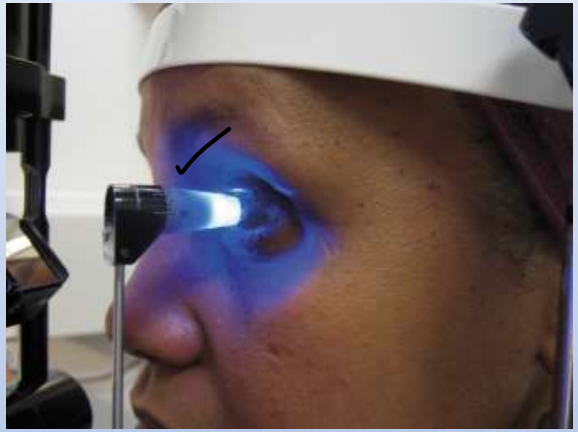
Glaucoma

- Aqueous Humor produced by Ciliary processes of Ciliary body
- Aqueous Humor – Nutrition to Cornea and Lens
- Aqueous Humor outflow major pathway – Trabecular meshwork pathway
- Aqueous Humor outflow minor pathway – Uveoscleral pathway
- Normal IOP: 10-21 mm Hg

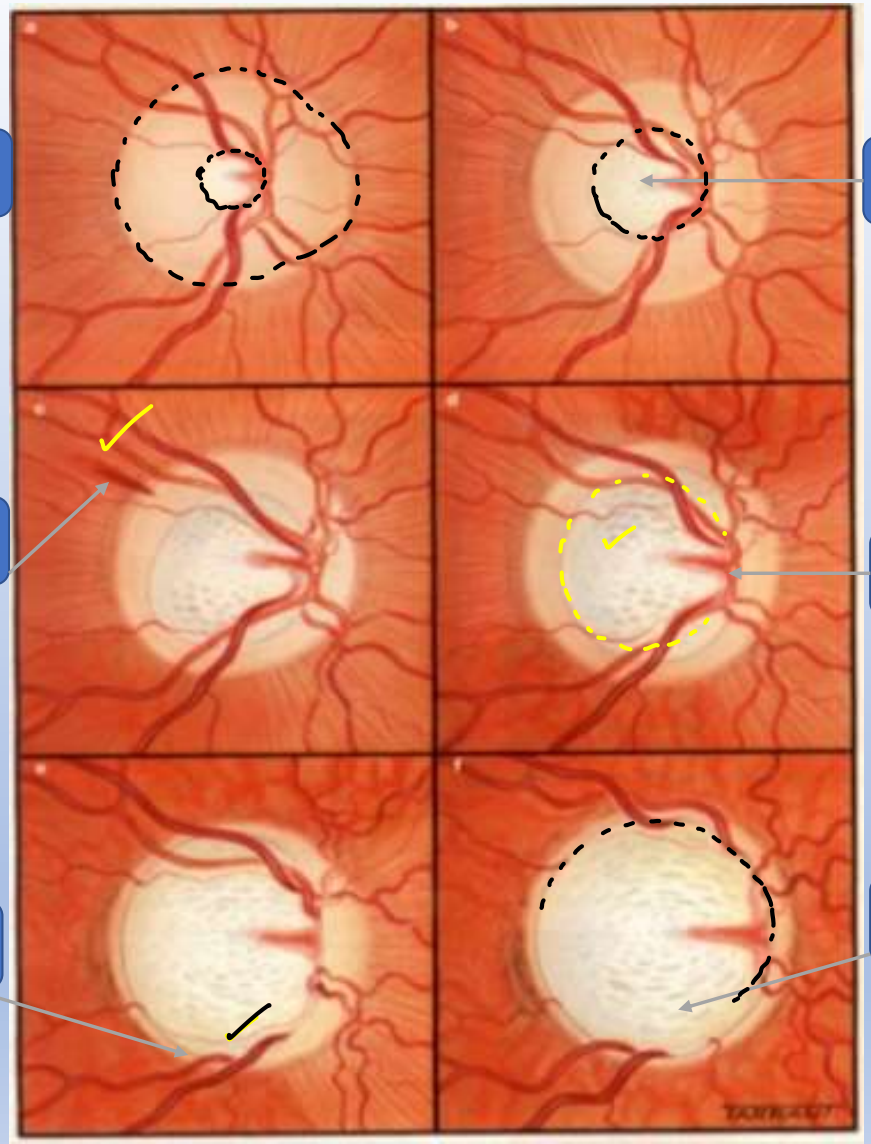


Goldmann Applanation Tonometry

measurement of IOP.



- Gonioscopy: examination of angle of AC
- Goniolenses: Zeiss, Keoppe



NORMAL OPTIC DISC

INCREASED C:D

SPLINTER HEMORRHAGE

NASALISATION OF VESSELS

BAYONETTING OF VESSELS

LAMINAR DOT SIGN

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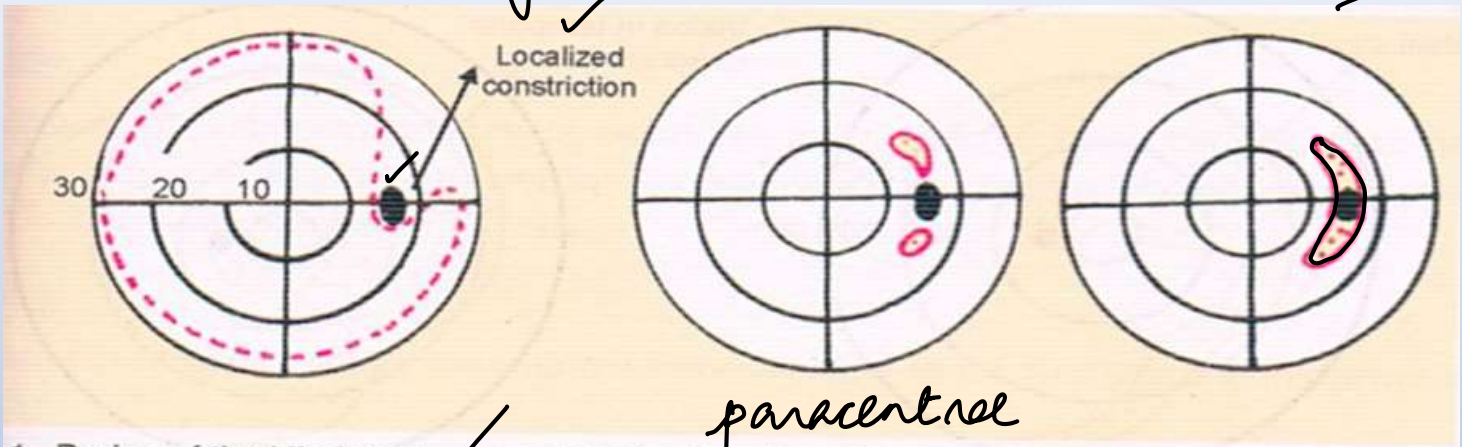
VISUAL FIELD DEFECTS IN GLAUCOMA

Bjerrum's area : 25-30° central

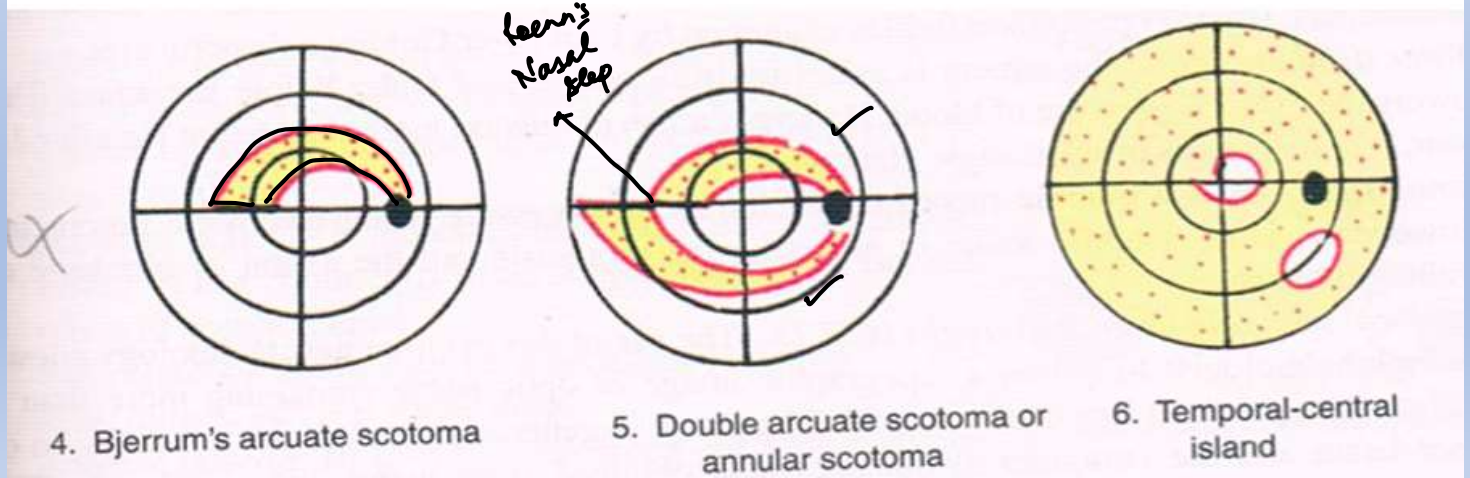


1. Baring of the blind spot

paracentral

2. Small scotomatous areas

3. Seidel's sign



4. Bjerrum's arcuate scotoma

5. Double arcuate scotoma or annular scotoma

6. Temporal-central island

- Buphthalmos – congenital glaucoma

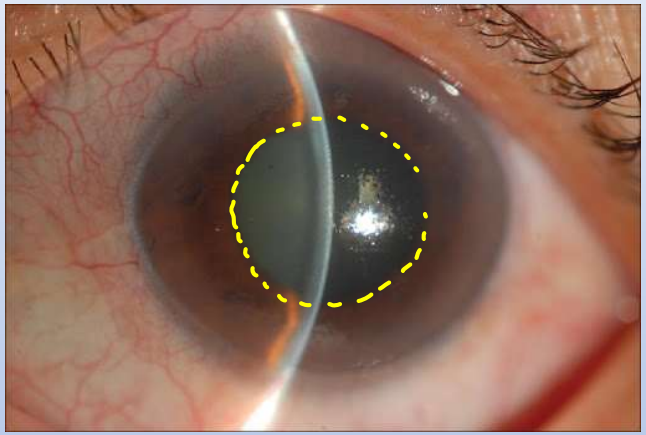
- Blepharospasm B
- Photophobia P
- Lacrimation L

- Haab's stria seen in Buphthalmos
  - Megalcornea, Hazy or clear
  - Bluish sclera

- Surgery of choice for congenital glaucoma with clear cornea – Goniotomy

- Surgery of choice for congenital glaucoma with hazy cornea – Trabeculotomy

Acute Angle Closure Glaucoma ✱



clinical Case

ocular emergency

after watching a movie.

- Painful red eye
- Fixed, mid-dilated oval pupil
- Shallow AC
- Stony hard eye due to high IOP 50-70 mmHg.

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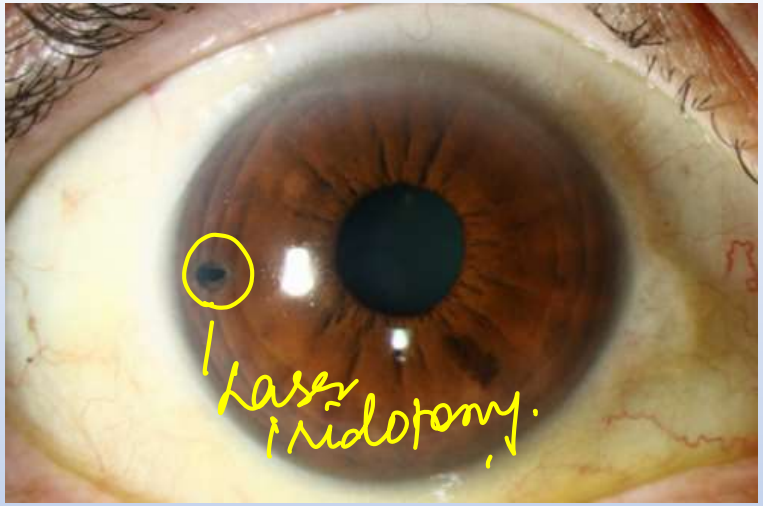
Acute Angle Closure Glaucoma: Management

- I/V Mannitol
- Acetazolamide

- Pilocarpine relieve pupil block

- **Laser Peripheral Iridotomy**



Nd Yag Laser.

Vogt'sTriad of ACG:

- Iris atrophy
- Glaucomflecken
- Pigment on endothelium

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- **Phacolytic Glaucoma**

↳ Morgagnian Cataract

antiglaucoma Rx + Cataract surgery

- **Phacomorphic Glaucoma**

↳ Intumescent mature cataract.

- **Malignant Glaucoma/ Aqueous Misdirection Syndrome**

. Ir
. AR is accumulating in vit cavity.
. IOPT, Shallow AC
Rx: Atropine.

- **Neovascular Glaucoma** due to Rubeosis Iridis

most common: Adv PDR
. CRVO
. Sickle cell disease.

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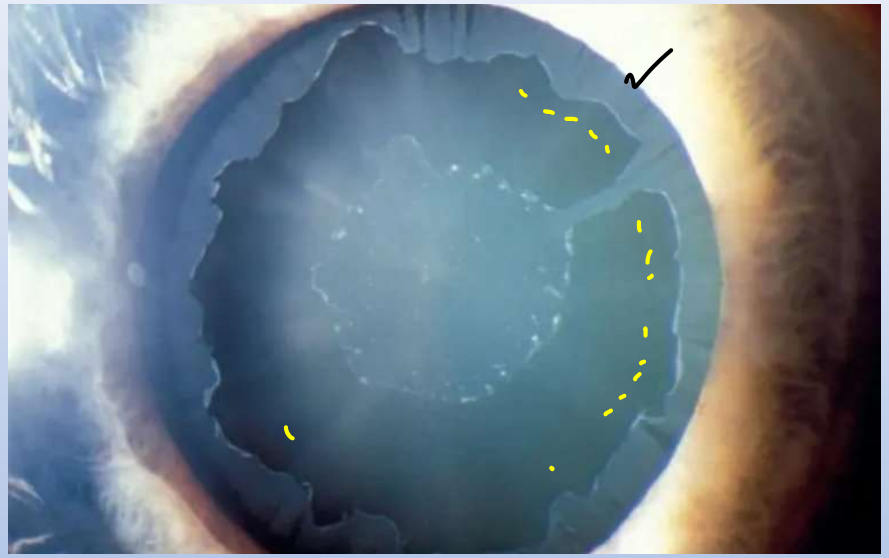
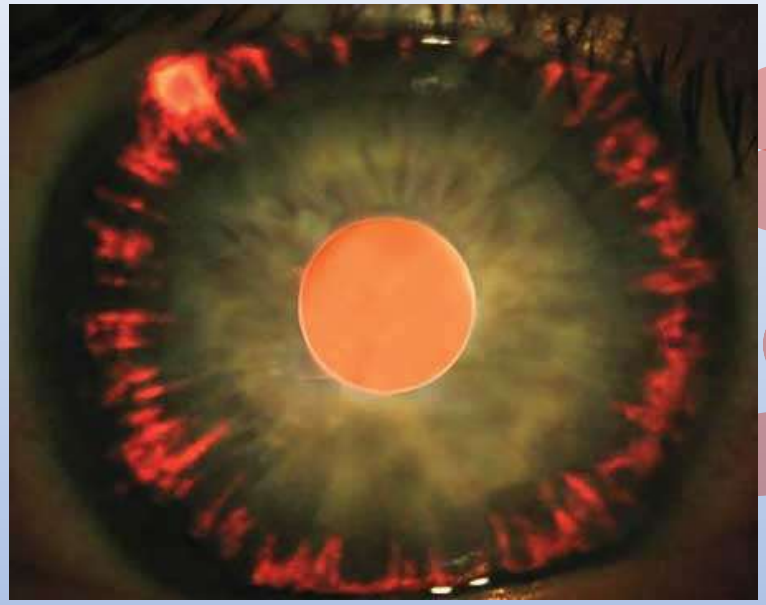
Pseudoexfoliation Glaucoma



Target sign.

Pigmentary Glaucoma

Iris pigment release



Transillumination defect

. Krukenberg spindle :

pigment on endothelium.

- First line anti-glaucoma drug or Drug of choice –

Prostaglandin analogues:

Latanoprost/Travoprost/Bimatoprost/Tafluprost

- Mechanism of action: Increased uveoscleral outflow
- Most common side effect of Prostaglandin analogues: Trichomegaly/Dark Iris
- Contraindicated in Uveitis and Cystoid Macular Edema

- Antiglaucoma drug contraindicated in Asthma, COPD, Heart block: Beta blockers Timolol.

- Antiglaucoma drug contraindicated in Children: Brimonidine
- Antiglaucoma drug causing sleep apnea: Brimonidine

- Drug of choice for Acute angle closure glaucoma: Pilocarpine
- Drug contraindicated in Acute angle closure glaucoma: Atropine

- Antiglaucoma drug causing retinal detachment: Pilocarpine
- Drug contraindicated in Uveitis with glaucoma: Pilocarpine, PG analogues

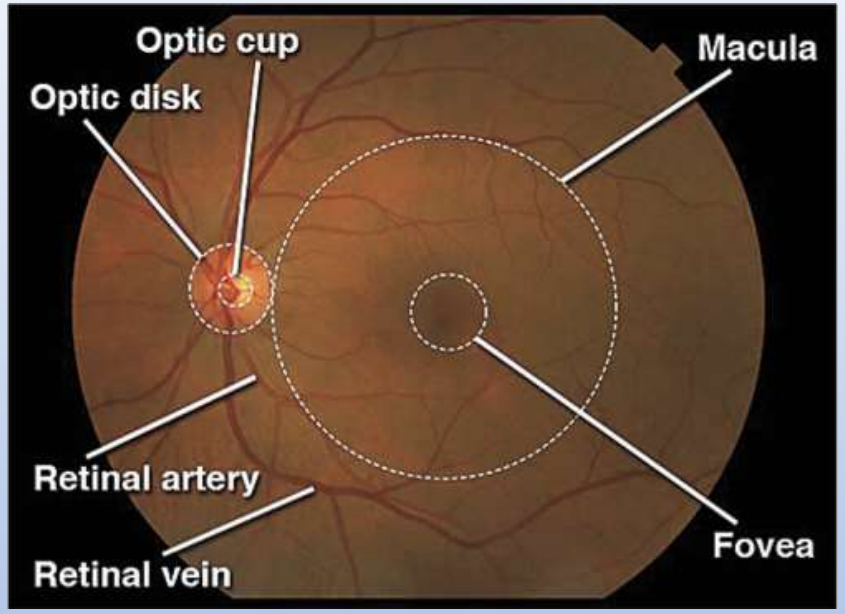
Netrasudil

- Rho-kinase inhibitor
- MOA: Increases Aqueous Outflow through Trabecular meshwork
- Side effects: Vortex Keratopathy

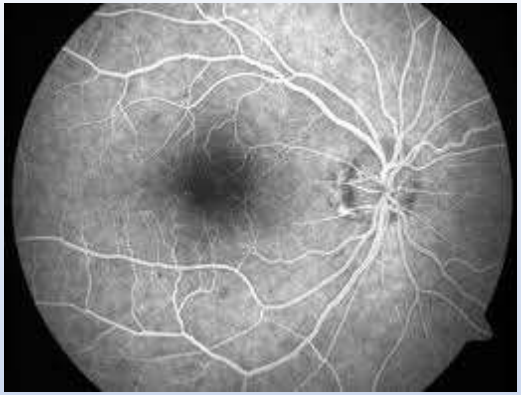
- Laser done for angle closure – Nd Yag Laser Iridotomy
- Laser done for open angle glaucoma – Laser Trabeculoplasty
- Surgery for Glaucoma – Trabeculectomy
- Seton surgery is Glaucoma valve surgery/Drainage implant

Ahmed valve.

Retina

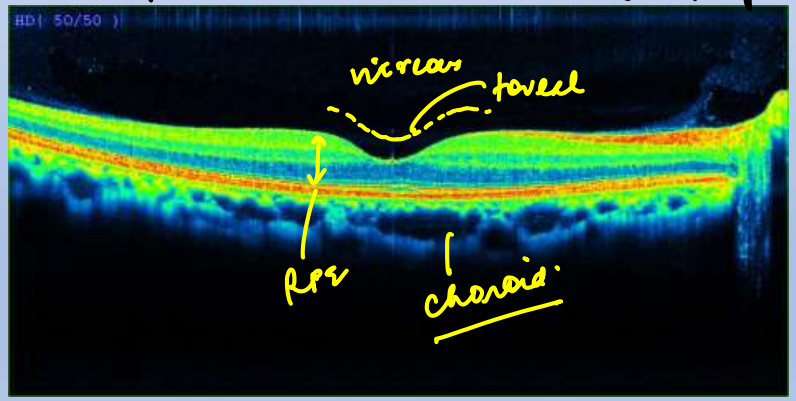

Optic cup
Optic disk
Macula
Retinal artery
Retinal vein
Fovea

Fundus Fluorescein Angiography



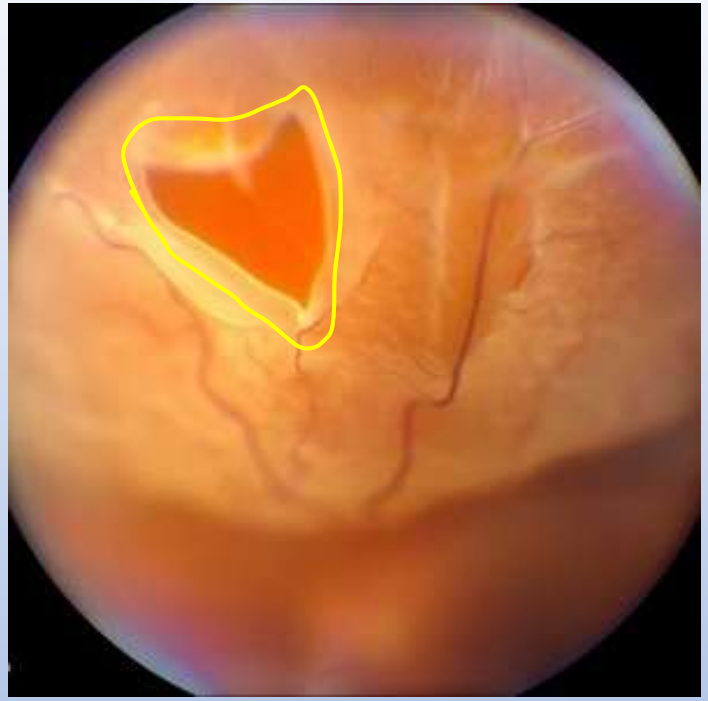
FFA

Optical Coherence Tomography OCT



macular area

- Retinal Detachment – separation of neurosensory retina from RPE
- Most common Retinal degeneration - lattice degeneration
- Rhegmatogenous RD – Due to retinal break
- Most dangerous retinal break – Horse shoe tear
- Most dangerous location of retinal break – Superotemporal



RRD

- Floaters
- Flashes of light
- Curtain like shadow in field of vision

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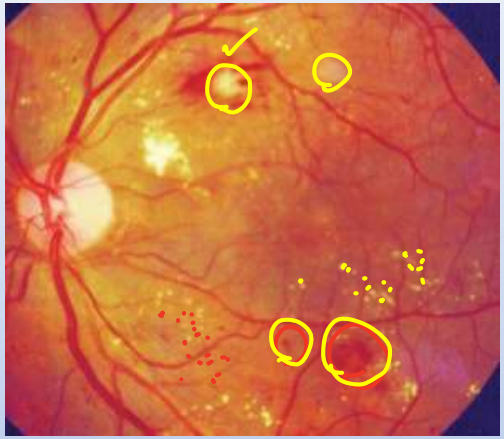
- Signs of Flashes/Photopsia/Curtain like shadow - Rhegmatogenous RD
- Shafer's sign/Tobacco dusting - Rhegmatogenous RD
- Most common cause of Tractional RD – Proliferative DR
- Shifting fluid is seen in – Exudative RD

Diabetic retinopathy

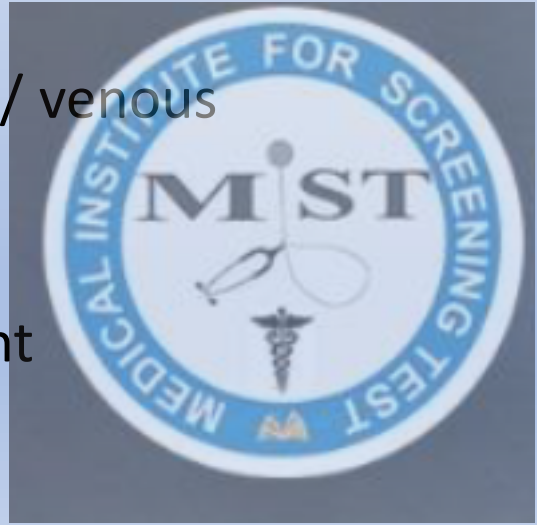
Non-proliferative DR NPDR



- Microaneurysms: earliest and hallmark sign
- Hard exudates: lipid
- Soft exudates: cotton wool spots
- Dot and blot haemorrhages
- IRMA and Venous beading
- Absence of new vessels

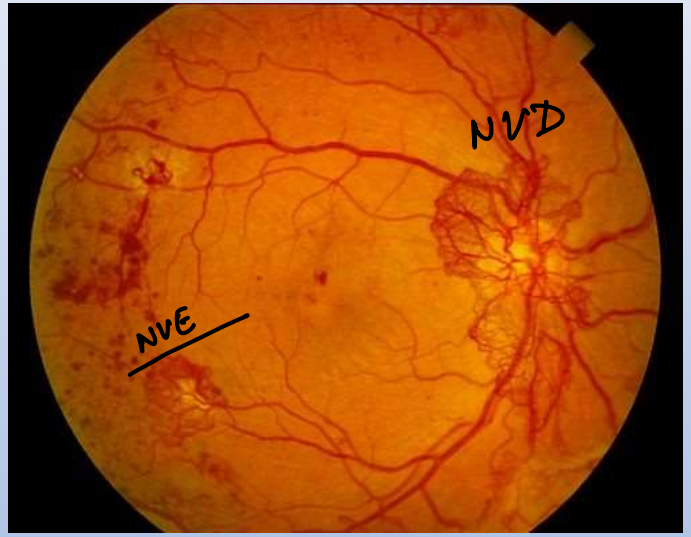
Classification of NPDR

- Mild NPDR – only microaneurysm present.
- Moderate NPDR – microaneurysms, retinal haemorrhages, hard exudates/cotton wool spots
- Severe NPDR – 4:2:1 rule – haemorrhages in 4 quadrants/ venous beading in 2 quadrants/ IRMA in one quadrant
- Very Severe NPDR – any two of severe NPDR signs present



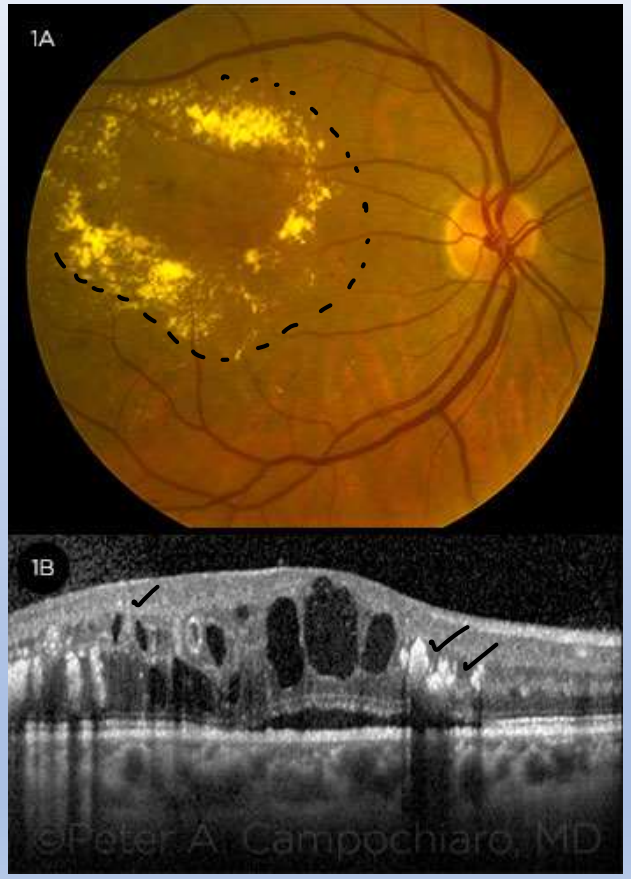
Proliferative DR

VEGF



- New vessels seen on disc and elsewhere on retina
- Neovascularisation

Diabetic macular edema



Hard exudates and retinal thickening
Most common cause of visual loss
Can occur with any stage of DR

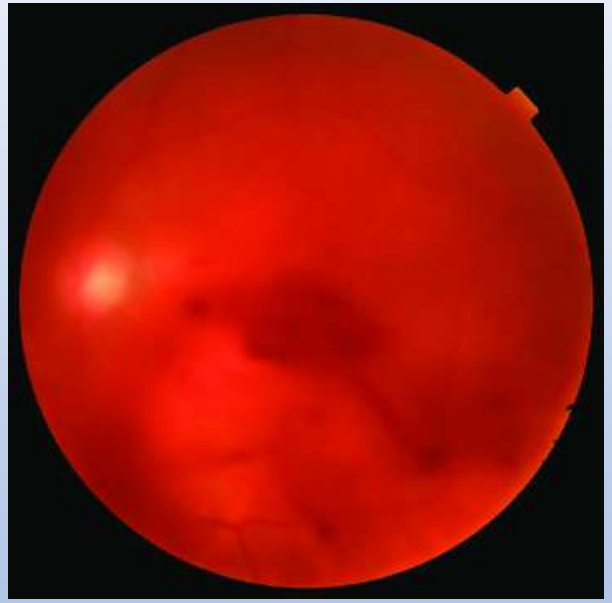
Diabetic Retinopathy: Treatment

- Pan retinal photocoagulation PRP laser - PDR
- Focal Laser or Grid laser – Diabetic Maculopathy
- Intravitreal Anti- VEGF injections
- Pars Plana Vitrectomy – Vitreous haemorrhage or TRD
- Observation with diabetic control for NPDR with no maculopathy

Examination Schedule for Diabetic Retinopathy

- Type 1 DM – 5 years after onset or after puberty
- Type 2 DM – At the time of diagnosis

Vitreous Haemorrhage :



Sudden painless loss of vision

Causes:

(1) Adv PDR

(2) CRVO

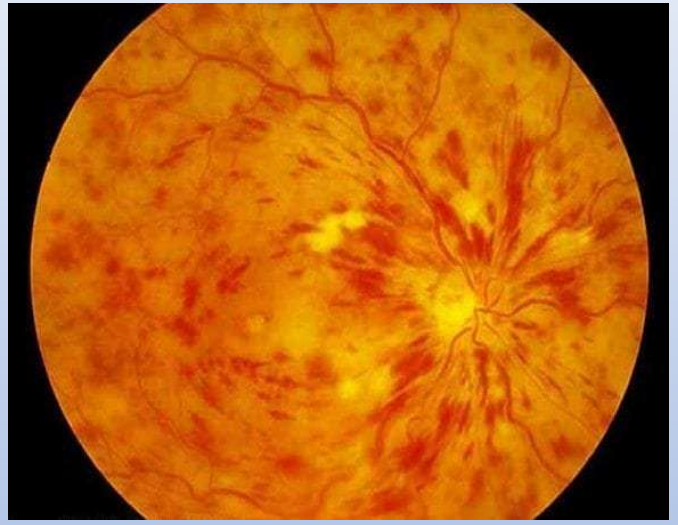
(3) Trauma

Eale's disease

Hypertensive Retinopathy: Keith & Wagner classification

- Grade 1: Mild generalized arteriolar attenuation
- Grade 2: Marked generalized and focal narrowing, A-V nipping, Salu's sign
- Grade 3: grade 2 + flame shaped hemorrhages, cotton wool spots, hard exudates, Bonnet sign, Gunn sign
- Grade 4: grade 3 + disc edema

Central Retinal vein occlusion CRVO



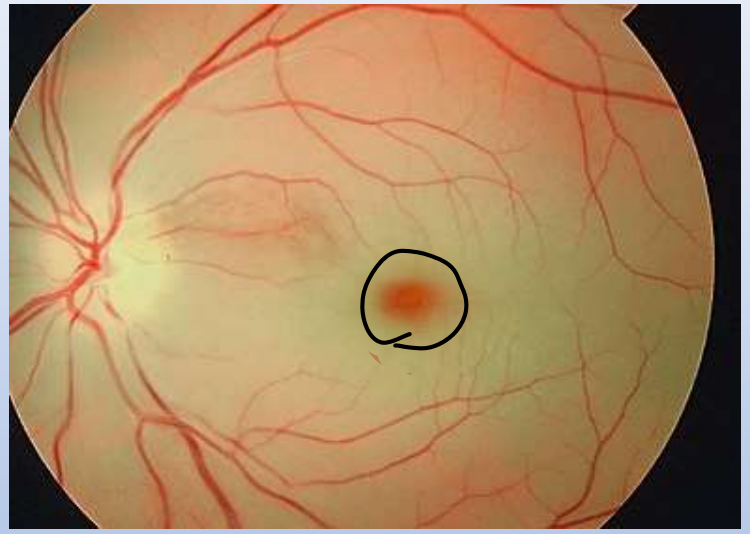
hypertensive
Sudden painless
decreased vision

- Tomato splash appearance/Blood in thunder appearance

- 90-day/100-day glaucoma

Branch Retinal vein occlusion BRVO

Central Retinal Artery Occlusion CRAO



CAD
present with
Sudden painless
loss of vision

- Pale retina
- Cherry red spot
- Cattle truck appearance of blood vessels

D/D Cherry red spot

- Christmas: CRVO
- Tree: Tay Sachs Disease ✓ 90-1.
- Never: Niemann Pick Disease
- Grows: Gaucher's disease
- Tall: Trauma(blunt)

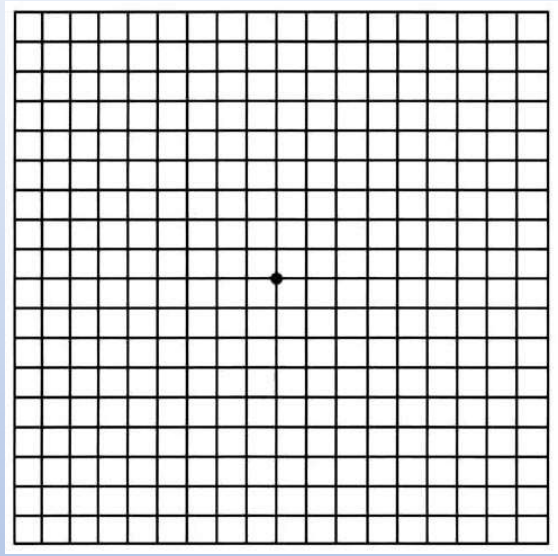
In

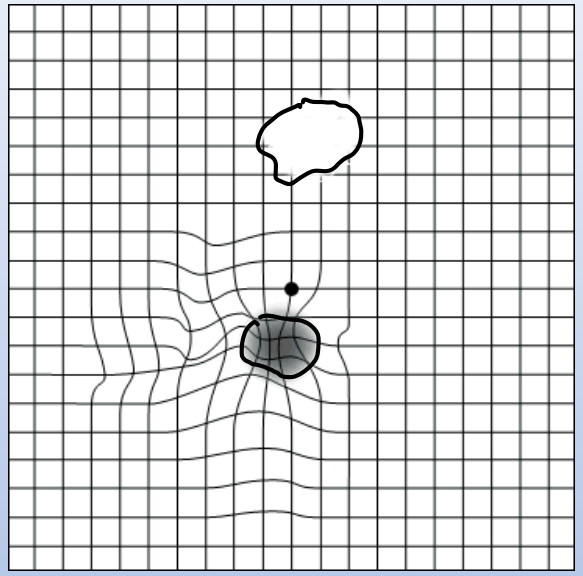
- Sand: Sandoff's disease

&

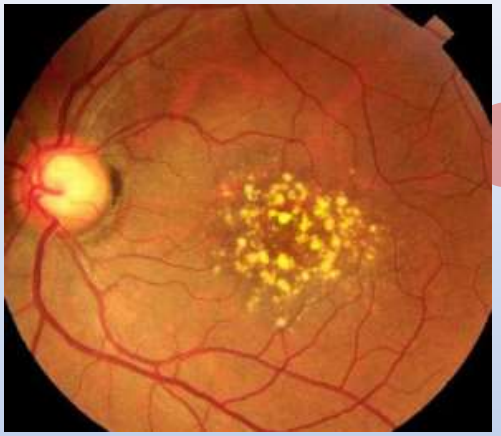
- Mud: Metachromatic leukodystrophy

Amsler Grid : to check for metamorphopsia

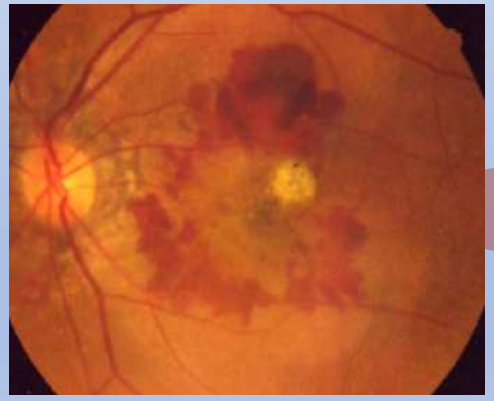




Age related macular degeneration ARMD

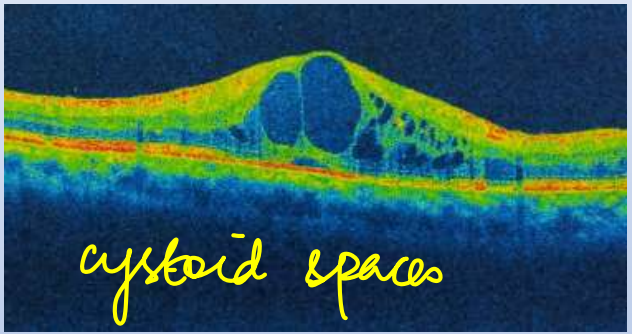


- Dry ARMD: Drusens
  lipofuscin
- Antioxidants



- Wet ARMD or exudative ARMD:
- CNVM
  choroidal Neovascular membrane
- Intravitreal anti VEGF Inj.

Cystoid Macular Edema



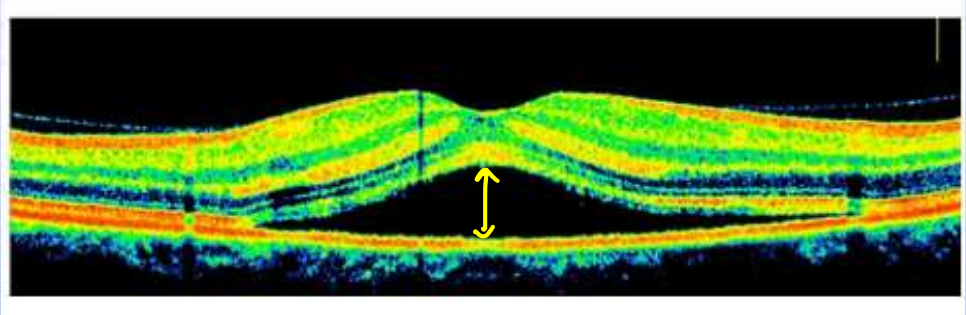
↓

(1) Uveitis
(2) DR, CRVO, BRVO
(3) Irvine Gass syndrome
(4) RP
(5) Niacin, PG analogues

Fundus Exam: Honeycomb appearance

FFA: Flower petal appearance

Central Serous Retinopathy



FFA :
smoke stack
or
Ink blot leakage

- young patients
- Type A personalities
- Steroid use
- Smoking

Retinitis Pigmentosa

loss of Rod photoreceptors

- Night blindness
- Difficulty in driving due to ring scotoma
- Pale disc, attenuated arterioles, bony spicule pigmentation
- ERG ★

electroretinogram

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Best Disease

Egg yolk appearance (lipofuscin deposit)

ERG (N)

EOG abnormal

electrooculogram

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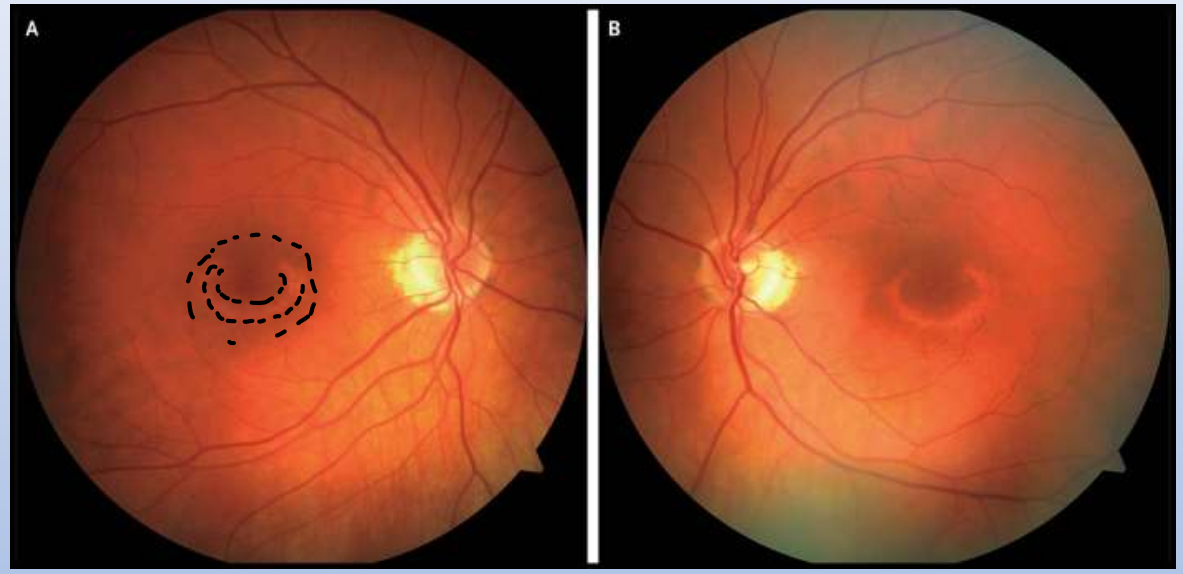
Bull's eye maculopathy



1. Chloroquine Toxicity
2. Hydrochloroquin "
3. Stargadt's disease

Retinoblastoma → most common intra-ocular malignancy of childhood.

- Retinoblastoma gene 13q 14
  RB1 gene. → Tumor suppressor gene.
- 40% Heritable, 60% Non-Heritable

- Most common Presentation
  ↓
  leucocoria

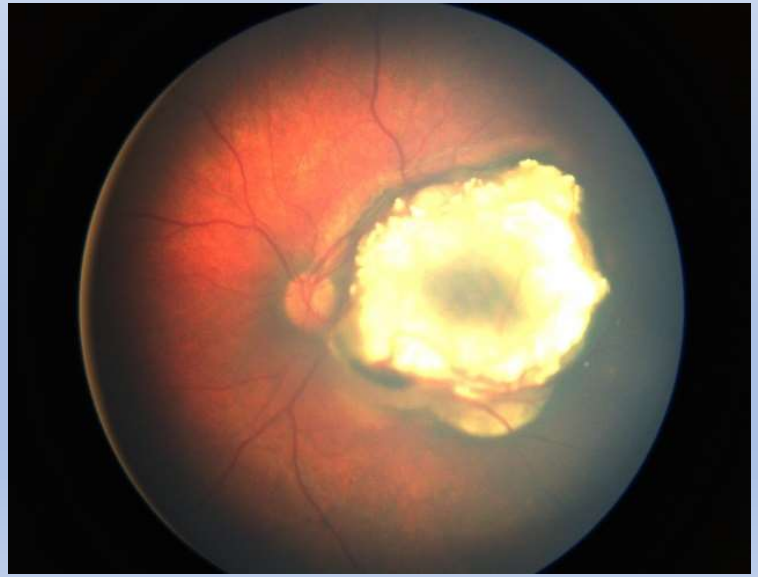
HERITABLE RETINOBLASTOMA	NON-HERITABLE RETINOBLASTOMA
GERMLINE MUTATION	SOMATIC MUTATION
1st mutation in Gamete ✓ 2nd mutation in Retinal Cell ✓	1st mutation in Retinal Cell ✓ 2nd mutation in Retinal Cell ✓
Multifocal , Bilateral tumor	Unifocal , Unilateral tumor
Passed to Offsprings	Can not be passed to Offsprings
Risk of Secondary tumors (Osteosarcoma) ✓	No Risk of Secondary tumors
Trilateral Retinoblastoma (Pinealoblastoma) ✓	-

Knudson 2 hit hypothesis

Retinoblastoma

Leucocorie

Squint.



Cauliflower like mass with calcification

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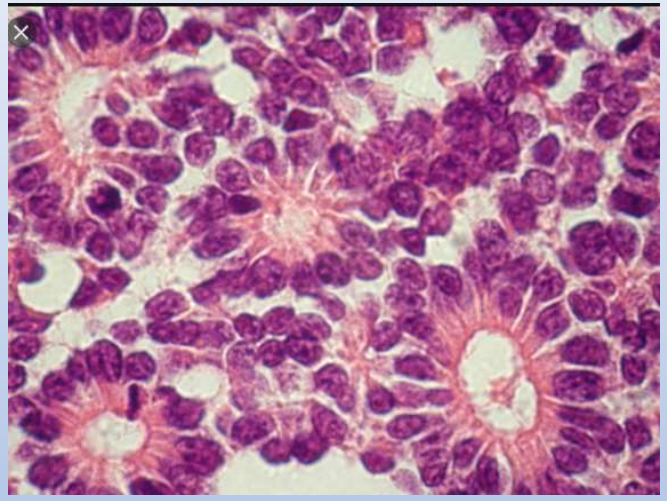
Retinoblastoma



Flexner wintersteiner Rosettes

Specific of RB

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Investigations:

- USG Bscan: Intraocular mass with calcification
- CT scan → avoided because of radiation
- MRI scan : IOC

Metastatic Spread:

Through optic nerve
(Direct spread)

Treatment

- Focal therapy- Laser, Cryotherapy, Thermotherapy, Brachytherapy
- Chemotherapy – Vincristine, Etoposide, Carboplatin
- Enucleation
- External Beam Radiotherapy

International Classification of Retinoblastoma (ICRB)

- Group A: Very low risk
- Group B: Low risk
- Group C: Moderate risk
- Group D: High risk

Focal therapy +/- chemotherapy

- Group E: Very high risk

Enucleation

↓

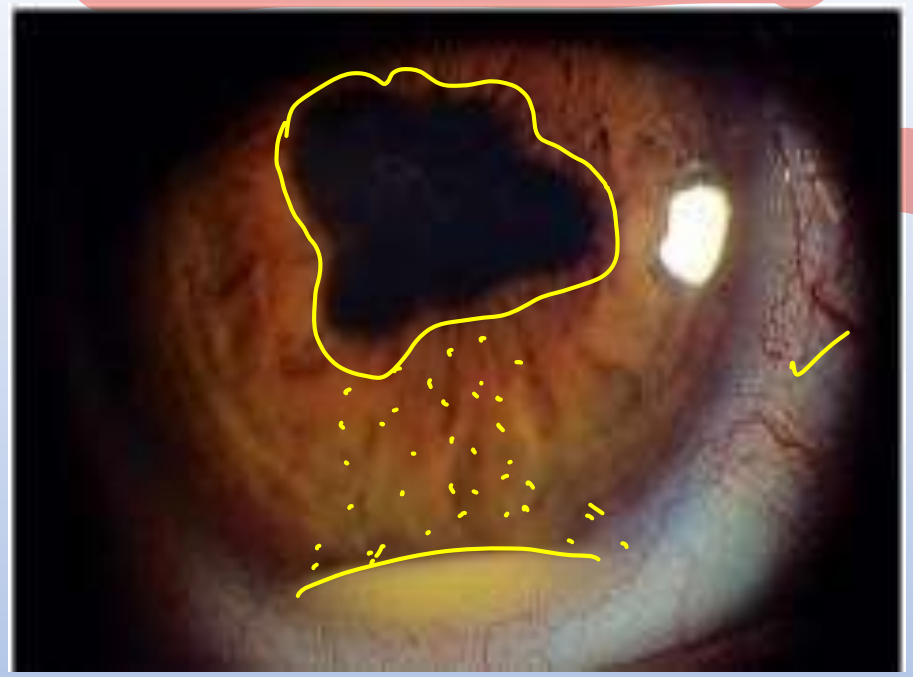
advanced tumors
>50% globe

Most common intraocular malignancy in adults : Choroidal melanoma.

Uveitis

Anterior uveitis:

Painful Red Eye
decreased vision



CF:

- circumcorneal congestion
- AC cells, Flare
- KPs
- Post synechiae
- hypopyon
- iris nodules

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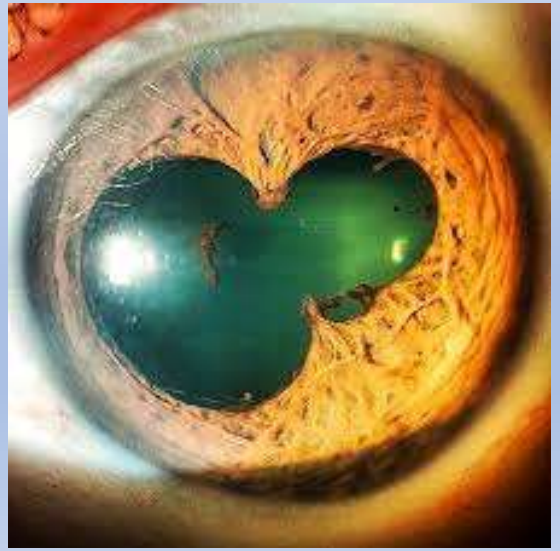
Mutton Fat
KPs:
granulomatous



360° synechia
Festooned
pupil.

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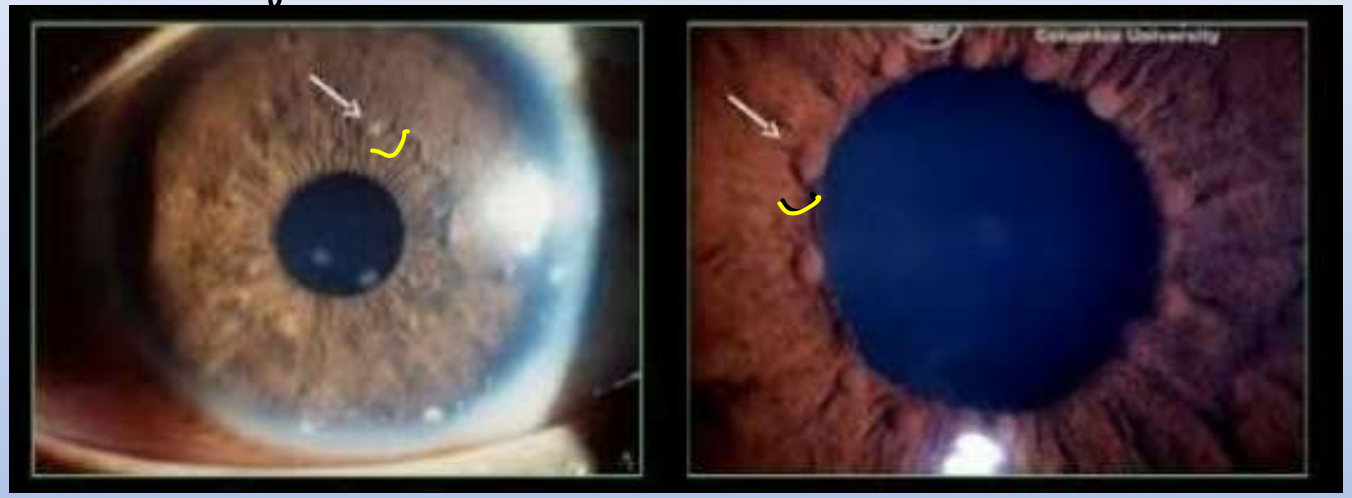
Iris Nodules: Busacca Nodules and Koeppe Nodules

granulomatous ant. uveitis



UVEITIS WITH ARTHRITIS

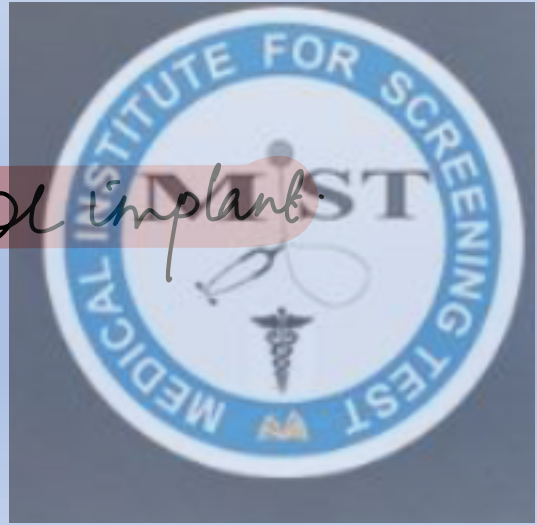
1. ANKYLOSING SPONDYLITIS → sacroiliitis.
2. REITER'S SYNDROME
3. PSORIATIC ARTHRITIS

HLA B27 assn

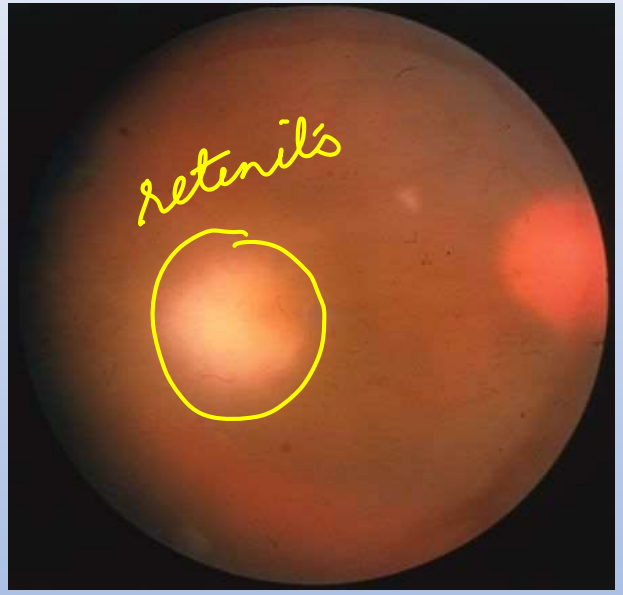
4. JUVENILE RHEUMATOIDARTHRITIS : avoid IOL implant

↓

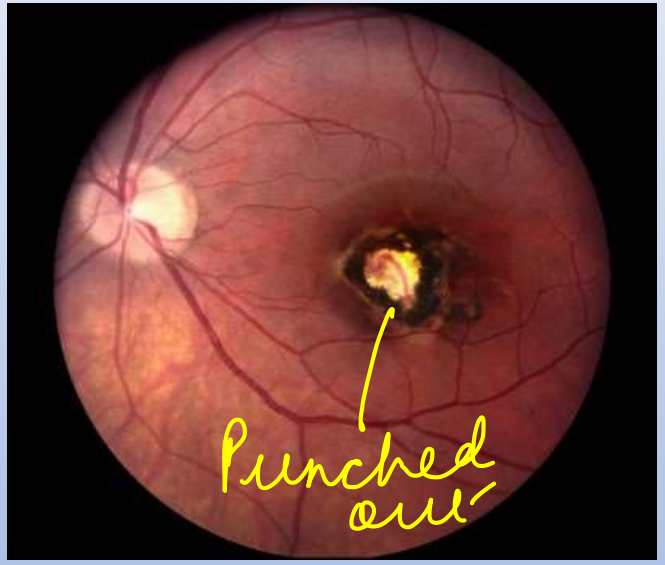
- Pauciarticular seronegative JRA
- Band shaped keratopathy



Toxoplasma

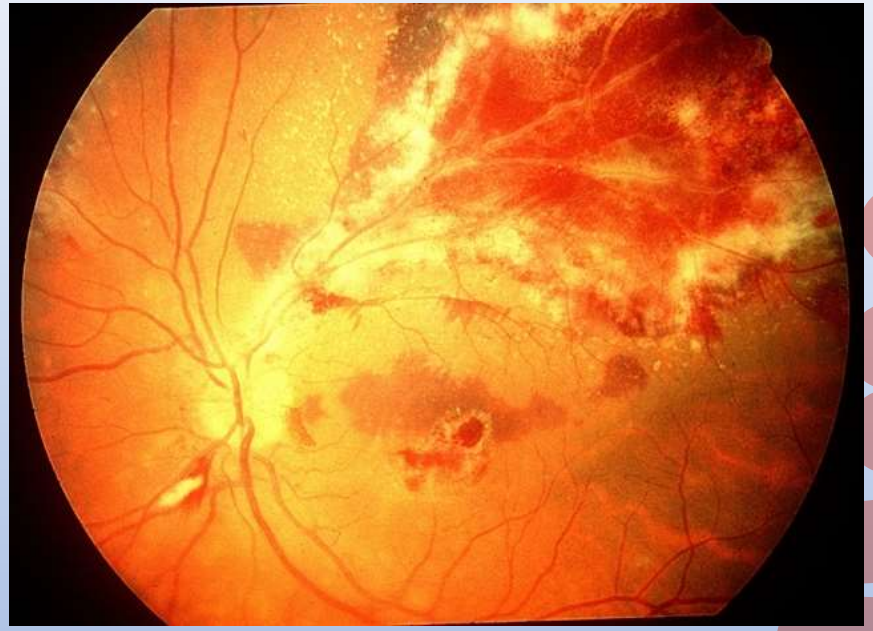


Headlight in Fog appearance



Macular scan

Clinical Case:



HIV

↓

opportunistic

:

CMV Retinitis

Pizza cheese appearance

Intravitreal ganciclovir

SYMPATHETIC OPHTHALMITIS

- Penetrating wound is the prerequisite
- Wounds involving Ciliary Body – Dangerous

- INJURED EYE – EXCITING EYE
- NORMAL EYE – SYMPATHEZING EYE

SYMPATHETIC OPHTHALMITIS

- Granulomatous Panuveitis in both eyes

- Onset – after 2 weeks of injury

- Earliest symptom – Weakness of Accommodation, Photophobia
- Earliest sign – Retrolental flare

Dalen Fuch's nodules

Treatment : Steroid therapy
Enucleation within 2 weeks
if it is severe eye injury

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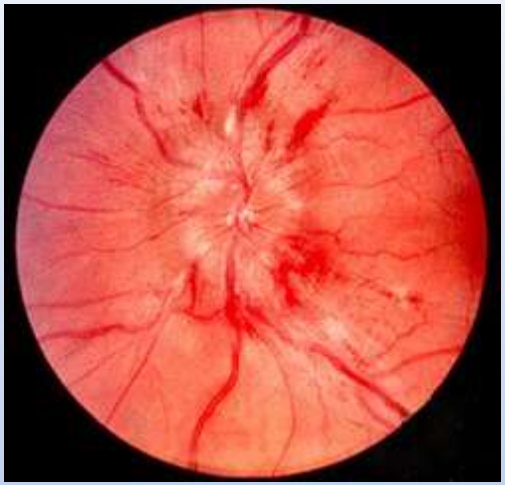
Neuro-ophthalmology

Optic Neuritis

- Decreased vision, Pain on ocular movements
- Defective color Vision Red green
- Marcus Gunn Pupil RAPD
- Central or centrocaecal scotoma
- Most common cause – Multiple sclerosis
  demyelinating.

Optic Neuritis

- Papillitis
  ↓
  children



Inv: VEP
MRI scan
⇒

- Retrobulbar Neuritis: Fundus examination is normal ✱
  adults
- Neuroretinitis → disc edema + macular edema.

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Toxic Optic Neuropathy

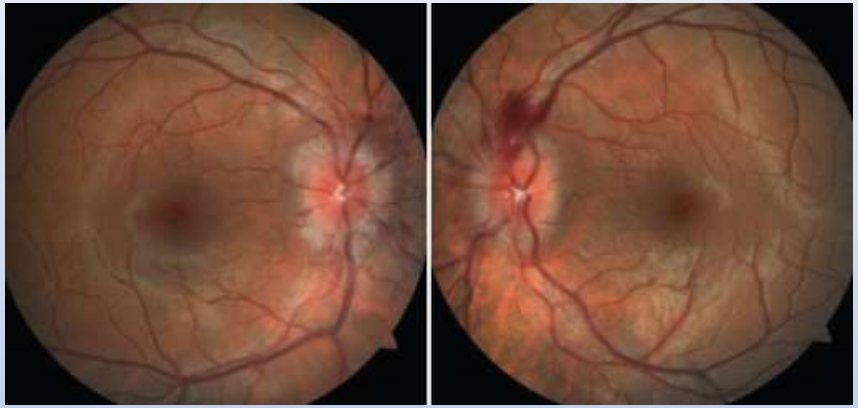
decreased vision, color vision def. RAPD

- Tobacco Alcohol amblyopia → Nutritional Amby: B12 def.
- Methyl alcohol poisoning

- Drug Toxicity:
  Ethambutol — ATT

Centrocaecal Scotome.

Papilledema : passive edema of optic disc due to raised intracranial pressure.



↓
vomiting
headache
6th N palsy

Early cases

- Amaurosis fugax — Transient blurring of vision
- Normal Pupil reactions
- Enlarged blind spots

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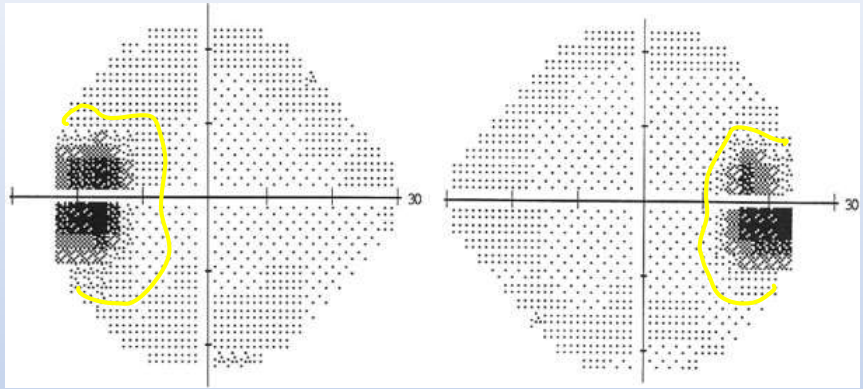

30
30

Idiopathic Intracranial Hypertension

Pseudotumor cerebri

- No ICSOL, normal ventricles.

- Risk Factors: Obesity, OCPs, Steroids, Vit A toxicity, Tetracyclines

females

Clinical case: Headache, Amaurosis fugax

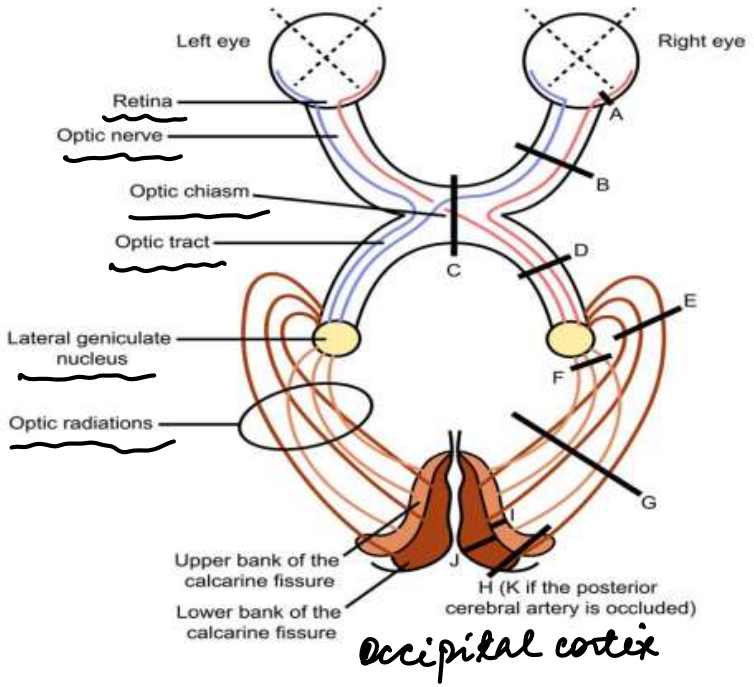
Diagnosis: Lumbar Puncture

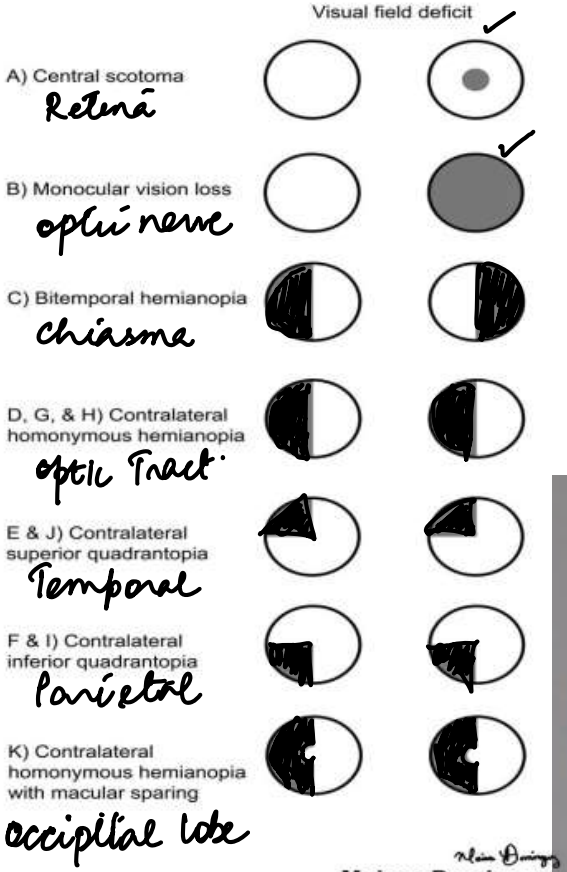
↓

Increased opening pressure of CSF.

Papilledema

Visual Field Defects





© Lineage

Moises Dominguez

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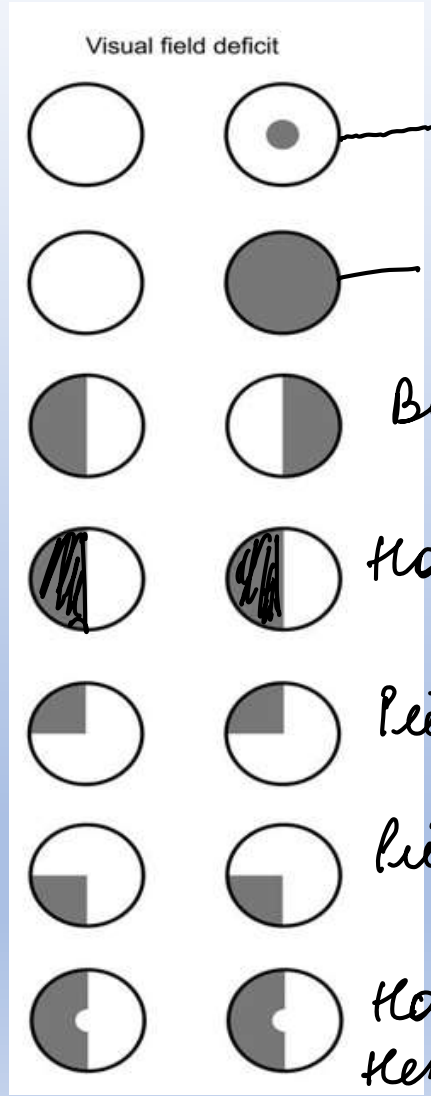
Visual field deficit
Retinā
optic nerve
Bitemporal hemianopia
optic chiasme
Homonymous hemianopia
optic Tract
Pie in sky
Temporal lobe
Pie on floor
Parietal lobe
Homonymous hemianopia with macular sparing
occipital lobe

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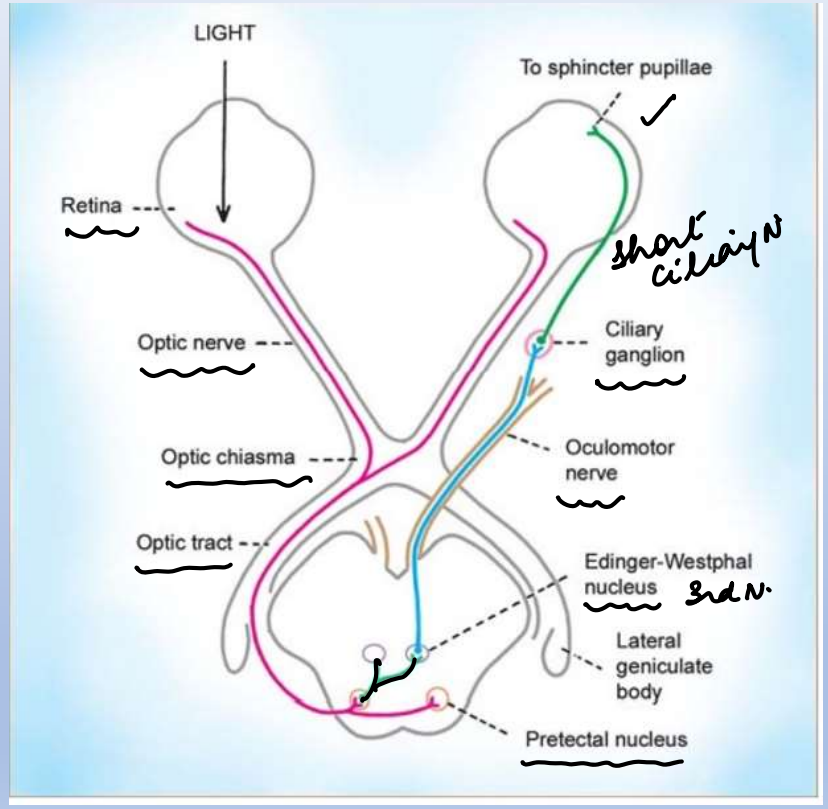
Optic Chiasma Lesions:

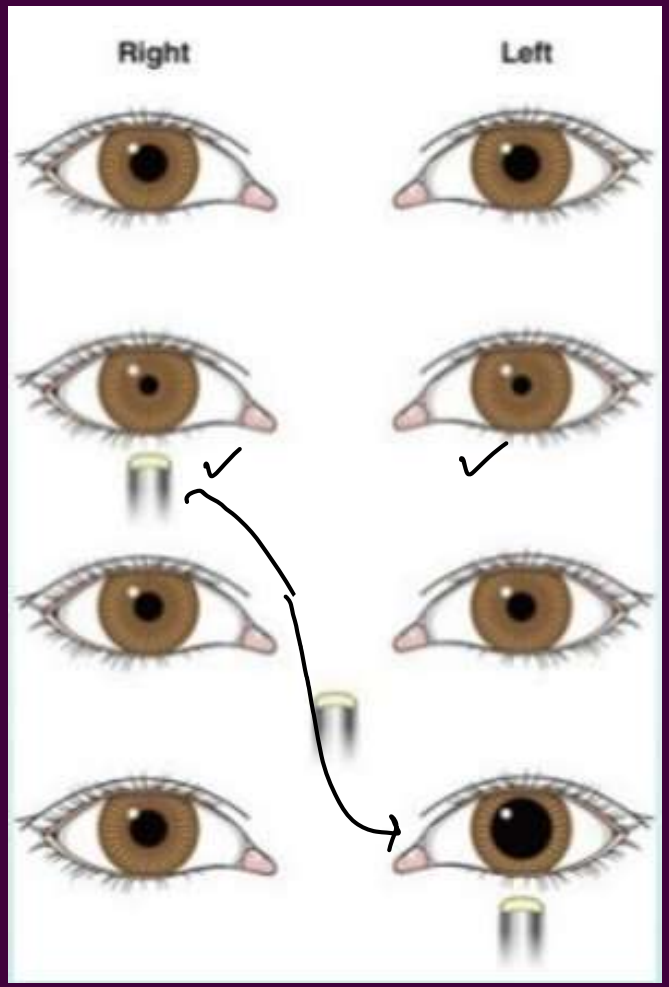
Pituitary Tumors – Pituitary Adenoma, Craniopharyngioma

↓
compresses central chiasme
Bitemporal hemianopia.

Pupillary Pathway

Light reflex





RAPD

or

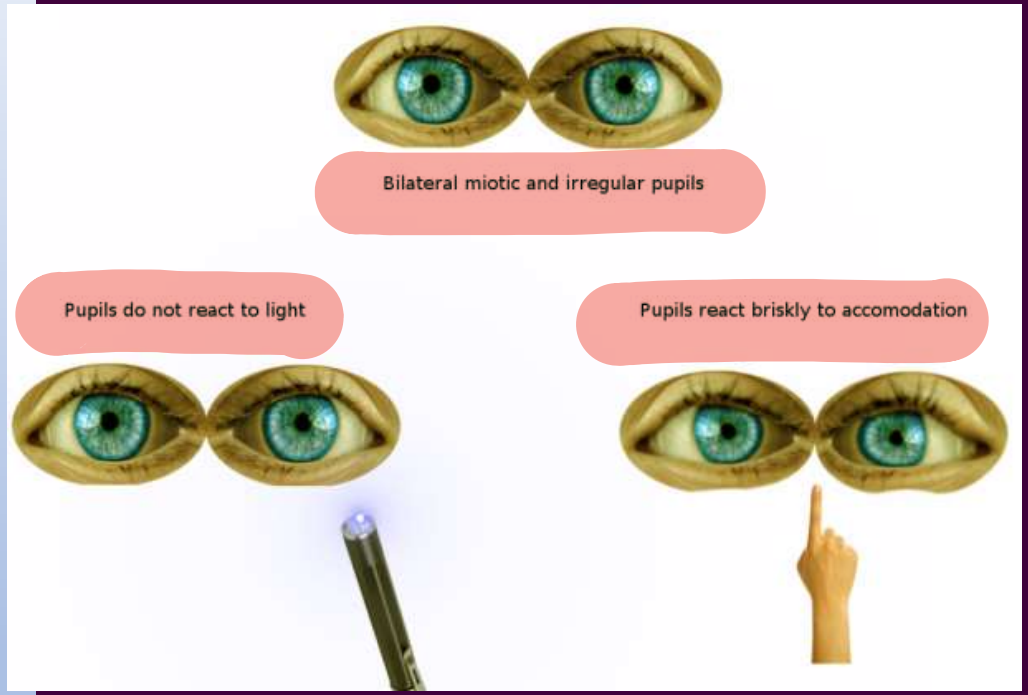
Marcus Gunn pupil.

- Optic Nerve lesions
- Severe Retinal CRAO RD.

Swinging Flash light

Argyll Robertson Pupil

A R P



Light reflex
- absent

- Accomodative Reflex present

Lesion :
- Pretectal
- Neurosyphilis
- DM.

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Adie's pupil

- unilateral
- Tonic pupil
- Light reflex (-)
  Near reflex (-)
  - pupil slow to Dilate

ciliary ganglion

Loss of Deep Tendon's Reflexes

HORNER SYNDROME

oculo sympathetic paresis

- mild Ptosis
- Miosis
- Anhydrosis
- Apparent enophthalmos
- Dilation lag.

Pancoast Tumor

↓

Apex of lung, cervical sympathetic chain.

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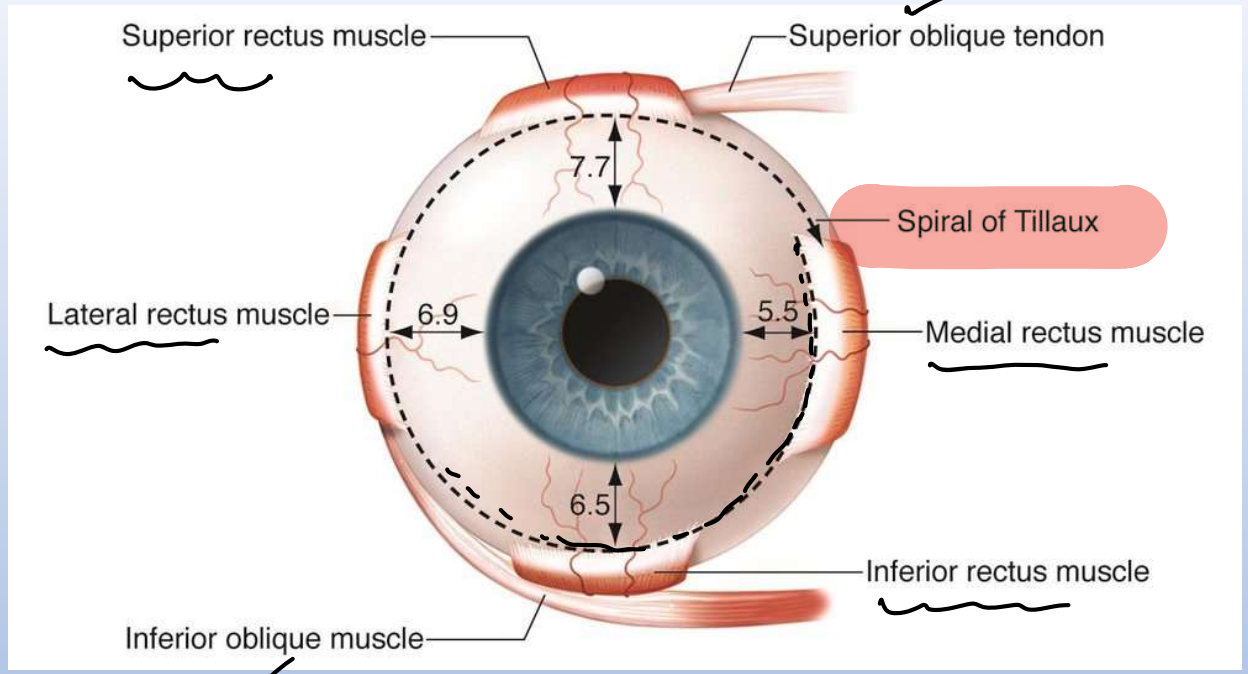
Squint



M I L S

MR is inserted closest to limbus

DR

LR

SR. Farthest from limbus.

Extraocular Muscles – Nerve Supply

- MR, IR, SR, IO – 3rd cranial nerve
- SO - 4th cranial nerve
- LR - 6th cranial nerve

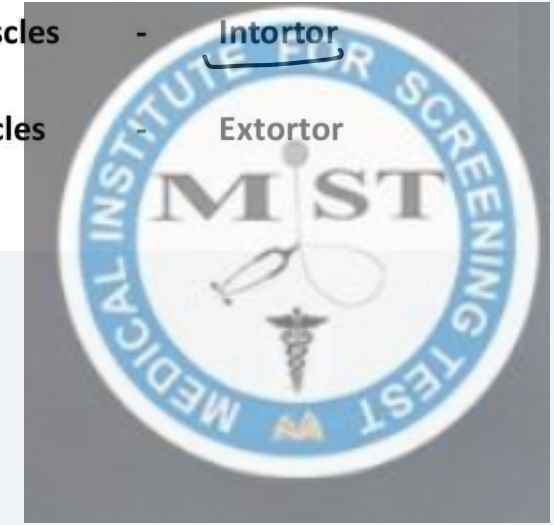
LR6 SO4 O3

Extraocular Muscles – Actions

MUSCLE	PRIMARY ACTION	SECONDARY ACTION	TERTIARY ACTION
MR	ADDUCTION	____	____
LR	ABDUCTION	____	____
SR	ELEVATION	INTORSION	ADDUCTION
IR	DEPRESSION	EXTORSION	ADDUCTION
SO	INTORSION	DEPRESSION	ABDUCTION
IO	EXTORSION	ELEVATION	ABDUCTION

Tips to remember

- **All obliques - Abductor**
- **All recti (except lateral rectus) - Adductor**
- **All superior muscles - Intortor**
  SR, SO
- **All inferior muscles - Extortor**
  IR, IO

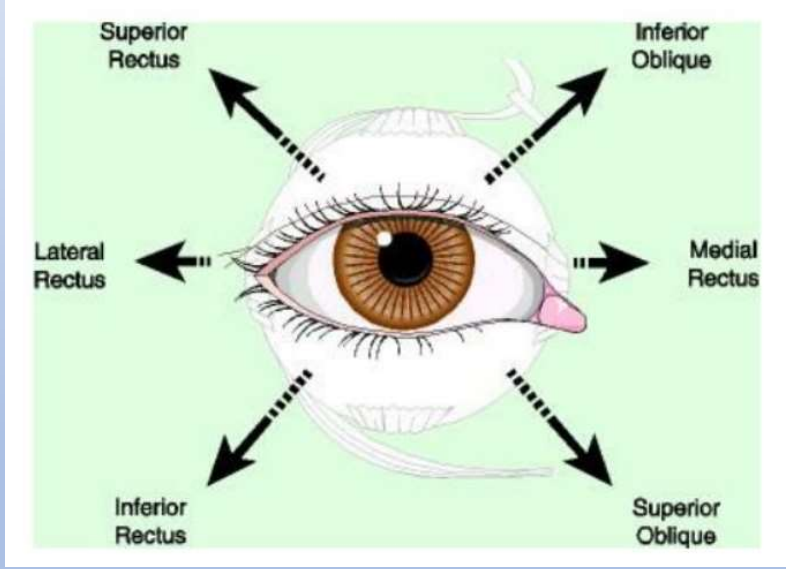


SIN RAD

IR: depresión, adduct, extorsión

Extraocular Muscles

- Synergist Muscles = muscle in same eye with same action



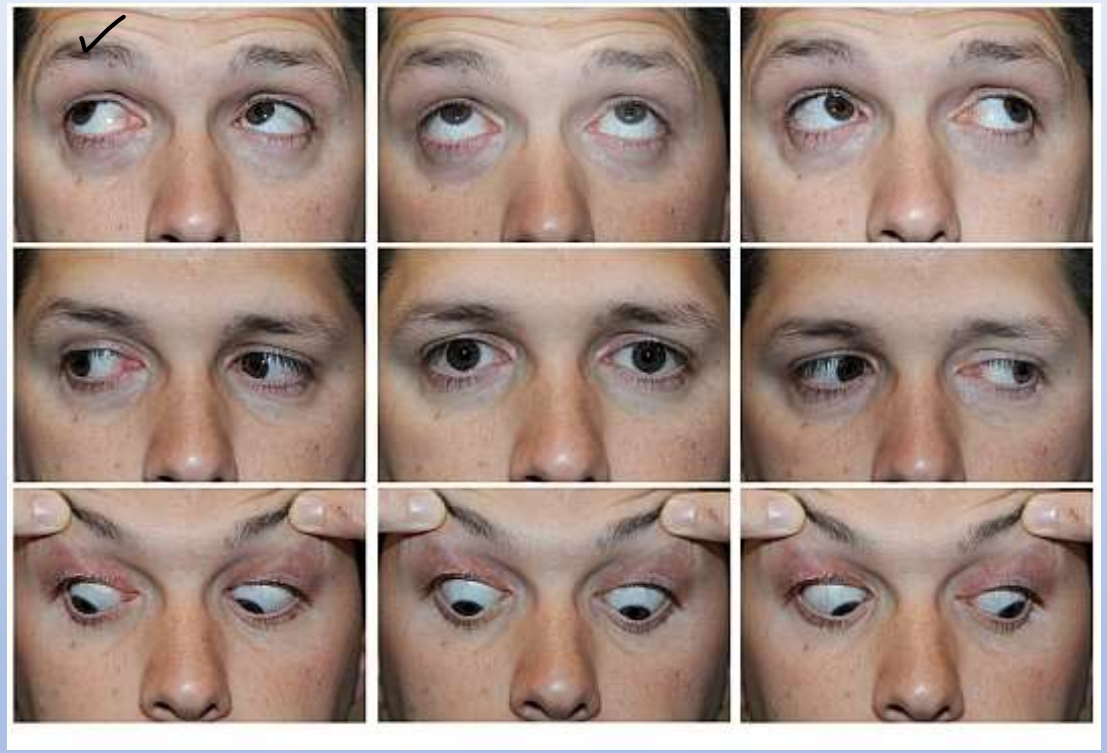
SR
IO
for elevation

Antagonist muscle MR ≠ LR

Extraocular Muscles: Yoke Muscles

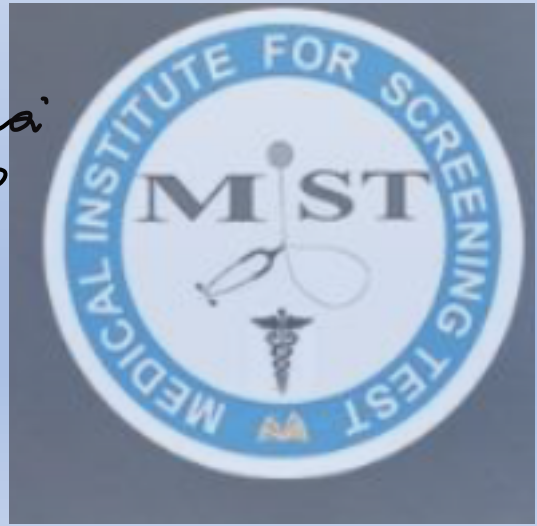
pair of muscle in each eye with movement in same direction

Dextro elevate



Levo depress
Left down

(L) IR
(R) SO



Extraocular Muscles

- Yoke Muscles

Dextroelevation : (R) up

(R) SR
↓ ↓
(L) IO

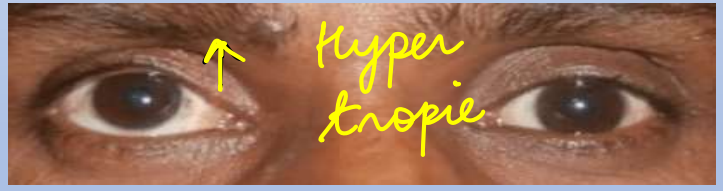
yoke

Strabismus : Ocular misalignment

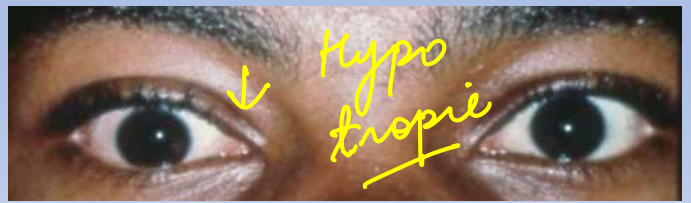
Latent Squint - Phoria Manifest Squint – Tropia

convergent squint

divergent squint



upward



downward deviation

Tests for Squint

- Hirschberg Test - Tropia
- Cover Test - Tropia
- Uncover Test- Phoria
- Alternate Cover Test
- Prism Bar Cover Test

Hirschberg Test

Corneal reflex : centre of pupil

Reflex is on
- pup. border 15° ≡ 30 PD
- mid Iris 30° ≡ 60 PD
- limbus 45° ≡ 90 PD

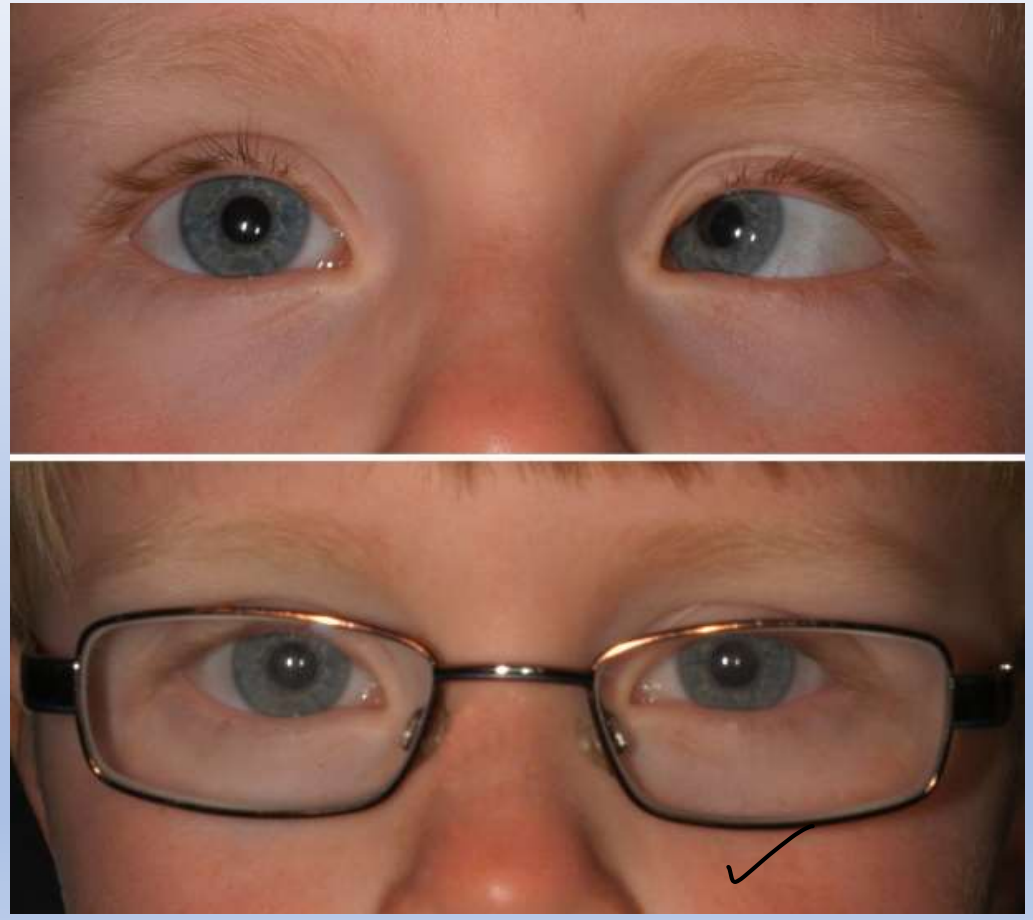
. Rules out pseudostrabismus

Accomodative Esotropia

Presents at age of 2-3 years, Most common type of squint in children

- **Refractive Type - High Hypermetropia, Normal AC/A ratio, Corrected by use of glasses.**
- **Non-Refractive Type – Minimum Hypermetropia, High AC/A ratio, Corrected by Bifocal glasses with near correction**
- **Mixed Type – Combination of High Hypermetropia and High AC/A ratio , Corrected by Bifocal glasses** with near correction

Accommodative Esotropia



Features of Paralytic Squint:

- Diplopia
- Eye movement restricted
- Compensatory head postures

6th N. palsy: LR:

left 6th N. palsy.

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3rd Nerve Palsy

- Most common cause – Ischemic (DM, HTN)
- Surgical Causes – Trauma, Tumors, Aneurysms

➢ Eye down & out

➢ Diplopia in all directions except down & out

➢ Ptosis (LPS)

➢ Pupillary Reactions

Ptosis
down & out
eye

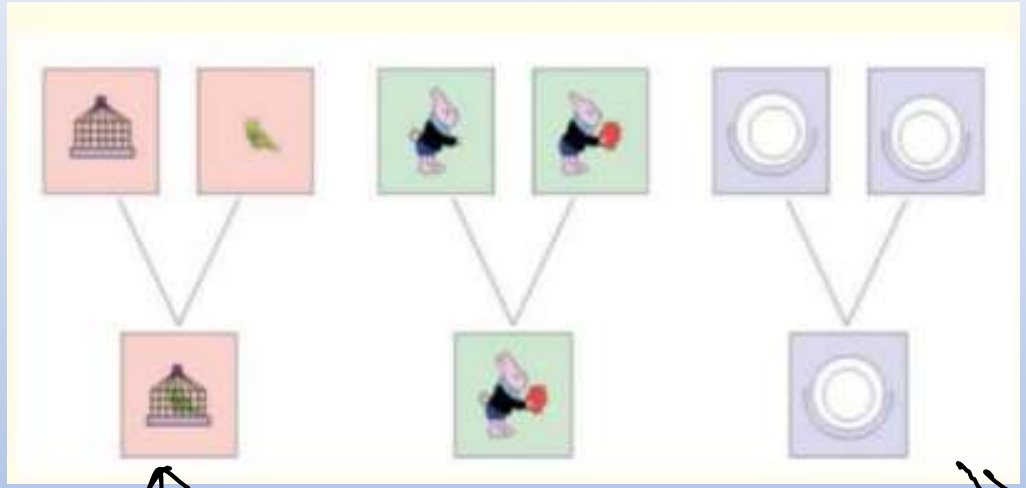
FOURTH NERVE PALSY

Superior oblique : congenital
- trauma.

- Hypertropia
- Head tilt to opposite direction
- Limited depression in adduction
- Diplopia: worse on looking down

Binocular single vision

Synaptophore



↑ Simultaneous macular perception

↓ Fusion

» Stereopsis

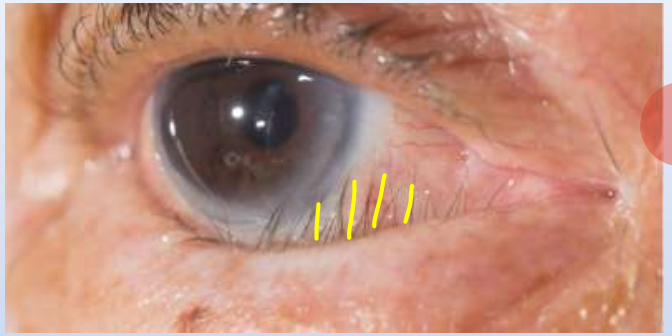
Titmus Fly Test : Stereopsis

Amblyopia: suppression of vision without any organic cause.

TREATMENT

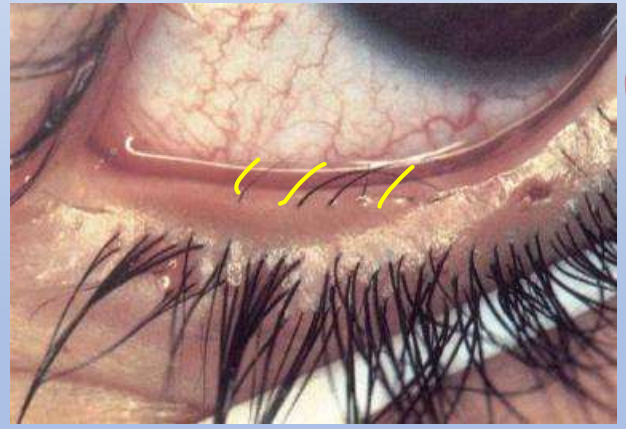
- **Occlusion therapy**
- **Penalisation**
- **Active Vision Therapy**

Lids and Adnexa



Trichiasis

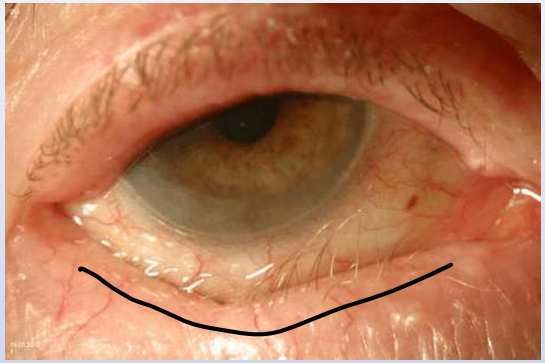
misdirected eye lash



Distichiasis

extra row of lashes

Entropion



inward rolling of lid margin

Surgeries – Jone's procedure

Weis procedure

Quickert procedure

Wheeler's procedure

Ectropion

outward rolling of eyelid margin

Surgeries:

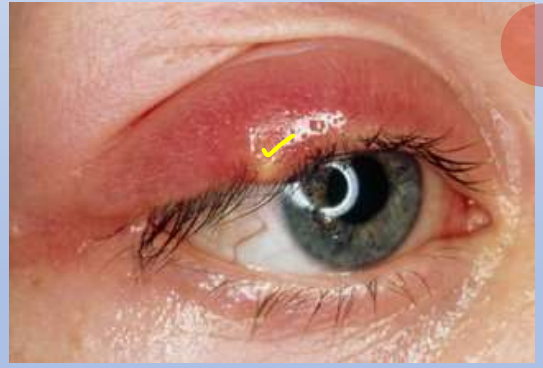
- Medial conjunctivoplasty
- Modified Kuhnt procedure
- V-Y procedure
- Lazy T procedure
- Z-plasty

Chalazion

meibomian glands

Painless nodule away from the lid margin

Incision & curretage



Stye

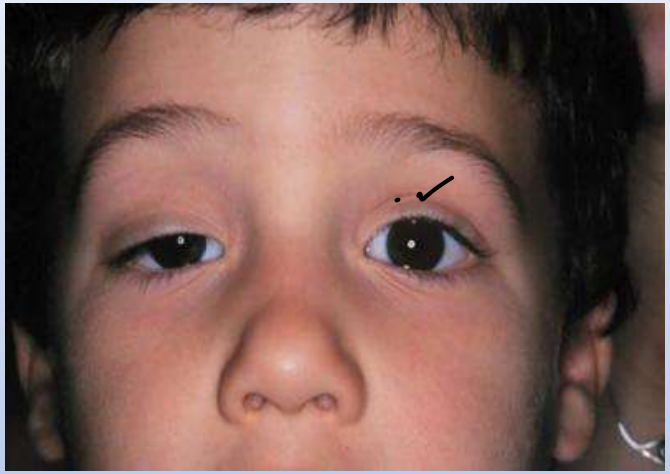
Painful nodular at the lid margin

Zeis glands

glands of moll

Ptosis

Drooping of lids



. congenital Ptosis
  ↳
. absence of lid crease
. Lid lag.

Surgery for Ptosis

- Fasanella servat procedure Mild Ptosis
- Levator resection Moderate Ptosis
- Frontalis sling surgery/Frontalis suspension Severe Ptosis

Complicated Ptosis

Marcus Gunn Synkinetic ptosis

(Marcus Gunn Jaw Winking Phenomenon)

Aberrant connection b/w third & fifth nerve

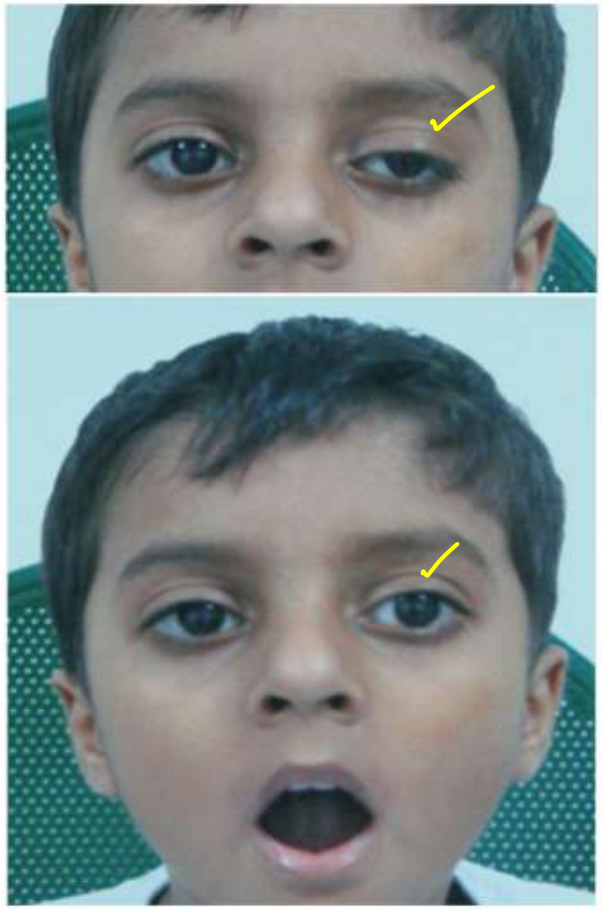


Figure 1: Patient with Marcus Gunn Jaw-Winking phenomenon. (A) 5year old child with unilateral upper eyelid ptosis as part of Marcus Gunn phenomenon. (B) Elevation of concomitant upper eyelid with mouth opening

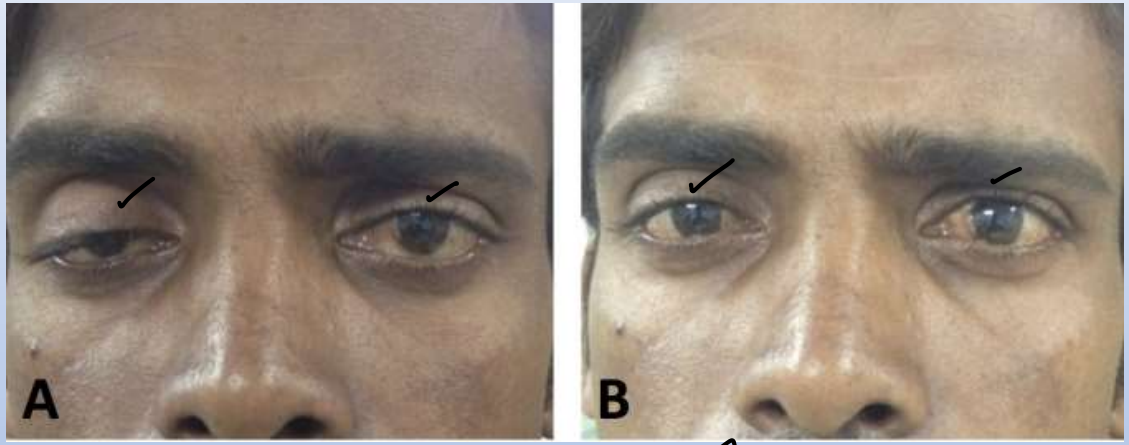
Myasthenia Gravis

Ptosis : Bilateral asymmetrical

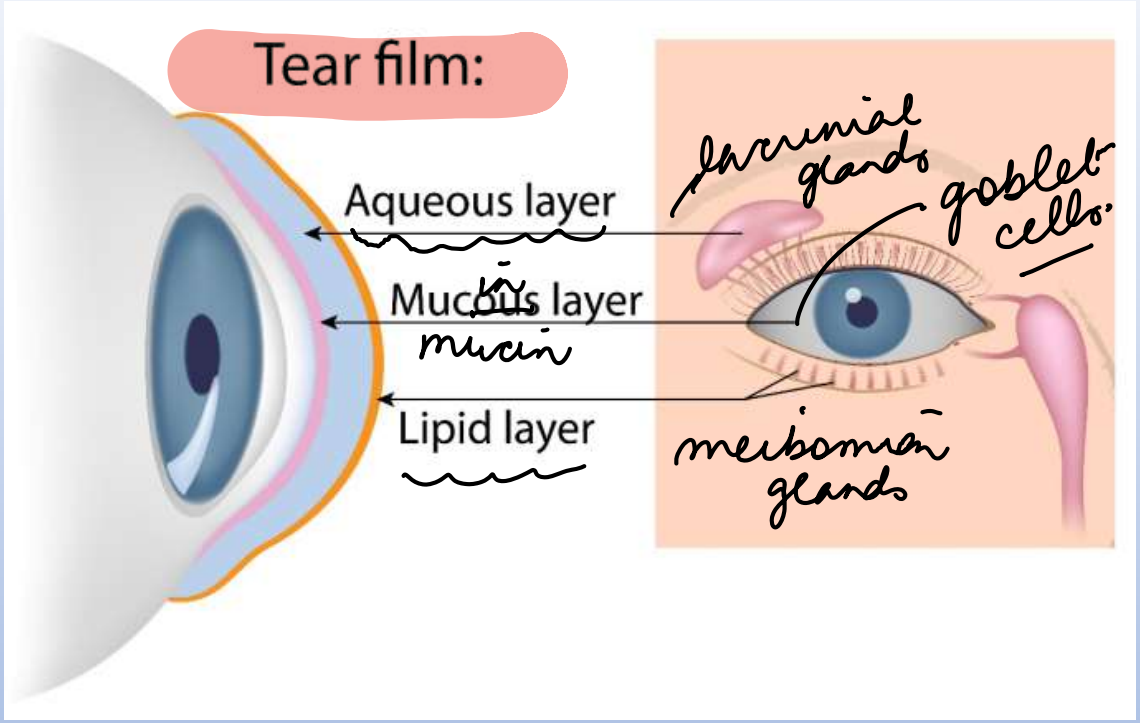
- Intermittent, worse by evening.
- Diplopie, variable

Myasthenia Gravis

Icepacks. empirone Test



Tensilon Test
Neostigmine


Tear film:
Aqueous layer
Mucous layer
Lipid layer
lacrimal glands
goblet cells
meibomian glands

Dry Eye

Schirmer's Test < 10 mm

Dry Eye.

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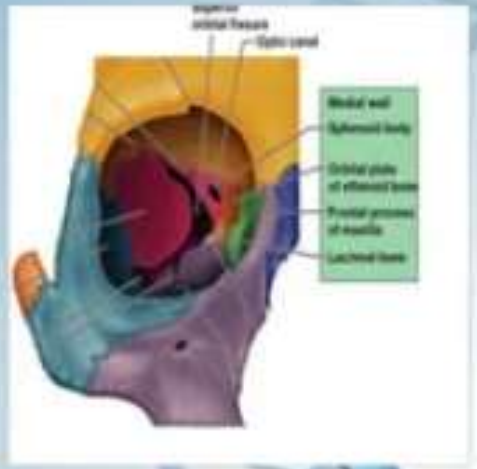
Applied anatomy: Lacrimal Sac

Medial Orbital Wall

- The medial wall is formed from front to back by the:

  - Frontal process of maxilla
  - Lacrimal bone
  - Orbital plate of the ethmoid bone
  - Body of the sphenoid bone.



↓
lacrimal fossa

Congenital NLDO

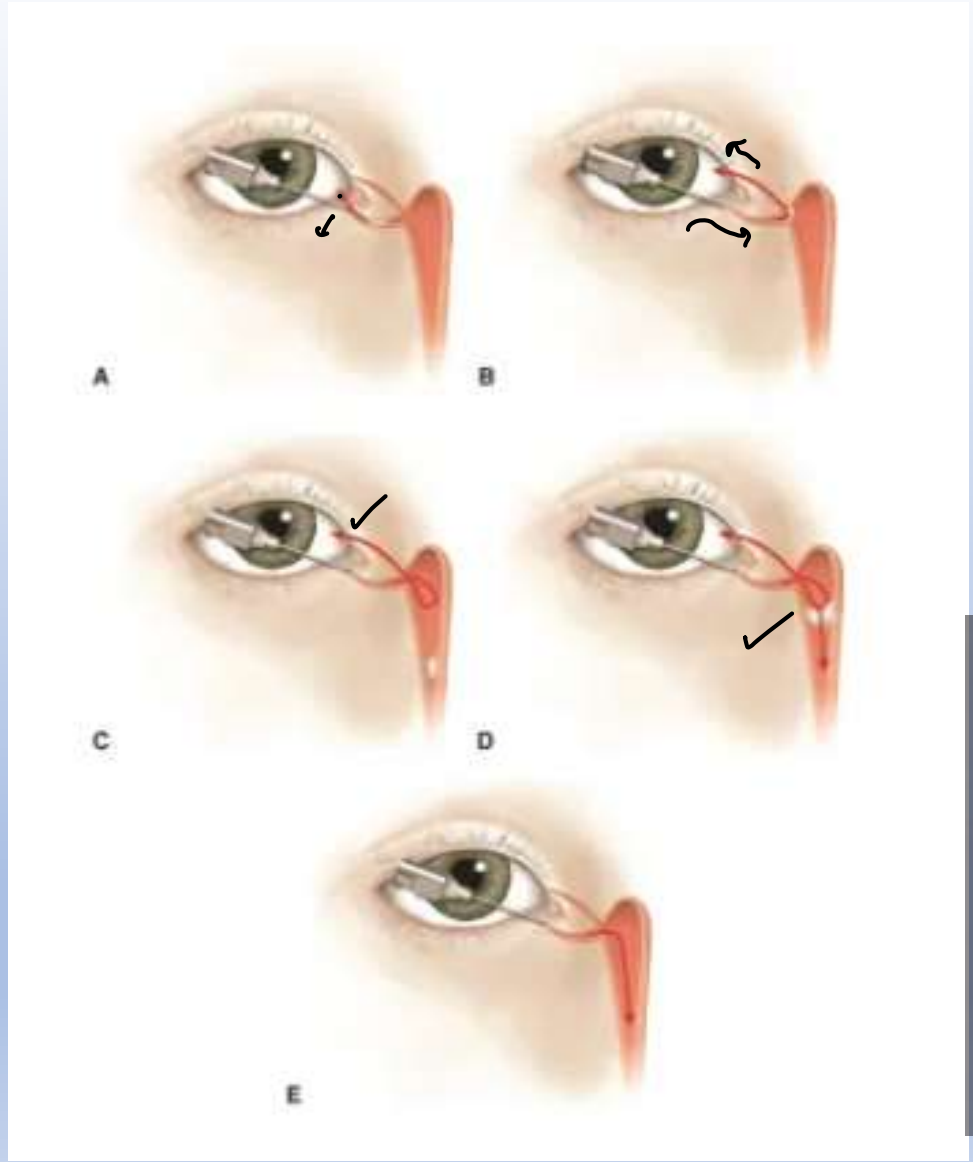
Failure of canalization of NLD by end of gestation.
Site of obstruction – Valve of Hasner

Rx Sac massage

- Criggler 's massage – till 9 months
- Syringing & probing – 9 months -4 year
- DCR after 4 years

DCR – anastomosis between lacrimal sac & middle meatus

Syringing



A, Complete canalicular obstruction: The cannula is advanced with difficulty, and irrigation fluid refluxes from the same canaliculus.

B, Complete common canalicular obstruction: A “soft stop” is encountered at the level of the common canaliculus, and irrigated fluid refluxes through the opposite punctum.

C, Complete nasolacrimal duct obstruction (NLDO): The cannula is easily advanced to the medial wall of the lacrimal sac, then a “hard stop” is felt, and irrigation fluid refluxes through the opposite punctum. Often, the refluxed fluid contains mucus and/or pus.

D, Partial NLDO: The cannula is easily placed, and irrigation fluid passes into the nose as well as refluxing through the opposite punctum.

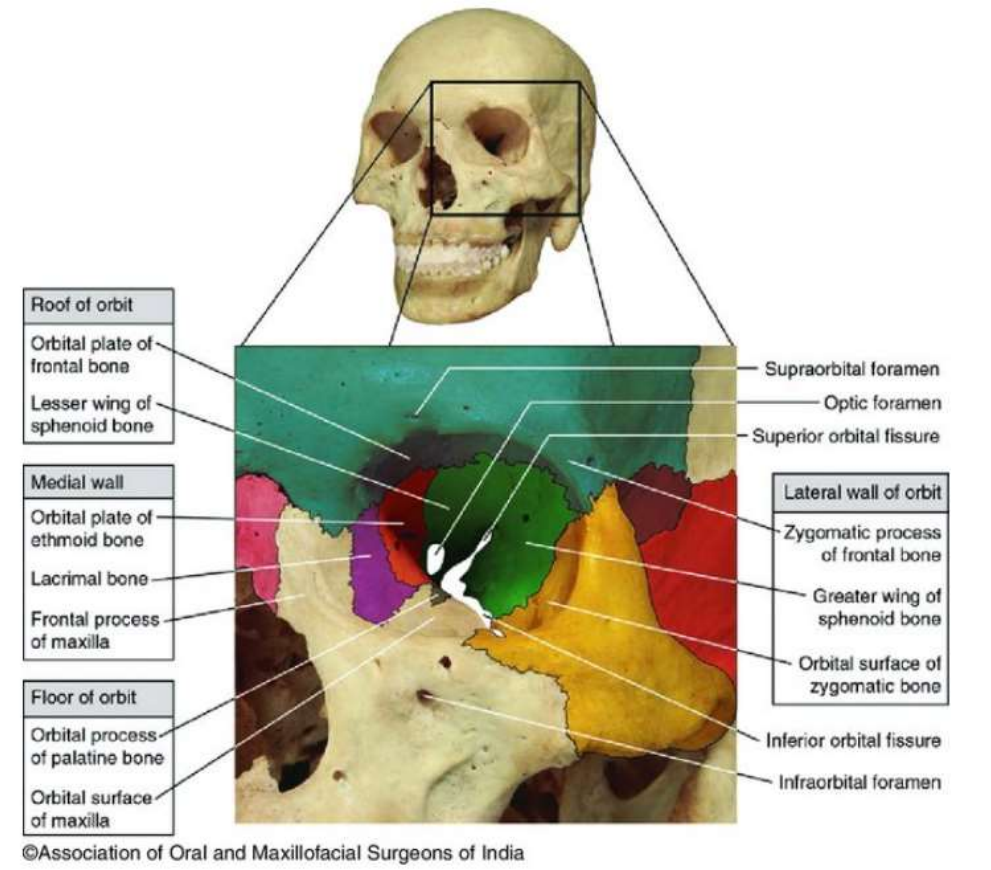
E, Patent lacrimal drainage system: The cannula is placed with ease, and most of the irrigation fluid passes into the nose

Dacryocystorhinostomy DCR:

1. Conventional External DCR for NLD obstruction
2. Endonasal DCR for NLD obstruction
3. Endocanalicular laser DCR for NLD obstruction: uses Holmium YAG laser
4. Conjunctivo-DCR with Jones tube for canalicular obstruction

Orbit

Orbital Walls



Hertel's Exophthalmometer

↳ measure Proptosis.

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Proptosis

- Most common cause of unilateral proptosis in children is Orbital Cellulitis
- Most common cause of unilateral proptosis in adults is Thyroid Ophthalmopathy
- Most common cause of bilateral proptosis in children is Neuroblastoma & Leukemia
- Most common cause of bilateral proptosis in adults is Thyroid Ophthalmopathy

Proptosis

- Intermittent Proptosis:

Orbital Varices – proptosis increases on coughing, sneezing, Valsalva

bending down

- Pulsatile Proptosis: Carotico-cavernous fistula, Meningocele

First Sign in Thyroid Ophthalmopathy – Lid Retraction/Dalrymple Sign

Most common/First muscle involved in thyroid eye disease – Inferior Rectus

Proptosis

optic N. compression

Orbital Cellulitis

ocular emergency

- Painful Proptosis
- Restricted Eye movements
- Conjunctival chemosis
- Decreased Vision, RAPD

Source of Infection
: Ethmoidal sinusitis

Hospitalized
: IV antibiotics
· ENT consult.

Orbital Tumors

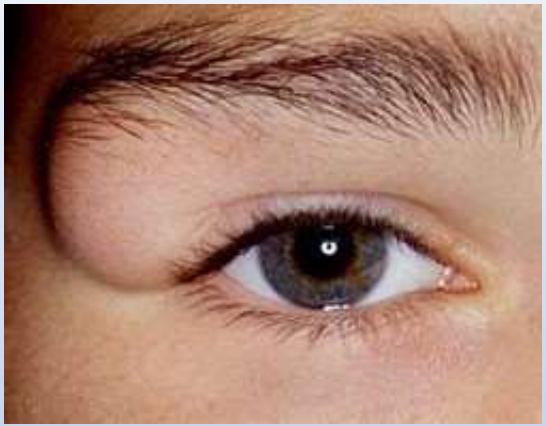
- Most common benign orbital tumor in adults is Cavernous Hemangioma

- Most common primary malignant orbital tumor in adults is Lymphoma

- Most common benign orbital tumor in children is Dermoid Cyst

- Most common primary malignant orbital tumor in children is Rhabdomyosarcoma

Dermoid cyst

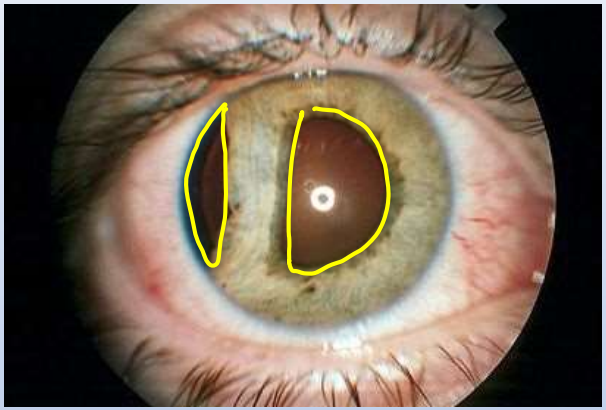


Plexiform neurofibroma of eyelid

S shaped deformity

Trauma

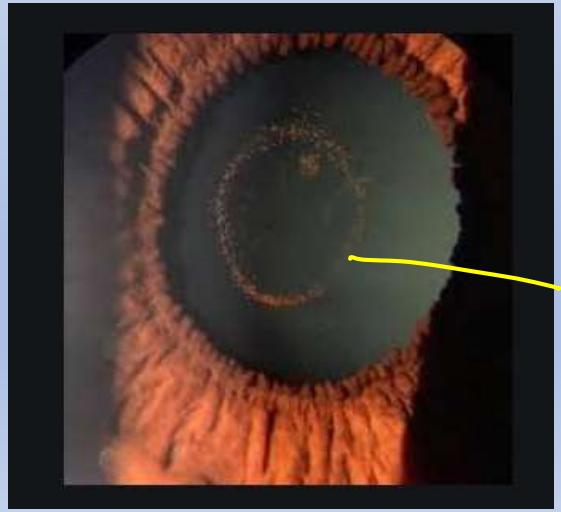
Blunt Trauma



Iridodialysis

D-shaped pupil

Hyphema is blood in anterior chamber.

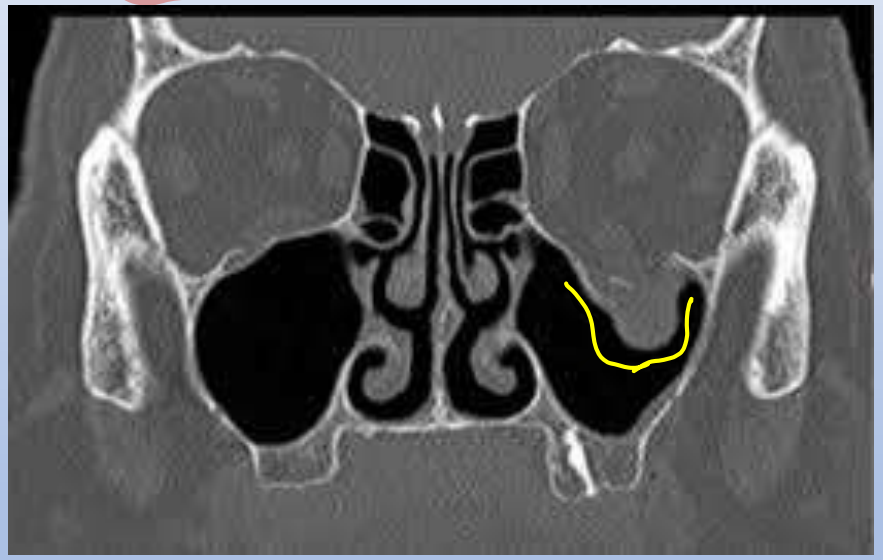


Roselle Cat.

vessels seg.

Orbital Blow out fracture

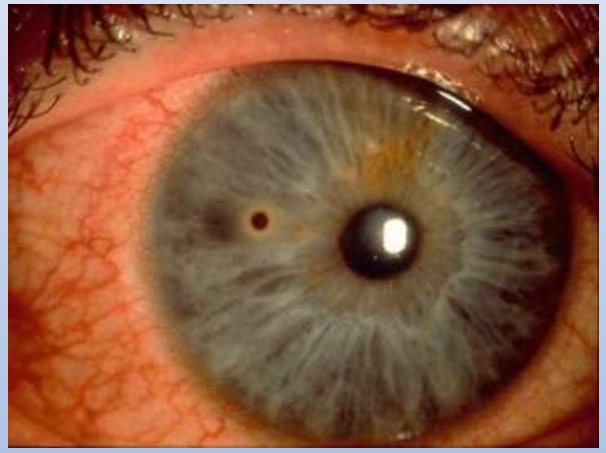
Floor of orbit



Tear drop sign

Foreign bodies:

Superficial FB removed with 26G needle.



: antibiotic
: Lubricaly

Foreign bodies:

Deep FB should be removed in OT

Foreign bodies:

Metallic FB:
Don't do MRI

Retained Intraocular foreign bodies:

metallic

- **Iron: Siderosis Bulbi**, iron deposits in cornea, lens, iris and retina.
- Can lead to retinal degeneration

- **Copper: Chalcosis**, KF ring and sunflower cataract

Chemical Injury

Alkali Injury : more severe : Lemic Injury.

Acid Injury : protein coagulation.

Immediate management: Thorough irrigation of eye with RL, normal saline till the pH returns normal.

Double Evert Lid: Remove residual chemical

Eye involvement

in

Systemic Diseases

Neurofibromatosis NF:

- Neurocutaneous syndrome
- Congenital disorders of CNS, Skin and Eyes
- 2 Types: NF-1 and NF-2
- NF-2: early onset bilateral cataracts

Ocular features of NF-1

- Lisch Nodules

iris hamartomas

- Optic nerve Glioma

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Sturge-Weber Syndrome

Hemangiomas – Conjunctival, Episcleral and Choroidal

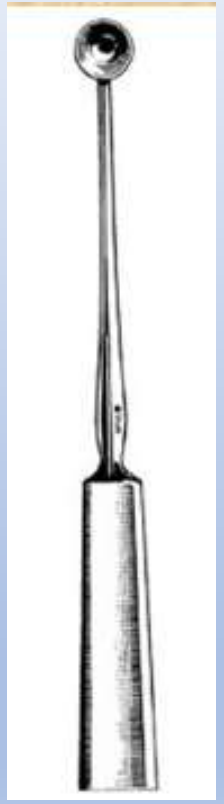
- Episcleral hemangiomas may be associated with unilateral glaucoma

Tuberous Sclerosis

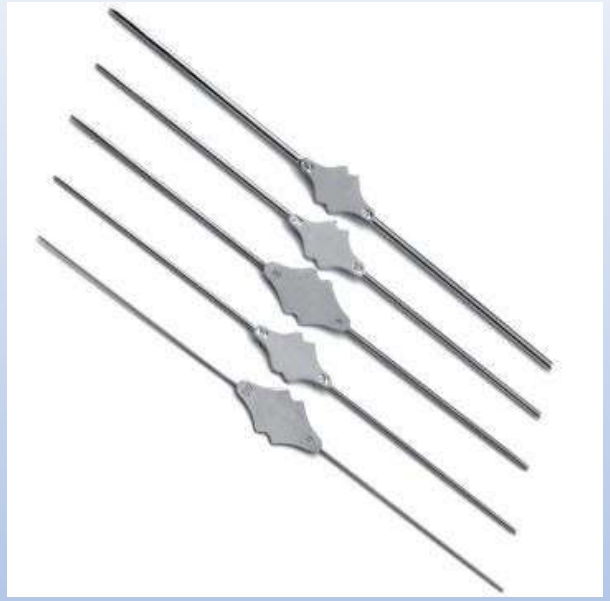
- Astrocytic Retinal Hamartomas
- Optic Nerve Hamartomas

Surgical Instruments

Chalazion Clamp & Scoop



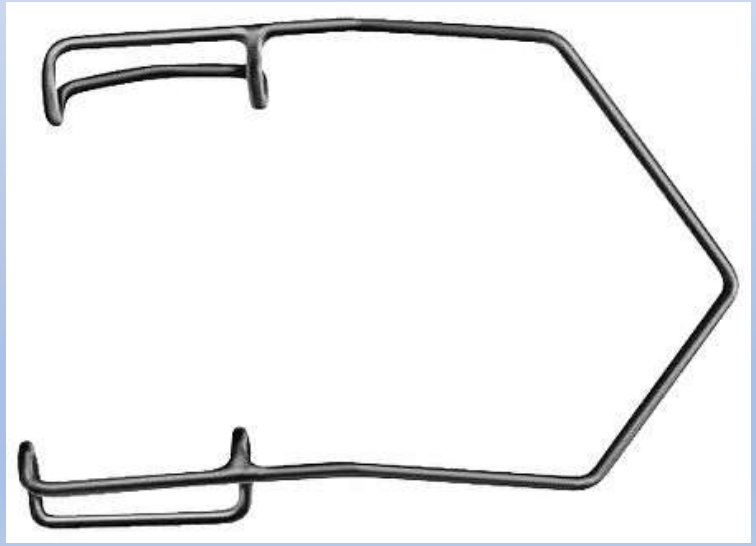
Lacrimal Probe Bowman



DCR
Kelly's
Bone
punch

Wire Speculum:

- To keep the lids open for intraocular surgeries



Superior Rectus Forceps :

- Used in ECCE and SICS to hold SR muscle
- Has wide teeth

Corneo-scleral Forceps : Lim's forceps

- Used in Cataract & Glaucoma surgery
- Has fine teeth

Needle Holder for suturing

McPherson Forceps:

- Used for corneoscleral suturing and to hold IOL
- Plain, no teeth

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For corneoscleral incisions

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Phaco Probe

Anaesthesia for Cataract surgery:

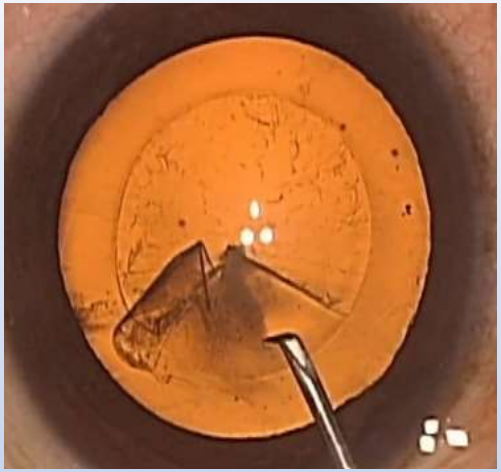
1. Local anaesthesia – Peribulbar or retrobulbar injection

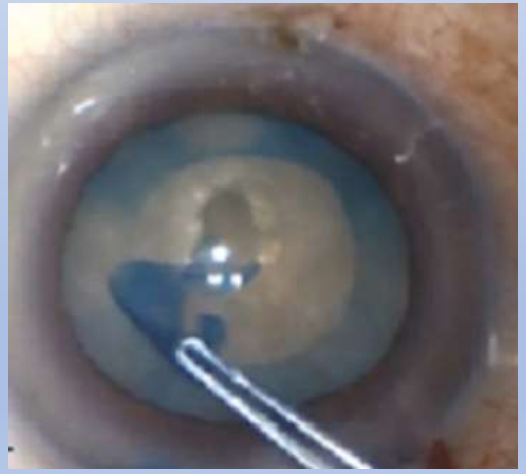
   DOC: 2% xylocaine/lignocaine + 0.5% Bupivaciane

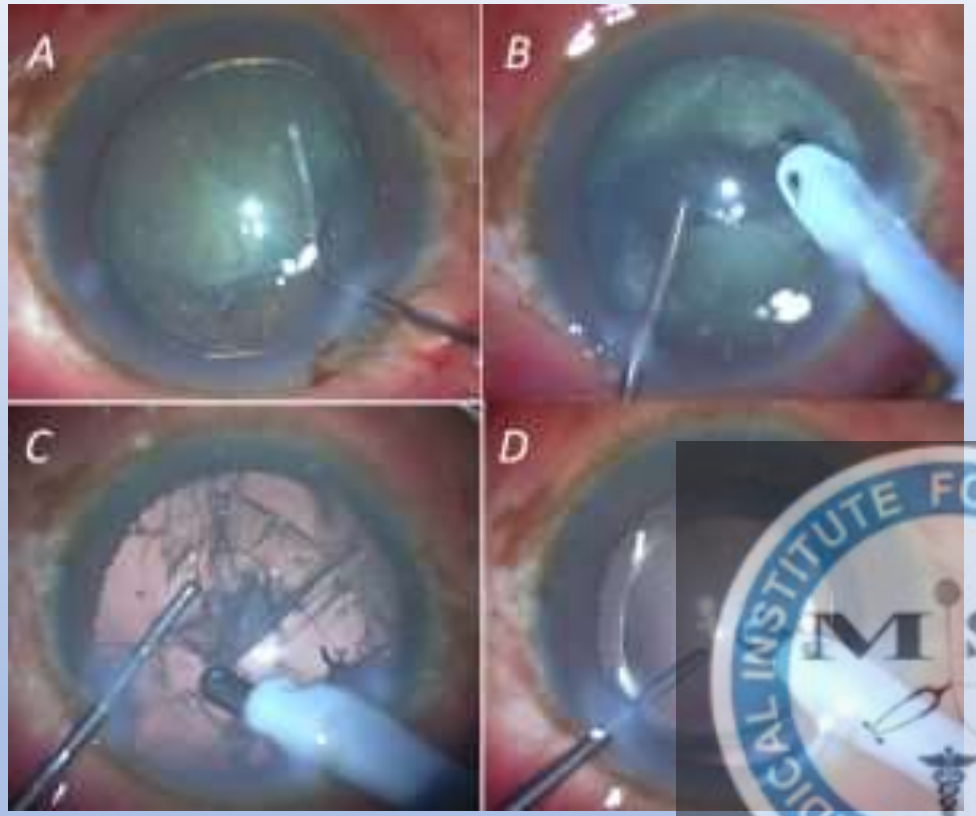
2. Topical anaesthesia –eye drops

   DOC: 0.5% Proparacaine

Viscoelastics: Sodium Hyaluronate, Methyl Cellulose, Chondroitin Sulphate



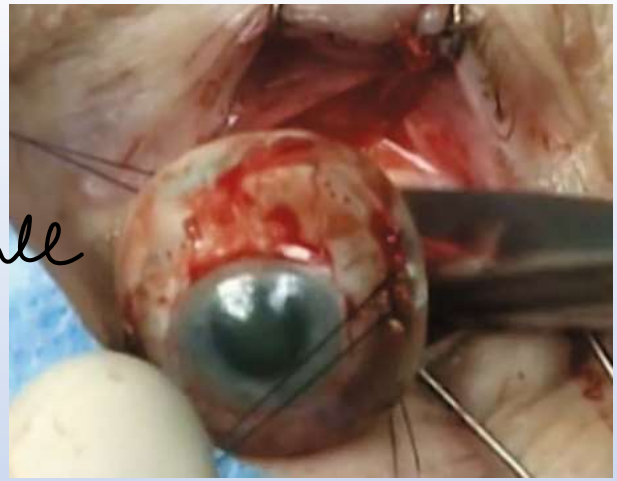



A
B
C
D

Enucleation

removal of eyeball

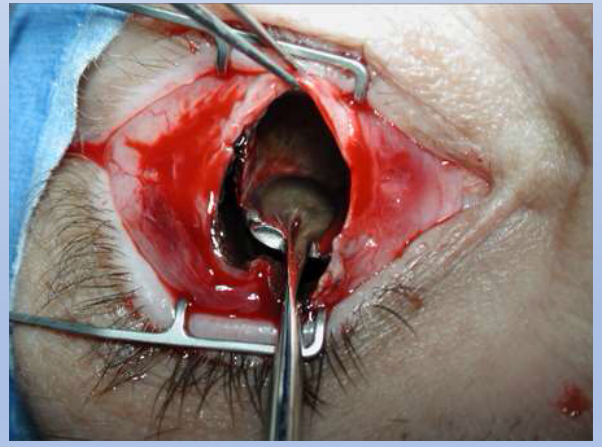
intraocular tumors



Evisceration

remove the middle and inner layers

Panophthalmitis



MEDICAL INSTITUTE FOR SCREENING TEST
MIST

LASERS in OPHTHALMOLOGY

- **Nd YAG LASER :** Capsulotomy, Iridotomy
- **EXCIMER LASER:** PRK, LASIK
- **FEMTOSECOND LASER:** SMILE, LASIK, Laser assisted Cataract surgery
- **Fd Nd YAG LASER:** Retinal Photocoagulation
- **DIODE LASER:** Retinal Photocoagulation, Thermotherapy
- **ARGON LASER:** Retinal Photocoagulation

All the Very Best